

ARROW FLOW MATCHING FOR REACTION MECHANISMS

Anonymous authors

Paper under double-blind review

ABSTRACT

Predicting a reaction mechanism means predicting where valence electrons go, and a prediction must conserve electrons and atoms exactly. We construct a generative model of one elementary step whose state is an integer bond–electron matrix and whose dynamics is a continuous-time Markov chain trained by discrete flow matching, and show that such an entrywise model has a defect no amount of training removes. A product measure supported on a level set of a non-trivial linear functional must be a point mass in every coordinate, so an entrywise model is compatible with exact electron conservation only if it is deterministic. We remove the defect by changing the chain’s elementary event. Arrow Flow Matching applies, instead of flipping one coordinate, one *move* from a four-element alphabet whose increments cancel identically, the matrix form of a chemist’s curly arrow, and it may terminate only at states an enumerated per-atom table accepts. Conservation becomes an algebraic identity and validity a mask on one softmax class, both holding with probability one, and every state the chain visits is a partially executed mechanism. Arrow Flow Matching attains validity and electron, atom and proton conservation of exactly one at every model size at equal step accuracy, and at its smallest width it already scores higher overall than every other model in the study at every width.

1 INTRODUCTION

Machine-designed routes are now run in the laboratory (Segler et al., 2018; Coley et al., 2019b; Mikulak-Klucznik et al., 2020), and learned models rank which reactions are worth running (Ahne-man et al., 2018; Dai et al., 2019; Xu et al., 2025). All reason about outcomes, while selectivity and byproducts are decided by the *mechanism* (Ren et al., 2026; Das et al., 2026), an ordered sequence of elementary steps. One step takes molecules to molecules over a *fixed* set of atoms and only the valence electrons move, so it redistributes small integers over the index set the *atom map* fixes. A model of that redistribution is a model of the mechanism, so its intermediate states, not only its answer, are readable chemistry. For machine learning it is a clean discrete testbed, with no continuous coordinates, no variable-length objects, and one exact, cheap, hard constraint.

Where the field stands. The dominant formulation treats a reaction as a string-to-string translation, mapping a reactant SMILES (Weininger, 1988) to a product SMILES (Nam & Kim, 2016; Schwaller et al., 2018; 2019; Tu & Coley, 2022). These models are accurate and remain the strongest baselines, but their output space contains no notion of an atom inventory or an electron budget, so conservation is not expressible in the space the decoder searches, so it fails exactly when the decoder does. A second family predicts *edits* on the molecular graph, a reaction centre and its local changes (Jin et al., 2017; Coley et al., 2019a; Sacha et al., 2021; Somnath et al., 2021; Chen & Jung, 2021), which fixes the atom multiset by construction but leaves the electron count unconstrained. A third family predicts the *mechanism* rather than the outcome. Kayala & Baldi (2012) rank elementary polar, pericyclic and radical steps between molecular orbitals, ELECTRO (Bradshaw et al., 2019) generates a reaction as a sequence of electron-path steps, Reactron (Chen et al., 2025) emits the arrow-pushing diagram of each step, and recent language models encode electron flow in an annotated SMILES (Neukomm et al., 2025). Mechanistic data at the scale these models need was assembled by Joung et al. (2025), whose FlowER transports the reactant bond–electron matrix (Dugundji & Ugi, 1973) to the product matrix by continuous flow matching (Lipman et al., 2023).

Why exactness is still open. Two of these models take the constraint seriously and neither delivers both halves of it. ELECTRO attains validity of exactly one because it never emits a molecule at all, only a path of electron movements over a structure that already exists. The price is coverage and accuracy. Its paths are linear and alternate bond removal with bond addition, which excludes the radical and concerted chemistry a mechanistic corpus contains, and in our study it reaches a top-1 step accuracy of 0.66 against 0.87 for the best model in the comparison. FlowER makes conservation a property of the *drift*. The target $\Delta B = B_p - B_r$ has zero weighted sum and centring the predicted field keeps it there along the whole trajectory. What the guarantee does not cover is the return trip to the integers. The endpoint of the differential equation lies off the lattice and is rounded, and the rounding is sum-preserving, so the total survives but the symmetry need not, an entry and its mirror image being rounded apart. That is one of the three invalid-output modes [Joung et al. \(2025\)](#) report. The other two lie outside the rounding altogether. An endpoint that over-redistributes carries negative entries before anything is rounded, and a symmetric, non-negative, correctly summed matrix can still seat more pairs on an atom than its shell holds. Nor does it reach the N local per-atom conditions, which are nowhere represented. Measured here, that model returns a valid molecule on 0.82 of its candidates. So a model exact on validity has been available, and a model exact on the global electron functional has been available, but not one exact on both while competitive in accuracy.

Contributions. We build a generative model of one elementary step twice over the same network body, changing only what its elementary event is, and show that the second version makes both guarantees hold with probability one at no cost in accuracy. Our contributions are the following.

- **Discrete FlowER** (§3, Appendix B). A discrete flow-matching model directly over integer bond-electron matrices, built from a Markov bridge pinned at the two data points, an endpoint head, a cross-entropy loss and an Euler sampler that resamples entries. Every intermediate state is an integer matrix, hence a partially executed mechanism, and decoding is the identity map.
- **An obstruction** (§4, Theorem 1). The head of any entrywise model is a product measure, the admissible matrices form a level set of a linear functional, and a product measure supported on such a level set is a point mass in every coordinate. Exact conservation and calibrated uncertainty are incompatible for this hypothesis class, whatever the data or the capacity.
- **Arrow Flow Matching** (§5). The elementary event becomes one move of a four-element alphabet lying in the kernel of the electron functional. Conservation is then an algebraic identity holding surely (Proposition 2), and gating the *stop* class by a table enumerated against the same valency checker the model is graded by makes the emitted state a molecule (Proposition 3). The mask costs $O(1)$ per candidate on a dense $(4, N, N)$ grid, and because the moves are sequential the model also admits a beam decoder and an exact trajectory likelihood, neither of which an entrywise head can supply.
- **Evidence** (§6). Across five model sizes and eight baselines, validity and electron, atom and proton conservation are exactly one at every size, while step accuracy matches the best model in the study and sits within 0.015 of the ceiling that branching in the data imposes on any model. The inductive bias also buys capacity. At the smallest width, 1.3M parameters, Arrow Flow Matching already has a higher overall score than any other model at any of the five widths, so the guarantees come with a substantially smaller network rather than at the price of a larger one.

A detailed discussion of related work, covering mechanistic reaction prediction, flow matching on continuous and discrete spaces, and constrained generative models, is in Appendix H.

2 THE OBJECT AND THE CONSTRAINT

All the chemistry this paper needs. A molecule is a labelled graph carrying a count on every vertex and every edge. Vertices are atoms labelled by element, an edge carries a *bond order* $\in \{1, 2, 3\}$ counting the electron *pairs* the two atoms share, and a vertex carries a *lone-pair count*, the electrons sitting on that atom alone ([Lewis, 1916](#)). Electrons move in pairs, and a chemist draws a step as curved arrows, each taking one pair from where it is to where it goes. An arrow has one tail and one head because departure and arrival are one event. Finally, each atom holds only so much, since carbon, nitrogen and oxygen accommodate at most four pairs in total, counting lone and shared, and hydrogen at most one. This is the *octet rule*, a capacity constraint on a vertex. These three facts

become, respectively, a state space, a move alphabet and a mask. Figure 3 in Appendix A.2 draws them on one real molecule, and Figure 4 in Appendix A.3 draws one step as a chemist’s arrows and as a matrix.

The matrix. Fix the N mapped atoms of one step and let $a = (a_1, \dots, a_N)$ record their elements. The atom map is a bijection between the reactant side and the product side, so a is the same vector on both of them and never changes. A set of molecules on these atoms is encoded by a symmetric matrix $B \in \text{Sym}_N := \{Z \in \mathbb{R}^{N \times N} : Z = Z^\top\}$, the *bond–electron matrix* of Dugundji & Ugi (1973), in which B_{ij} for $i \neq j$ is the bond order of the pair $\{i, j\}$ and B_{ii} is the number of lone-pair electrons on atom i . The units deliberately differ on and off the diagonal, pairs off it and electrons on it. This is inherited from the data format and is the source of every factor of two below. Since B is symmetric it is fixed by its upper triangle, so write

$$\mathcal{E} := \{(i, j) : i \leq j\}, \quad \mathcal{D} := \{(i, i)\}, \quad \mathcal{O} := \mathcal{E} \setminus \mathcal{D},$$

and write Z_e for the entry at $e \in \mathcal{E}$. An off-diagonal *position* $e \in \mathcal{O}$ is one position that the matrix layout prints twice, at (i, j) and at (j, i) . Write B_r, B_p for the reactant and product matrices and $\Delta := B_p - B_r$. Figure 3 in Appendix A.2 carries one reactant through the structure a chemist draws, the matrix the model is handed, and the quantity that must not change.

Two per-atom quantities, and one global one. Let V_a be the group valence of element a (its outer-shell electron count as a neutral free atom, so $V_C = 4$, $V_O = 6$, $V_H = 1$) and n_i^H the number of hydrogens on atom i left implicit. Define the *bonding-pair* and *total pair* counts at atom i , and the formal charge:

$$\beta_i(Z) = \sum_{j \neq i} Z_{ij} + n_i^H, \quad c_i(Z) = \frac{1}{2} Z_{ii} + \beta_i(Z), \quad q_i = V_{a_i} - Z_{ii} - \sum_{j \neq i} Z_{ij} - n_i^H, \quad (1)$$

the subtracted terms in q_i being the electrons assigned to atom i , and the $\frac{1}{2}$ in c_i converting the diagonal from electrons to pairs. The valence-shell occupancy of atom i in electrons is $Z_{ii} + 2\beta_i = 2c_i$, so the octet rule reads

$$Z_{ii} + 2\beta_i(Z) \leq 8 \iff c_i(Z) \leq 4. \quad (2)$$

This is a *cap on the number of pairs touching a nucleus*, hydrogen’s duet rule being $c_i \leq 1$. The total valence-electron count is a *linear* functional of the matrix. The diagonal being in electrons, the off-diagonal in pairs, and each unordered pair printed twice, the weights are $w_e = 1$ on $\mathcal{D}$ and $w_e = 2$ on $\mathcal{O}$, and

$$\mathcal{N}(Z) := \sum_{e \in \mathcal{E}} w_e Z_e = \sum_i Z_{ii} + 2 \sum_{i < j} Z_{ij} = \sum_{i,j} Z_{ij}, \quad \mathcal{N}(\Delta) = 0. \quad (3)$$

The off-diagonal weight 2 being exactly the multiplicity of the symmetric layout, $\mathcal{N}$ is nothing but the sum of the whole matrix. And $\mathcal{N}(\Delta) = 0$, equivalently $\mathcal{N}(B_p) = \mathcal{N}(B_r)$, is the one piece of physics the model must respect.

What a solution must provide, and what is checked. Given B_r we require a conditional distribution $p_\theta(\cdot \mid B_r)$ that can be both *sampled* and *scored*, since a downstream search over pathways consumes a ranked list. A candidate $\hat{B}$ is useful only if it is a molecule, and being a molecule is four conditions and nothing more. Does $\hat{B}$ carry the reactant’s multiset of heavy atoms, its count of hydrogens, and its total valence-electron count, and is it *chemically valid*, meaning that the electrons it assigns satisfy the valency rules of the elements involved, so that every atom’s shell occupancy and formal charge are ones that element admits? Everything in this paper is aimed at making the answer to all four be “yes” with probability one. The first two are automatic, the atom map fixing the vertex set, and the third reduces to $\mathcal{N}$ (Appendix A.1). Hence

$$\text{a perfect score on all four checks} \iff \Pr [\mathcal{N}(\hat{B}) = \mathcal{N}(B_r)] = \Pr [\hat{B} \text{ is chemically valid}] = 1. \quad (4)$$

The second probability is the hard one to write down. Valency is a rule per element, not a single equation on the matrix, which is why §5 discharges it by an enumerated table rather than by algebra, and why that table is built against the same valency checker the experiments score with. Training data are steps with empirical law $\pi(x_0, x_1)$, writing $x_0 = B_r$ and $x_1 = B_p$ throughout, these names anticipating the roles of start and end of a process in time, and the element vector a conditions everything and is suppressed. Appendices A.2 and A.3 give a tutorial-length development of this section, covering the three descriptions of a molecule side by side, the per-atom quantities worked out entry by entry on Figure 3, and one complete step with its difference matrix, for a reader who wants the encoding in more detail than the main text carries.

3 DISCRETE FLOW MATCHING OVER THE MATRIX

Every entry of B is a small non-negative integer (Figure 3), so the first model keeps the lattice and follows the discrete flow-matching recipe (Campbell et al., 2024; Gat et al., 2024) literally. We call it **Discrete FlowER** and summarise it here. Appendix B gives its detailed derivation and description, namely the path, the generator, the marginal rate, the loss, the sampler, the network parameterisation and the algorithm. The state space is the box

$$\mathcal{V}_{\text{off}} = \{0, 1, 2, 3\}, \quad \mathcal{V}_{\text{dia}} = \{0, \dots, 14\}, \quad \mathcal{V}_e = \begin{cases} \mathcal{V}_{\text{off}}, & e \in \mathcal{O} \\ \mathcal{V}_{\text{dia}}, & e \in \mathcal{D} \end{cases} \quad \mathcal{X} = \prod_{e \in \mathcal{E}} \mathcal{V}_e \quad (5)$$

and the source of the process is the conditioning object itself, so the path is a Markov bridge pinned at the two data points $x_0 = B_r$ and $x_1 = B_p$, and no prior appears. Each position independently holds its reactant value with probability $1 - \kappa_t$ and its product value otherwise, with a schedule $\kappa_0 = 0$, $\kappa_1 = 1$, and we use $\kappa_t = t$,

$$p_t(x | x_0, x_1) = \prod_{e \in \mathcal{E}} \left[(1 - \kappa_t) \delta_{x_e, (x_0)_e} + \kappa_t \delta_{x_e, (x_1)_e} \right] \quad (6)$$

The rate generating the marginal path is the posterior mean of the rate generating this conditional path, and it is fixed by the posterior distribution of the endpoint at each position. The network therefore predicts the product value at every position, and is trained by a plain cross-entropy at states drawn from the bridge,

$$p_\theta(\nu | x, t) = \prod_{e \in \mathcal{E}} p_\theta^e(\nu_e | x, t), \quad p_\theta^e(\nu_e | x, t) = \frac{\exp \ell_\theta(x, t)_{e, \nu_e}}{\sum_{\nu' \in \mathcal{V}_e} \exp \ell_\theta(x, t)_{e, \nu'}}, \quad (7)$$

$$\mathcal{L}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0, 1]} \mathbb{E}_{(x_0, x_1) \sim \pi} \mathbb{E}_{x_t \sim p_t(\cdot | x_0, x_1)} \left[- \sum_{e \in \mathcal{E}} \log p_\theta^e((x_1)_e | x_t, t) \right], \quad (8)$$

whose minimiser recovers the exact marginal generator. Sampling is an Euler scheme that at each of S steps resamples every position from its endpoint distribution with a probability rising from $1/S$ to 1 and otherwise leaves it untouched, and the emitted candidate is the per-position arg-max, whose exact log-probability the factorised head supplies at a single sample. Every intermediate state is an integer matrix, so decoding is the identity map and nothing is rounded, projected or repaired, which is where this model departs from Continuous FlowER (Joung et al., 2025) (Figure 6 in Appendix B.9). The model is complete, trainable and accurate, and §6 reports it as the direct baseline. What it never does is couple two positions, and §4 shows what that costs.

4 WHY AN ENTRYWISE MODEL CANNOT BE VALID

The model of §3 factorises over positions at *two* places, its training path (6) and its head (7), and each factorisation costs it (3). Appendix C.2 shows the cost on the path, whose electron count is right in mean but has strictly positive variance. Here we show it on the head.

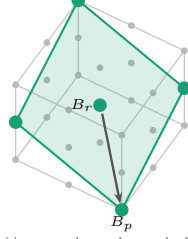
The head. Write $n := \mathcal{N}(B_r)$. The matrices carrying the right electron count form the *conservation sheet* $\mathcal{H}_n := \{x \in \prod_e \mathcal{V}_e : \sum_e w_e x_e = n\}$, the lattice points of the box (5) that one linear equation cuts out. Every molecule the model may emit lies on it, B_r and B_p included. Its one relevant feature is that it is a *slice* of the box and not a box itself, whereas the support of a product measure is always a box, the product of the supports of its factors.

Theorem 1 (a product measure cannot live on the conservation sheet). *Let $\mu = \bigotimes_{e \in \mathcal{E}} \mu_e$ be a product measure on $\prod_e \mathcal{V}_e$ with $\text{supp } \mu \subseteq \mathcal{H}_n$ for some fixed n . Then every μ_e is a point mass.*

The proof is given in Appendix G. Figure 1(a,b) draws the two shapes it compares. A *factorised head* is therefore compatible with exact conservation only if it is deterministic in every position. Any calibrated uncertainty anywhere in the matrix, which is the purpose of a generative model and what Remark 5 sets out to keep, places mass off the sheet. This is a property of the hypothesis class, not of the fit, so no quantity of data or capacity removes it, and since the chemically valid structures sit inside a level set of $\mathcal{N}$, the same argument applies to validity. In §6, the model of §3 conserves electrons on 96.0% of its candidates and is chemically valid on 97.6% of them. Figure 10

(a) what is allowed is a *slice*

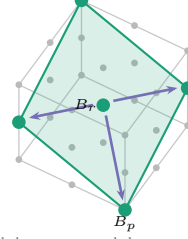
$$x_{11} + x_{44} + 2x_{14} = n$$



Three positions over three values each, the rest of the matrix frozen. All 27 points are matrices, only the five teal ones conserve, so $\mathcal{H}_n$ is the rhombus. Panels (b) and (c) both run from B_r to B_p and differ only in *what one draw may change*.

(b) one head per position spans a *box*

Each position chooses on its own, so one draw moves along an edge, changing a single coordinate and shifting $\mathcal{N}$ by $w_e \neq 0$. The reachable set is the orange box, six of whose eight corners are off the rhombus. No choice of probabilities deletes a corner, which is Theorem 1.

(c) one head over moves stays *inside*

The head chooses among whole moves, each in $\ker \mathcal{N}$, so one draw changes all three coordinates at once and every violet arrow lands back *on* the rhombus. The spread is as wide as in (b), yet every draw conserves. This is what §5 does.

Figure 1: Theorem 1, drawn on the three positions of Figure 6, together with the way out that §5 takes. A box with more than one point has an edge, an edge moves exactly one coordinate, and $\mathcal{N}$ is strictly monotone along it, so the only box that fits inside the sheet is a single point, which is (a) against (b). Panel (c) is the contrast the rest of the paper builds. Changing the elementary event from “move one coordinate” to “move a whole element of $\ker \mathcal{N}$ ” makes the support of the head a subset of the sheet of any size one likes, so calibrated spread and exact conservation stop being alternatives. Figure 7 in Appendix C.2 shows the cost of (b) on the training path rather than on the head.

in Appendix C.6 draws the two ends the theorem leaves, the deterministic head it permits and the spread it forbids, as the marginals and the electron count each induces.

The dichotomy, and the only way out of it. The theorem is not an argument against factorised heads in general. It is an argument that *exactness and spread cannot coexist while the elementary event moves one coordinate*, because a support closed under moving one coordinate at a time is a box, and a box inside a slice is a point. Read that way it says what to change. Neither the training nor the architecture is at fault, so neither is where the repair goes. The one object left is the elementary event, and if it is replaced by something whose increments already lie in $\ker \mathcal{N}$, then the support closed under it is a subset of the sheet and may be as large as one likes. That is Figure 1(c), and it is exactly what §5 does. Appendix C.5 explains why imposing the constraint only at decoding, by projecting or filtering the output, is no substitute.

5 ARROW FLOW MATCHING, CHANGING THE ELEMENTARY EVENT

§4 left one thing to change and said what has to be true of the replacement. The elementary event must have its increments inside $\ker \mathcal{N}$, and it must be named by few enough indices that the set of legal ones stays a fixed dense grid rather than a ragged list. This section exhibits such an event and calls the model it defines **Arrow Flow Matching**. Everything else, the network body, the bridge, the cross-entropy and the iterate-and-condition sampler, is carried over from §3 untouched, and Table 2 in Appendix F sets the two constructions side by side. Appendix D.1 reads a bond–electron matrix as a lattice gas, in which a curly arrow is the hop of one electron pair, and that reading motivates the construction below.

One pair, wherever it sits. How much a pair changes an entry depends on where it sits. On the diagonal, which counts electrons, a pair is two units, and off it, which counts pairs, it is one unit. Define

$$g_e := \frac{2}{w_e} = \begin{cases} 2, & e \in \mathcal{D} \text{ (two electrons make a lone pair)} \\ 1, & e \in \mathcal{O} \text{ (one unit of bond order is a pair)} \end{cases} \quad \text{so} \quad w_e g_e = 2 \quad \forall e \in \mathcal{E}, \quad (9)$$

the identity $w_e g_e = 2$ saying that one pair is worth two electrons *wherever it sits*. An arrow transfers one pair from a source site e^- to a sink site e^+ :

$$A = g_{e^+} \mathbf{1}_{e^+} - g_{e^-} \mathbf{1}_{e^-} \implies \mathcal{N}(A) = g_{e^+} w_{e^+} - g_{e^-} w_{e^-} = 2 - 2 = 0, \quad (10)$$

where $\mathbf{1}_e \in \text{Sym}_N$ is the symmetric unit at position e , a 1 at e and zeros elsewhere, which off the diagonal means a 1 at (i, j) and at (j, i) , those being one position and not two, so that $\mathcal{N}(\mathbf{1}_e) = w_e$. The implication follows from linearity of $\mathcal{N}$ and (9) at both sites. Conservation is therefore not

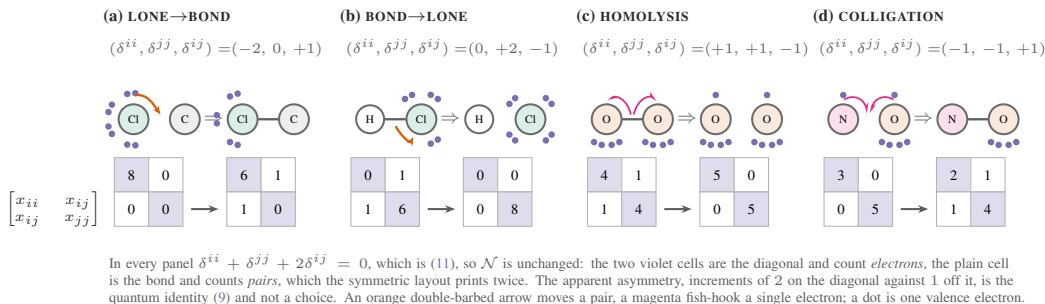


Figure 2: The move alphabet, drawn twice: as the chemist’s arrows on the smallest fragment exhibiting each type, and as the three matrix entries each type touches, with the values they take before and after. Columns (a) and (b) are arrows in the sense of (10), one whole pair leaving one site and arriving at another; (c) and (d) move a single electron each way. Every move is named by its type and an ordered atom pair, which is what lets the head be a dense $(4, N, N)$ tensor and the admissibility test (13) five interval comparisons. Columns (a) and (b) are inverse to one another with their atoms swapped, and so are (c) and (d); by Remark 7 that is why no reachable state has an empty candidate list.

a constraint imposed on the arrow. *It is an identity forced by the arrow having a tail and a head carrying the same quantum*, the mathematical content of the chemist’s drawing convention. Nothing about which sites they are, which molecule it is, or how large N is enters. Figure 12 in Appendix D.2 writes (10) out on one real hop, with both weights and both quanta shown at the two sites.

Four move types, each named by an ordered atom pair. The concrete alphabet has *four* types, and it is chosen that way so that every move is named by a type τ and an ordered pair of atoms (i, j) , so the model’s candidate set is then the dense grid $\{1, \dots, 4\} \times [N] \times [N]$, four scores per ordered atom pair, with no ragged list. Write $\delta_\tau^{ii}, \delta_\tau^{jj}, \delta_\tau^{ij}$ for the increments a move applies to x_{ii}, x_{jj} and $x_{ij} = x_{ji}$, and every move changes those three entries and no others (Figure 2).

type τ	what it does	δ^{ii}	δ^{jj}	δ^{ij}	requires
LONE→BOND	a lone pair on i becomes a bond to j	-2	0	+1	$x_{ii} \geq 2, x_{ij} \leq 2$
BOND→LONE	the bond i - j breaks, the pair onto j	0	+2	-1	$x_{ij} \geq 1$
HOMOLYSIS	the bond i - j breaks, one electron each	+1	+1	-1	$x_{ij} \geq 1$
COLLIGATION	one electron on each of i, j pairs up	-1	-1	+1	$x_{ii}, x_{jj} \geq 1, x_{ij} \leq 2$

The conservation identity is now visible by inspection, since for each row

$$\mathcal{N}(A_\tau) = \delta_\tau^{ii} + \delta_\tau^{jj} + 2\delta_\tau^{ij} = 0, \quad (11)$$

the factor 2 being the off-diagonal weight of (3). The first two rows are arrows in the sense of (10), with $e^- = (i, i), e^+ = (i, j)$ and $e^- = (i, j), e^+ = (j, j)$, so for them (11) is (10) written out. The last two move single electrons, and are needed whenever a step changes a diagonal entry by an odd amount, which pair moves cannot do. Radical chemistry does this routinely, and a mechanistic corpus contains a small but irreducible fraction of such steps (a few per cent in the data, there always a single metal–phosphorus bond breaking with one electron to each atom, or forming from two such electrons, Appendix D.4). The two pair moves are ordered in (i, j) and the two single-electron moves are symmetric, so the latter are offered on the upper triangle only and no physical move is ever two classes of the same softmax. Two properties of the alphabet are used below and established in Appendix D. The chemist’s third arrow type, the *bond migration* from bond (i, j) to an adjacent bond (j, k) , is named by an atom *triple* and would make the candidate set ragged, but it is the composition of a BOND→LONE and a LONE→BOND and so is not needed (Remark 6), and the alphabet is closed under inversion, so *no reachable state has an empty candidate list* (Remark 7).

The physical sector, and the two guarantees. Collecting the conditions of §2, the matrices a model may legitimately emit are

$$\mathcal{C} = \left\{ x \in \mathbb{Z}_{\geq 0}^{\mathcal{E}} : \underbrace{\mathcal{N}(x) = \mathcal{N}(B_r)}_{\text{one global linear equation}} \wedge \underbrace{x_{ij} \leq 3 \forall i \neq j}_{\text{site capacity}} \wedge \underbrace{(a_i, x_{ii}, \beta_i(x)) \in \mathcal{T} \forall i}_{N \text{ local conditions}} \right\} \quad (12)$$

where $\mathcal{T}$ is a table of admissible per-atom local states, and both endpoints of every training pair lie in $\mathcal{C}$. *The decomposition is the structural point of the construction, one global condition and N local ones.* The global one is discharged by the *algebra* of the moves, the local ones by an $O(1)$ *mask*.

Proposition 2 (conservation). *Let $(x^{(k)})_{k \geq 0}$ be any chain whose every jump is a move of the alphabet. Then $\mathcal{N}(x^{(k)}) = \mathcal{N}(x^{(0)})$ surely, not in expectation, and irrespective of how the jumps are chosen.*

The proof is an induction on k using linearity of $\mathcal{N}$ and (11) (Appendix G). Since $\mathcal{N}(x^{(k)})$ is a constant random variable, its entire law is that of $\mathcal{N}(B_r)$, so every moment is exact and the variance is zero, against the entrywise path, whose mean is right and whose variance is strictly positive (Appendix C.2). A move is *admissible* at a state when the three entries it touches stay inside the ranges the representation can express. Writing D for the largest lone-pair count and $\bar{\beta}$ for the largest bonding-pair count the table covers,

$$\mathfrak{A}(x) = \left\{ (\tau, i, j) : \begin{array}{l} 0 \leq x_{ii} + \delta_{\tau}^{ii} \leq D, \quad 0 \leq x_{jj} + \delta_{\tau}^{jj} \leq D, \quad 0 \leq x_{ij} + \delta_{\tau}^{ij} \leq 3, \\ 0 \leq \beta_i(x) + \delta_{\tau}^{ij} \leq \bar{\beta}, \quad 0 \leq \beta_j(x) + \delta_{\tau}^{ij} \leq \bar{\beta} \end{array} \right\} \quad (13)$$

These are five interval tests on five scalars, whatever the move type and whatever N . Reading them against the table above recovers each type’s stated requirement, $0 \leq x_{ii} - 2$ being “ i must have a lone pair to give” and $x_{ij} + 1 \leq 3$ being “the bond is not already triple”. Figure 14 in Appendix E.1 draws the whole of $\mathfrak{A}(x)$ at one real reactant, four grids of $N \times N$ cells with the live classes marked, which is the object the softmax below is normalised over. What is *not* in (13) is the table $\mathcal{T}$, which instead gates the move that *terminates* the chain.

Proposition 3 (validity). *Let a chain start at $x^{(0)} = B_r$, take jumps in $\mathfrak{A}(x^{(k)})$, and be permitted to terminate only at a state x with $(a_i, x_{ii}, \beta_i(x)) \in \mathcal{T}$ for every i . Then the state at which it terminates lies in $\mathcal{C}$.*

The proof is in Appendix G, and its brevity is the point. Validity has been converted from a property verified after the fact into a *precondition of the emitting move*. Note what this is not. It is not repair, and no candidate is ever generated and then filtered, projected or rounded. A state outside $\mathcal{C}$ is simply not a state the chain is able to leave from, so it never becomes a candidate at all.

The table $\mathcal{T}$ is enumerated, not written down. A valency rule invented on paper guarantees validity only against itself, whereas (4) asks for validity against the checker the answers will be graded by. So $\mathcal{T}$ is built by exhaustive enumeration *against that checker*. For every element a , lone-pair count d and bonding-pair count b , a probe molecule realising that local state is submitted, and the triple (a, d, b) is kept only if it survives both halves of (4). Over 65 elements the table has 9580 rows. For carbon the enumeration reproduces the octet $c_i \leq 4$ exactly, a check on the method rather than an input to it. At run time $\mathcal{T}$ is a flat boolean tensor indexed by (a_i, x_{ii}, β_i) . Appendices E.3 and E.4 give the probe construction and the two refinements that testing, rather than thinking, showed to be necessary.

Why the table gates only the stop move. Suppose $\mathcal{T}$ were imposed at *every* move. A single move changes $c_i = \frac{1}{2}x_{ii} + \beta_i$ by at most one at either atom, so an atom at its capacity, the normal state of a saturated main-group atom with $c_i = 4$, cannot accept a pair unless it simultaneously loses one, and the octet rule becomes a *dynamical selection rule* that can lock a step completely. Such jams occur at every pericyclic step, the Diels–Alder reaction being the canonical one. Enforcing $\mathcal{T}$ at every move would therefore need an alphabet of every connected block of up to five arrows, state-dependent and ragged, whereas enforcing it on the *stop* move alone costs nothing that (4) asks for and makes 100% of steps expressible as an ordered sequence of single moves through representable matrices. The cost is conceptual. An intermediate state is a well-formed bond–electron matrix carrying the right number of electrons, but need not be a state the reconstruction would accept. It is a *book-keeping* intermediate, and what the construction guarantees, all it needs to guarantee, is that the chain can only *leave* from a state that is a molecule. Appendix E.6 works the Diels–Alder jam out in full.

The bridge over moves. We now repeat §3 with the elementary event changed from “flip one position” to “apply one move”, so that the network is trained on exactly the states the sampler will visit. *Decomposition:* factorise $\Delta = B_p - B_r$ into moves the way a chemist factorises a step into curly arrows, a factorisation that always exists, since every degree-preserving transformation of this kind decomposes into cyclic electron-pushing diagrams (Flamm et al., 2023). Expand each bond migration into the two moves of Remark 6 and taking each odd-parity change by HOMOLYSIS or COLLIGATION, and write $\Delta = \sum_{k=1}^m A_k$. *Ordering:* the m moves are a *multiset*, their sum being Δ in any order, but every intermediate must satisfy (13), since a step that both forms and breaks the

same bond can drive its order to four if the two are taken the wrong way round. Order them greedily, at each point taking any remaining move admissible at the running state. Greedy is a heuristic and not a theorem, but one with no observed failures. It finds an admissible order for every step it is given, the rare awkward case being a step whose moves have to be taken in one particular order to keep every intermediate representable (Appendix F.2). It is deterministic, so it is computed once, offline, and cached. *The path:* writing $A_{(1)}, \dots, A_{(m)}$ for the ordered moves, the bridge is the family of *prefixes*

$$x_K = B_r + \sum_{r \leq K} A_{(r)} \quad \text{with} \quad A_{(r)} \in \mathfrak{A}(x_{r-1}) \quad \forall r, \quad x_0 = B_r, \quad x_m = B_p. \quad (14)$$

Drawing the whole order *before* taking prefixes is what makes every prefix admissible by construction. A greedy walk chosen on the fly could paint itself into a corner, a prefix of a pre-drawn order cannot. Training draws $K \sim \text{Uniform}\{0, \dots, m\}$, the natural counting process on a path of $m + 1$ points. The role played in (6) by κ_t is played here by K/m , and the analogue of “an entry has already switched” is “a move has already been applied”. A step with $m = 0$, in which the reactant *is* the product and the pathway terminates, contributes only $K = 0$, and such steps are a sizeable minority of any mechanistic corpus, one per pathway. Elementary steps are short, a handful of arrows at most, so m is small. Figures 18 and 19 in Appendix F.2 show the whole of this paragraph on two real steps.

The head, one masked softmax over moves and stopping. The network body is unchanged from §3. It takes the joint (reactant, current) code of every matrix entry, which records where the chain started and where it stands so the encoder can see what it has already done, carried per atom pair off the diagonal and per atom on it, then relative self-attention over atoms, giving one vector h_i per atom. Only the head changes. For each of the four move types τ it scores the pair (i, j) by a small bilinear product of h_i and h_j plus a bias learned for the pair’s entry code, filling the dense $(4, N, N)$ logit tensor announced above, and one *stop* logit from the pooled molecule, the three stages being drawn beside one another in Figure 20 of Appendix F.3,

$$p_\theta(\cdot \mid x, K) = \text{softmax}\left(\left\{\ell_\theta(\tau, i, j)\right\}_{(\tau, i, j) \in \mathfrak{A}(x)} \cup \left\{\ell_\theta(\emptyset) \text{ if } x \in \mathcal{C}\right\}\right), \quad (15)$$

every other class being driven to $-\infty$ *before* the normalisation, exactly as in constrained decoding for language (Hokamp & Liu, 2017; Post & Vilar, 2018; Willard & Louf, 2023). That is how Propositions 2 and 3 hold at the level of the *parameterisation* rather than of the sampler. An inadmissible move is not a move given low probability, it is a class the distribution does not contain, and by Remark 7 the admissible set is never empty, so (15) is always well defined.

Loss. The trajectory likelihood factorises over its positions, so maximum likelihood on the bridge is again plain cross-entropy, namely (8) with the endpoint target replaced by the next elementary event:

$$\mathcal{L}(\theta) = -\mathbb{E}_{(B_r, B_p) \sim \pi} \mathbb{E}_{K \sim \text{Unif}\{0, \dots, m\}} \left[\log p_\theta(y_K \mid x_K, K) \right], \quad y_K = \begin{cases} A_{(K+1)}, & K < m, \\ \emptyset, & K = m. \end{cases} \quad (16)$$

The target is admissible by construction, with $A_{(K+1)} \in \mathfrak{A}(x_K)$ by the ordering and $B_p \in \mathcal{C}$ by (12), so it always carries finite mass under the mask and the loss is finite for every example. Stopping is an ordinary class of an ordinary softmax, competing with the moves, and *the model learns when the mechanism is over by the same mechanism it learns what to do next*. Remark 5 applies verbatim. Appendix F.5 explains why the construction is kept in discrete time.

Sampling, ranking and cost. Candidates are produced by rolling the chain forward. Each hypothesis carries its own running matrix, and at each position the state is re-encoded, (15) is formed, and the hypothesis either stops or applies one move:

$$\begin{aligned} x^{(0)} &= B_r, & y^{(k)} &\sim p_\theta(\cdot \mid x^{(k)}, k), & x^{(k+1)} &= \begin{cases} x^{(k)} + A_{y^{(k)}}, & y^{(k)} \neq \emptyset, \\ \text{stop}, & y^{(k)} = \emptyset, \end{cases} \\ s(\hat{B}) &= \sum_k \log p_\theta(y^{(k)} \mid x^{(k)}, k). \end{aligned} \quad (17)$$

where a drawn class $y = (\tau, i, j)$ names the matrix $A_y = \delta_\tau^{ii} \mathbf{1}_{ii} + \delta_\tau^{jj} \mathbf{1}_{jj} + \delta_\tau^{ij} \mathbf{1}_{ij}$ it adds, the row of the alphabet table read at that atom pair, so the subscript on A is the class the softmax emitted and

nothing more. There is no decoding stage, and *no rounding, no projection and no repair*. The score $s(\hat{B})$ is the exact log-probability of the trajectory, so candidates can be ranked by score as well as by how often they were drawn. The model also admits a *beam* decoder, which the entrywise model of §3 does not (Appendix F.4). A hypothesis that has not stopped within S positions emits the last state at which it *was* allowed to stop, which always exists because $x^{(0)} = B_r \in \mathcal{C}$. Building the mask costs one diagonal, one row sum and four broadcast comparisons, against a body quadratic in N , so the constraint is also the cheapest part of the model (Appendix E.8). Table 2 in Appendix F sets the two constructions side by side.

6 EXPERIMENTS

The experiments answer three questions, in this order. Do the two guarantees hold in practice, exactly and at every model size, and do they hold for no other model? Is anything paid for them in predictive accuracy, measured against the same network body under the entrywise elementary event and against seven further baselines? And how much capacity does the construction need, since a constraint that has to be learned costs parameters and one that is built in should not?

Datasets. We use the mechanistic dataset released with Continuous FlowER (Joung et al., 2025), in which expert-curated mechanistic templates are applied to USPTO reactions to impute an ordered pathway of elementary steps for each recorded transformation. We take the released splits unchanged (Appendix I.1). A pathway is terminated by an identity step, in which the product equals the reactant, and 18.5% of steps are of that kind. Every hydrogen is mapped explicitly, so $n_i^H = 0$ throughout and β_i is a plain row sum.

Models. We compare nine models, re-implemented in one framework and trained here on the same data with the same optimiser schedule at matched parameter counts, so the numbers are ours and not the originating papers’. Four predict a product string or graph, the Molecular Transformer (Schwaller et al., 2019), Graph2SMILES (Tu & Coley, 2022) and the two mechanistic language models MechT5 and MechLLaMa of Neukomm et al. (2025). Two predict the mechanism, ELECTRO’s electron-path model (Bradshaw et al., 2019) and Reactron (Chen et al., 2025), which we run with that paper’s own weight-shared message-passing stack rather than our body. Three are flow models over the bond–electron matrix and share one network body, one featurisation and every architecture hyper-parameter, so the only difference between them is the elementary event, which is Continuous FlowER (Joung et al., 2025), Discrete FlowER (§3) and Arrow Flow Matching (§5).

Metrics. Every model emits a ranked list of at most ten candidates, each rebuilt into a molecule, with duplicates merged and a short list padded out with failures. Six quantities are reported, and Figure 22 in Appendix J.2 shows all six. **Step accuracy at k** is the fraction of steps whose recorded product, compared as a canonical atom-map-free string, is among the k highest-ranked candidates. **Pathway accuracy at k** rolls the model out recursively from a pathway’s initial reactants, each prediction feeding the next step, and counts a pathway correct when some branch reaches its recorded final product. **Teacher-forced pathway accuracy at k** is the stricter conjunction in which every step of a pathway is given its recorded reactant and must rank the recorded product within k . **Validity** is the fraction of candidates that are chemically valid, and **electron, atom and proton conservation** the fractions carrying the reactant’s valence-electron count, heavy-atom multiset and hydrogen count, a chemically invalid candidate failing all four rather than being dropped. The **overall score** is the geometric mean of top-1 step accuracy, top-1 pathway accuracy, validity and the mean of the three conservation checks, geometric rather than arithmetic so that a collapsed pillar cannot be bought back with surplus on the others. Because the data branch, a deterministic decoder cannot satisfy every recorded answer at $k = 1$. The resulting ceiling is drawn at $k = 1$ in every accuracy panel, and Appendix J.1 derives it and gives the sampling budgets these numbers are computed at.

Settings. All models are trained for 100 epochs with AdamW, a 1000-step linear warmup and cosine decay to a tenth of the peak rate, gradient clipping at 1.0, and early stopping on validation loss. Every model is trained at five widths spanning roughly 1.2M to 21M parameters, and the checkpoint with the best validation loss is evaluated. Appendix I gives the per-model architecture and optimiser hyper-parameters, the width ladder, and the decoding budgets. Every model is decoded by sampling with frequency ranking, the one protocol all nine models can fill, and Appendix J reports the other four regimes.

Table 1: All nine models at the smallest width, roughly 1.3M parameters, under sampling with frequency ranking. Path@1 is the recursive roll-out, and the last four columns are validity and electron, heavy-atom and proton conservation, each averaged over the candidate budget. The score column is the geometric-mean composite of §6. In each column the 1st, 2nd and 3rd best entry is coloured, ties sharing a place. The top-1 ceiling induced by branching in the data is 0.8840. Figure 22 in Appendix J.2 plots the same comparison.

model	params	top-1	top-3	top-5	top-10	path@1	valid.	elec.	atom	prot.	score
Molecular Transformer	1.29 M	0.6974	0.8069	0.8273	0.8358	0.4640	0.9203	0.7828	0.7772	0.7531	0.6922
Graph2SMILES	2.36 M	0.7656	0.8866	0.9035	0.9103	0.7340	0.9717	0.8539	0.8515	0.8400	0.8250
MechT5	3.77 M	0.8329	0.9315	0.9386	0.9396	0.6913	0.9539	0.9529	0.9539	0.9539	0.8507
MechLLaMa	2.05 M	0.7765	0.8674	0.8736	0.8741	0.4517	0.8707	0.8694	0.8707	0.8707	0.7180
ELECTRO	1.27 M	0.6616	0.7588	0.7653	0.7656	0.2887	1.0000	0.9952	1.0000	1.0000	0.6608
Reactron	1.48 M	0.8068	0.9125	0.9178	0.9180	0.7803	0.9763	0.9752	0.9763	0.9763	0.8801
Continuous FlowER	1.30 M	0.8435	0.9598	0.9689	0.9694	0.8583	0.8211	0.8133	0.8211	0.8211	0.8352
Discrete FlowER (§3)	1.26 M	0.8730	0.9738	0.9832	0.9847	0.9500	0.9761	0.9596	0.9761	0.9761	0.9415
Arrow Flow Matching (§5)	1.27 M	0.8692	0.9784	0.9858	0.9862	0.9303	1.0000	1.0000	1.0000	1.0000	0.9483

Results. Table 1 reports the comparison at the smallest width, and Appendix J.2 plots it (Figure 22) together with the sweep over all five widths (Figure 23). *The guarantees hold.* Validity and electron, atom and proton conservation are 1.0000000 to seven decimal places for Arrow Flow Matching at every one of the five widths. ELECTRO also reaches validity 1, by construction, but its electron conservation is 0.9952 and its step accuracy about three quarters of the best. The two entrywise flow models are the weakest on conservation, the continuous one losing it in the rounding of its off-lattice endpoint and the discrete one exactly as Theorem 1 says it must. *The guarantees are not paid for in accuracy.* Arrow Flow Matching is within 0.004 of the best top-1 step accuracy at every width, the best model at every $k \geq 3$ at the smallest width, and 0.015 below the 0.8840 ceiling that branching in the data imposes on any model. Its accuracy is flat in capacity over a $15\times$ range, so the guarantee is not bought with capacity. *The bias replaces capacity.* Its overall score at 1.3M parameters, 0.9483, is higher than that of any other model at any width, the nearest being 0.9415 for Discrete FlowER at the same width and 0.9344 for the Molecular Transformer at sixteen times the budget. It is behind only on the recursive pathway metric, 0.9303 against 0.9500 for Discrete FlowER, since a roll-out compounds the top-1 rate over many steps, a narrow deficit given that a pathway search consumes ranked lists in which every invalid candidate takes a slot. The guarantee is stated against a specific validity checker, and Appendix J.3 records the two qualifications this entails.

7 LIMITATIONS

(i) The bridge needs an offline decomposition and ordering of every training step, which the entrywise model does not. (ii) A product can be reached by several admissible orders of the same moves, so the trajectory score (17) is a lower bound on $\log p_\theta(\hat{B} \mid B_r)$, where the entrywise score is exact. (iii) Intermediate states conserve electrons and are representable, but need not be molecules, since the table gates only the stop move. (iv) The four move types cover this dataset completely, but genuinely three-centre elementary events would need the alphabet extended, though (11) is all that is required of new increments.

8 CONCLUSION

We showed that a discrete flow-matching model of a mechanistic step that factorises over the positions of the bond–electron matrix cannot put its mass where the chemistry is without collapsing to a point mass, and that changing only the elementary event removes the obstruction. With one move of a four-element alphabet in the kernel of the electron functional as the event, Arrow Flow Matching makes conservation an algebraic identity and validity a table lookup gating one softmax class, every state the chain occupies is a partially executed mechanism, and it reaches the best overall score in the study already at its smallest width. Several directions remain open. The alphabet can be extended to three-centre and general single-electron events, with (11) as the only requirement on the new increments. Validity conditions that are not local, such as those of aromatic rings, lie beyond any per-atom table and call for a mask over larger substructures. Summing the trajectory likelihood over the admissible orders of one multiset of moves would give the exact product likelihood rather than the lower bound (17), and might close the remaining gap on the recursive pathway metric. Fi-

nally, the principle is not specific to chemistry. Wherever a hard constraint is a linear functional of a discrete state, as for mass, charge or flow conservation on graphs, an elementary event in the kernel of that functional together with a cheap local mask should give the same guarantees, and testing this beyond reaction mechanisms is the natural next step.

REPRODUCIBILITY STATEMENT

The construction is given in full in §5 and Appendix B, with the derivations that the main text compresses, the network parameterisation and a summary algorithm in Appendices B.10 and B.11, the enumeration of $\mathcal{T}$ and the ordering procedure in Appendices E.3 and F.2, and every proof in Appendix G. Appendix I lists the data splits, the architecture and optimiser hyper-parameters of all nine models at all five widths, the schedule and the decoding budgets. The dataset is the public mechanistic dataset of Joung et al. (2025). Appendix J reports every evaluation regime rather than the one in the main text.

AI USE STATEMENT

Generative AI tools were used for coding assistance, editing prose, preparing figures and tables, and running the experiments. They were not used to generate research ideas but were used to prove them theoretically. All AI-assisted output was reviewed by the authors, and we take responsibility for the final content of this work.

REFERENCES

- D. T. Ahneman, J. G. Estrada, S. Lin, S. D. Dreher, A. G. Doyle. *Predicting reaction performance in C–N cross-coupling using machine learning*. Science 360(6385):186–190, 2018. <https://doi.org/10.1126/science.aar5169>
- K. Ahmed, S. Teso, K.-W. Chang, G. Van den Broeck, A. Vergari. *Semantic Probabilistic Layers for Neuro-Symbolic Learning*. Advances in Neural Information Processing Systems 35 (NeurIPS), 2022. arXiv:2206.00426, <https://arxiv.org/abs/2206.00426>
- M. S. Albergo, E. Vanden-Eijnden. *Building Normalizing Flows with Stochastic Interpolants*. International Conference on Learning Representations (ICLR), 2023. arXiv:2209.15571, <https://arxiv.org/abs/2209.15571>
- M. S. Albergo, N. M. Boffi, E. Vanden-Eijnden. *Stochastic Interpolants: A Unifying Framework for Flows and Diffusions*. Preprint, arXiv:2303.08797, 2023. <https://arxiv.org/abs/2303.08797>
- J. Austin, D. D. Johnson, J. Ho, D. Tarlow, R. van den Berg. *Structured Denoising Diffusion Models in Discrete State-Spaces*. Advances in Neural Information Processing Systems 34 (NeurIPS), 2021. <https://papers.nips.cc/paper/2021/hash/958c530554f78bcd8e97125b70e6973d-Abstract.html>
- J. Bradshaw, M. J. Kusner, B. Paige, M. H. S. Segler, J. M. Hernández-Lobato. *A Generative Model For Electron Paths*. International Conference on Learning Representations (ICLR), 2019. arXiv:1805.10970, <https://arxiv.org/abs/1805.10970>
- A. Campbell, J. Benton, V. De Bortoli, T. Rainforth, G. Deligiannidis, A. Doucet. *A Continuous Time Framework for Discrete Denoising Models*. Advances in Neural Information Processing Systems 35 (NeurIPS), 2022. arXiv:2205.14987, <https://arxiv.org/abs/2205.14987>
- A. Campbell, J. Yim, R. Barzilay, T. Rainforth, T. Jaakkola. *Generative Flows on Discrete State-Spaces: Enabling Multimodal Flows with Applications to Protein Co-Design*. Proceedings of the 41st International Conference on Machine Learning (ICML), PMLR 235:5453–5512, 2024. <https://proceedings.mlr.press/v235/campbell124a.html>
- S. Chen, Y. Jung. *Deep Retrosynthetic Reaction Prediction using Local Reactivity and Global Attention*. JACS Au 1(10):1612–1620, 2021. <https://doi.org/10.1021/jacsau.1c00246>
- R. T. Q. Chen, Y. Lipman. *Flow Matching on General Geometries*. International Conference on Learning Representations (ICLR), 2024. arXiv:2302.03660, <https://arxiv.org/abs/2302.03660>
- S. Chen, K. S. Park, T. Kim, S. Han, Y. Jung. *Predicting Chemical Reaction Outcomes Based on Electron Movements Using Machine Learning*. Preprint, arXiv:2503.10197, 2025. <https://arxiv.org/abs/2503.10197>
- J. K. Christopher, S. Baek, F. Fioretto. *Constrained Synthesis with Projected Diffusion Models*. Advances in Neural Information Processing Systems 37 (NeurIPS), 2024. arXiv:2402.03559, <https://arxiv.org/abs/2402.03559>

- C. W. Coley, R. Barzilay, T. S. Jaakkola, W. H. Green, K. F. Jensen. *Prediction of Organic Reaction Outcomes Using Machine Learning*. ACS Central Science 3(5):434–443, 2017. <https://doi.org/10.1021/acscentsci.7b00064>
- C. W. Coley, W. Jin, L. Rogers, T. F. Jamison, T. S. Jaakkola, W. H. Green, R. Barzilay, K. F. Jensen. *A graph-convolutional neural network model for the prediction of chemical reactivity*. Chemical Science 10(2):370–377, 2019. <https://doi.org/10.1039/C8SC04228D>
- C. W. Coley, D. A. Thomas III, J. A. M. Lummiss, J. N. Jaworski, C. P. Breen, V. Schultz, T. Hart, J. S. Fishman, et al. *A robotic platform for flow synthesis of organic compounds informed by AI planning*. Science 365(6453):eaax1566, 2019. <https://doi.org/10.1126/science.aax1566>
- H. Dai, C. Li, C. W. Coley, B. Dai, L. Song. *Retrosynthesis Prediction with Conditional Graph Logic Network*. Advances in Neural Information Processing Systems 32 (NeurIPS), 2019. <https://proceedings.neurips.cc/paper/2019/hash/0d2b2061826a5df3221116a5085a6052-Abstract.html>
- M. Das, A. Hoque, M. Baranwal, R. B. Sunoj. *DeepMech: a machine learning framework for chemical reaction mechanism prediction*. Chemical Science 17:15745–15761, 2026. <https://doi.org/10.1039/D6SC02809H>
- C. Duan, Y. Du, H. Jia, H. J. Kulik. *Accurate transition state generation with an object-aware equivariant elementary reaction diffusion model*. Nature Computational Science 3(12):1045–1055, 2023. <https://doi.org/10.1038/s43588-023-00563-7>
- C. Duan, G.-H. Liu, Y. Du, T. Chen, Q. Zhao, H. Jia, C. P. Gomes, E. A. Theodorou, H. J. Kulik. *Optimal transport for generating transition states in chemical reactions*. Nature Machine Intelligence 7:615–626, 2025. <https://doi.org/10.1038/s42256-025-01010-0>
- J. Dugundji, I. Ugi. *An algebraic model of constitutional chemistry as a basis for chemical computer programs*. Topics in Current Chemistry 39:19–64, 1973. <https://doi.org/10.1007/BFb0051317>
- P. Esser, S. Kulal, A. Blattmann, R. Entezari, J. Müller, H. Saini, Y. Levi, D. Lorenz, A. Sauer, F. Boesel, D. Podell, T. Dockhorn, Z. English, K. Lacey, A. Goodwin, Y. Marek, R. Rombach. *Scaling Rectified Flow Transformers for High-Resolution Image Synthesis*. Proceedings of the 41st International Conference on Machine Learning (ICML), 2024. arXiv:2403.03206, <https://arxiv.org/abs/2403.03206>
- N. Fishman, L. Klärner, V. De Bortoli, E. Mathieu, M. Hutchinson. *Diffusion Models for Constrained Domains*. Transactions on Machine Learning Research (TMLR), 2023. arXiv:2304.05364, <https://arxiv.org/abs/2304.05364>
- C. Flamm, S. Müller, P. F. Stadler. *Every atom-atom map can be explained by electron pushing diagrams*. Preprint, arXiv:2311.13492, 2023. <https://arxiv.org/abs/2311.13492>
- I. Gat, T. Remez, N. Shaul, F. Kreuk, R. T. Q. Chen, G. Synnaeve, Y. Adi, Y. Lipman. *Discrete Flow Matching*. Advances in Neural Information Processing Systems 37 (NeurIPS), 2024. https://proceedings.neurips.cc/paper_files/paper/2024/hash/f0d629a734b56a642701bba7bc8bb3ed-Abstract-Conference.html
- T. Gneiting, A. E. Raftery. *Strictly Proper Scoring Rules, Prediction, and Estimation*. Journal of the American Statistical Association 102(477):359–378, 2007. <https://doi.org/10.1198/016214506000001437>
- J. Ho, A. Jain, P. Abbeel. *Denosing Diffusion Probabilistic Models*. Advances in Neural Information Processing Systems 33 (NeurIPS), 2020. arXiv:2006.11239, <https://arxiv.org/abs/2006.11239>
- C. Hokamp, Q. Liu. *Lexically Constrained Decoding for Sequence Generation Using Grid Beam Search*. Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL), 2017. arXiv:1704.07138, <https://arxiv.org/abs/1704.07138>
- P. Holderrieth, M. Havasi, J. Yim, N. Shaul, I. Gat, T. Jaakkola, B. Karrer, R. T. Q. Chen, Y. Lipman. *Generator Matching: Generative modeling with arbitrary Markov processes*. International Conference on Learning Representations (ICLR), 2025 (Oral). arXiv:2410.20587, <https://arxiv.org/abs/2410.20587>
- E. Hoogetboom, D. Nielsen, P. Jaini, P. Forré, M. Welling. *Argmax Flows and Multinomial Diffusion: Learning Categorical Distributions*. Advances in Neural Information Processing Systems 34 (NeurIPS), 2021. <https://papers.nips.cc/paper/2021/hash/67d96d458abdef21792e6d8e590244e7-Abstract.html>
- I. Igashov, A. Schneuing, M. Segler, M. Bronstein, B. Correia. *RetroBridge: Modeling Retrosynthesis with Markov Bridges*. International Conference on Learning Representations (ICLR), 2024. arXiv:2308.16212, <https://arxiv.org/abs/2308.16212>
- W. Jin, C. W. Coley, R. Barzilay, T. Jaakkola. *Predicting Organic Reaction Outcomes with Weisfeiler-Lehman Network*. Advances in Neural Information Processing Systems 30 (NIPS), 2017. arXiv:1709.04555, <https://arxiv.org/abs/1709.04555>
- J. F. Joung, M. H. Fong, N. Casetti, J. P. Liles, N. S. Dassanayake, C. W. Coley. *Electron flow matching for generative reaction mechanism prediction*. Nature 645(8079):115–123, 2025. <https://doi.org/10.1038/s41586-025-09426-9>; preprint arXiv:2502.12979, <https://arxiv.org/abs/2502.12979>

- M. A. Kayala, P. Baldi. *ReactionPredictor: Prediction of Complex Chemical Reactions at the Mechanistic Level Using Machine Learning*. Journal of Chemical Information and Modeling 52(10):2526–2540, 2012. <https://doi.org/10.1021/ci3003039>
- S. Kim, J. Woo, W. Y. Kim. *Diffusion-based generative AI for exploring transition states from 2D molecular graphs*. Nature Communications 15:341, 2024. <https://doi.org/10.1038/s41467-023-44629-6>
- G. N. Lewis. *The Atom and the Molecule*. Journal of the American Chemical Society 38(4):762–785, 1916. <https://doi.org/10.1021/ja02261a002>
- Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, M. Le. *Flow Matching for Generative Modeling*. International Conference on Learning Representations (ICLR), 2023. arXiv:2210.02747, <https://arxiv.org/abs/2210.02747>
- Y. Lipman, M. Havasi, P. Holderrieth, N. Shaul, M. Le, B. Karrer, R. T. Q. Chen, D. Lopez-Paz, H. Ben-Hamu, I. Gat. *Flow Matching Guide and Code*. Preprint, arXiv:2412.06264, 2024. <https://arxiv.org/abs/2412.06264>
- X. Liu, C. Gong, Q. Liu. *Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow*. International Conference on Learning Representations (ICLR), 2023. arXiv:2209.03003, <https://arxiv.org/abs/2209.03003>
- G.-H. Liu, A. Vahdat, D.-A. Huang, E. A. Theodorou, W. Nie, A. Anandkumar. *I²SB: Image-to-Image Schrödinger Bridge*. Proceedings of the 40th International Conference on Machine Learning (ICML), 2023. arXiv:2302.05872, <https://arxiv.org/abs/2302.05872>
- G.-H. Liu, T. Chen, E. A. Theodorou, M. Tao. *Mirror Diffusion Models for Constrained and Watermarked Generation*. Advances in Neural Information Processing Systems 36 (NeurIPS), 2023. arXiv:2310.01236, <https://arxiv.org/abs/2310.01236>
- A. Lou, S. Ermon. *Reflected Diffusion Models*. Proceedings of the 40th International Conference on Machine Learning (ICML), 2023. arXiv:2304.04740, <https://arxiv.org/abs/2304.04740>
- A. Lou, C. Meng, S. Ermon. *Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution*. Proceedings of the 41st International Conference on Machine Learning (ICML), 2024 (Oral). arXiv:2310.16834, <https://arxiv.org/abs/2310.16834>
- B. Mikulak-Klucznik, P. Gołębiowska, A. A. Bayly, O. Popik, T. Klucznik, S. Szymkuć, E. P. Gajewska, P. Dittwald, O. Staszewska-Krajewska, W. Beker, T. Badowski, K. A. Scheidt, K. Molga, J. Mlynarski, M. Mrksich, B. A. Grzybowski. *Computational planning of the synthesis of complex natural products*. Nature 588:83–88, 2020. <https://doi.org/10.1038/s41586-020-2855-y>
- J. Nam, J. Kim. *Linking the Neural Machine Translation and the Prediction of Organic Chemistry Reactions*. Preprint, arXiv:1612.09529, 2016. <https://arxiv.org/abs/1612.09529>
- T. A. Neukomm, Z. Jončev, P. Schwaller. *Teaching Language Models Mechanistic Explainability Through MechSMILES*. Preprint, arXiv:2512.05722, 2025. <https://arxiv.org/abs/2512.05722>
- M. Post, D. Vilar. *Fast Lexically Constrained Decoding with Dynamic Beam Allocation for Neural Machine Translation*. Proceedings of NAACL-HLT, 2018. arXiv:1804.06609, <https://arxiv.org/abs/1804.06609>
- G. Landrum et al. *RDKit: Open-source cheminformatics*. <https://www.rdkit.org>
- K. Ren, K. Tang, Y. Zhao, L. Zhang, J. Du, Q. Meng, Q. Liu. *Reactive machine learning potential for accelerating transition state search in organic synthesis*. Nature Communications 17:6253, 2026. <https://doi.org/10.1038/s41467-026-72945-0>
- V. Sabanza Gil, A. M. Bran, M. Franke, R. Schlama, J. S. Luterbacher, P. Schwaller. *Holistic chemical evaluation reveals pitfalls in reaction prediction models*. Preprint, arXiv:2312.09004, 2023. <https://arxiv.org/abs/2312.09004>
- M. Sacha, M. Błaż, P. Byrski, P. Dąbrowski-Tumański, M. Chromiński, R. Loska, P. Włodarczyk-Pruszyński, S. Jastrzębski. *Molecule Edit Graph Attention Network: Modeling Chemical Reactions as Sequences of Graph Edits*. Journal of Chemical Information and Modeling 61(7):3273–3284, 2021. <https://doi.org/10.1021/acs.jcim.1c00537>
- S. S. Sahoo, M. Arriola, Y. Schiff, A. Gokaslan, E. Marroquin, J. T. Chiu, A. Rush, V. Kuleshov. *Simple and Effective Masked Diffusion Language Models*. Advances in Neural Information Processing Systems 37 (NeurIPS), 2024. arXiv:2406.07524, <https://arxiv.org/abs/2406.07524>
- P. Schwaller, T. Gaudin, D. Lányi, C. Bekas, T. Laino. *“Found in Translation”: predicting outcomes of complex organic chemistry reactions using neural sequence-to-sequence models*. Chemical Science 9(28):6091–6098, 2018. <https://doi.org/10.1039/C8SC02339E>
- P. Schwaller, T. Laino, T. Gaudin, P. Bolgar, C. A. Hunter, C. Bekas, A. A. Lee. *Molecular Transformer: A Model for Uncertainty-Calibrated Chemical Reaction Prediction*. ACS Central Science 5(9):1572–1583, 2019. <https://doi.org/10.1021/acscentsci.9b00576>
- M. H. S. Segler, M. Preuss, M. P. Waller. *Planning chemical syntheses with deep neural networks and symbolic AI*. Nature 555:604–610, 2018. <https://doi.org/10.1038/nature25978>

- N. Shaul, I. Gat, M. Havasi, D. Severo, A. Sriram, P. Holderrieth, B. Karrer, Y. Lipman, R. T. Q. Chen. *Flow Matching with General Discrete Paths: A Kinetic-Optimal Perspective*. International Conference on Learning Representations (ICLR), 2025 (Oral). arXiv:2412.03487, <https://arxiv.org/abs/2412.03487>
- Y. Shi, V. De Bortoli, A. Campbell, A. Doucet. *Diffusion Schrödinger Bridge Matching*. Advances in Neural Information Processing Systems 36 (NeurIPS), 2023. arXiv:2303.16852, <https://arxiv.org/abs/2303.16852>
- J. Shi, K. Han, Z. Wang, A. Doucet, M. K. Titsias. *Simplified and Generalized Masked Diffusion for Discrete Data*. Advances in Neural Information Processing Systems 37 (NeurIPS), 2024. arXiv:2406.04329, <https://arxiv.org/abs/2406.04329>
- V. R. Somnath, C. Bunne, C. W. Coley, A. Krause, R. Barzilay. *Learning Graph Models for Retrosynthesis Prediction*. Advances in Neural Information Processing Systems 34 (NeurIPS), 2021. arXiv:2006.07038, <https://arxiv.org/abs/2006.07038>
- Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, B. Poole. *Score-Based Generative Modeling through Stochastic Differential Equations*. International Conference on Learning Representations (ICLR), 2021 (Oral). arXiv:2011.13456, <https://arxiv.org/abs/2011.13456>
- H. Stark, B. Jing, C. Wang, G. Corso, B. Berger, R. Barzilay, T. Jaakkola. *Dirichlet Flow Matching with Applications to DNA Sequence Design*. Proceedings of the 41st International Conference on Machine Learning (ICML), PMLR 235, 2024. arXiv:2402.05841, <https://arxiv.org/abs/2402.05841>
- A. Tong, K. Fatras, N. Malkin, G. Hugué, Y. Zhang, J. Rector-Brooks, G. Wolf, Y. Bengio. *Improving and generalizing flow-based generative models with minibatch optimal transport*. Transactions on Machine Learning Research (TMLR), 2024. arXiv:2302.00482, <https://arxiv.org/abs/2302.00482>
- Z. Tu, C. W. Coley. *Permutation Invariant Graph-to-Sequence Model for Template-Free Retrosynthesis and Reaction Prediction*. Journal of Chemical Information and Modeling 62(15):3503–3513, 2022. <https://doi.org/10.1021/acs.jcim.2c00321>
- P. Tuo, J. Chen, J. Li. *Flow matching for reaction pathway generation*. Nature Communications 17:8769, 2026. <https://doi.org/10.1038/s41467-026-75654-w>; preprint arXiv:2507.10530, <https://arxiv.org/abs/2507.10530>
- C. Vignac, I. Krawczuk, A. Siraudin, B. Wang, V. Cevher, P. Frossard. *DiGress: Discrete Denoising Diffusion for Graph Generation*. International Conference on Learning Representations (ICLR), 2023. arXiv:2209.14734, <https://arxiv.org/abs/2209.14734>
- D. Weininger. *SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules*. Journal of Chemical Information and Computer Sciences 28(1):31–36, 1988. <https://doi.org/10.1021/ci00057a005>
- B. T. Willard, R. Louf. *Efficient Guided Generation for Large Language Models*. Preprint, arXiv:2307.09702, 2023. <https://arxiv.org/abs/2307.09702>
- J. Xu, Z. Zhang, T. Friedman, Y. Liang, G. Van den Broeck. *A Semantic Loss Function for Deep Learning with Symbolic Knowledge*. Proceedings of the 35th International Conference on Machine Learning (ICML), PMLR 80:5502–5511, 2018. <https://proceedings.mlr.press/v80/xu18h.html>
- L.-C. Xu, M.-J. Tang, J. An, F. Cao, Y. Qi. *A unified pre-trained deep learning framework for cross-task reaction performance prediction and synthesis planning*. Nature Machine Intelligence 7:1561–1571, 2025. <https://doi.org/10.1038/s42256-025-01098-4>
- R. Yadav, Q. Yan, G. Wolf, A. J. Bose, R. Liao. *Retro SynFlow: Discrete Flow Matching for Accurate and Diverse Single-Step Retrosynthesis*. Advances in Neural Information Processing Systems (NeurIPS), 2025. arXiv:2506.04439, <https://arxiv.org/abs/2506.04439>

756	APPENDIX CONTENTS	
757		
758		
759	A Extended background, the objects and the checks	17
760	A.1 The task	17
761	A.2 Three descriptions of one molecule	17
762	A.3 One step, checked entry by entry	18
763		
764		
765	B Discrete FlowER, derivation, parameterisation and algorithm	18
766	B.1 State space, coupling and path	18
767	B.2 The generator	19
768	B.3 The conditional rate is a legitimate generator, and it generates the bridge	20
769	B.4 The marginal rate is a posterior expectation	20
770	B.5 What to parameterise, and the loss	20
771	B.6 The minimiser of the loss	21
772	B.7 Relation to the evidence bound	21
773	B.8 Why a proper scoring rule, drawn	21
774	B.9 Sampling, and what decoding costs	22
775	B.10 Network parameterisation	23
776	B.11 The algorithm	24
777		
778		
779		
780		
781		
782		
783	C The obstruction in detail	24
784	C.1 What exactly the entrywise baseline is	24
785	C.2 The path is exact in mean and not surely	25
786	C.3 The proof of Theorem 1, drawn	25
787	C.4 One trajectory each	25
788	C.5 Repairing at the end is not the same as being right	25
789	C.6 The two ends of the dichotomy	26
790		
791		
792		
793		
794	D The move alphabet	27
795	D.1 The lattice-gas dictionary	27
796	D.2 The arrow identity, written out	28
797	D.3 What is not in the alphabet, and why nothing is lost	29
798	D.4 Single-electron moves	29
799		
800		
801		
802	E Admissibility, the table and the jam	30
803	E.1 Admissible moves, worked at a real reactant	30
804	E.2 The table $\mathcal{T}$, one inequality and 65 numbers	31
805	E.3 How the capacities are measured	31
806	E.4 What the sweep returns	32
807	E.5 A third correction, in the reconstruction itself	33
808		
809		

810	E.6	What one move does to a capacity, and the jam	33
811	E.7	Where the table is checked	34
812	E.8	Realisation on a GPU	34
813			
814			
815	F	The bridge, the head and decoding	35
816	F.1	Decomposition, cutting a step into arrows	36
817	F.2	Ordering	36
818	F.3	The head, stage by stage	36
819	F.4	Two ways a decoder can go wrong	37
820	F.5	Why not a continuous-time formulation	39
821			
822			
823			
824	G	Proofs	39
825			
826	H	Related work	39
827			
828			
829	I	Experiment settings	41
830	I.1	Data	41
831	I.2	Training procedure	41
832	I.3	Decoding budgets	41
833			
834			
835	J	Further experiments	42
836	J.1	What makes the comparison fair, and what the five regimes measure	42
837	J.2	The main comparison and the size sweep, plotted	43
838	J.3	Two qualifications	43
839	J.4	The other four regimes at the smallest width	44
840	J.5	The size sweep under the other four regimes	44
841			
842			
843			
844			
845			
846			
847			
848			
849			
850			
851			
852			
853			
854			
855			
856			
857			
858			
859			
860			
861			
862			
863			

A EXTENDED BACKGROUND, THE OBJECTS AND THE CHECKS

This appendix develops §2 at tutorial length. Nothing here is needed to follow the construction. It is written for a reader who wants the encoding, the per-atom quantities and the conservation law in more detail than the main text carries.

A.1 THE TASK

A reaction mechanism is an ordered sequence of elementary steps. One elementary step is a pair of molecule sets, a reactant side and a product side, together with an *atom map*, a bijection between the atoms of the two sides. Because the map is a bijection, both sides carry one indexing $1, \dots, N$, and no atom is created or destroyed: the nuclei stay put, and what changes is where the valence electrons sit. Given B_r we require a conditional distribution $p_\theta(\cdot \mid B_r)$ from which candidates can be drawn and scored. There is a further requirement, and it is the subject of this paper: a candidate is useful only if it is a *molecule*. The emitted matrix must be reconstructible into a chemical structure, and it must carry the same number of valence electrons as the reactant. A model that is accurate but occasionally emits a five-valent carbon has not solved the task. It has solved a relaxation of it. Of the four checks of §2, the heavy atoms and the hydrogens are automatic, the atom map fixing the vertex set, and the electron count reduces to $\mathcal{N}$, since summing q_i over i and using $Z_{ij} = Z_{ji}$ gives $\sum_i (V_{a_i} - q_i) = \sum_i Z_{ii} + 2 \sum_{i < j} Z_{ij} + \sum_i n_i^H = \mathcal{N}(Z) + \sum_i n_i^H$, whose last term is fixed by the atom map.

A.2 THREE DESCRIPTIONS OF ONE MOLECULE

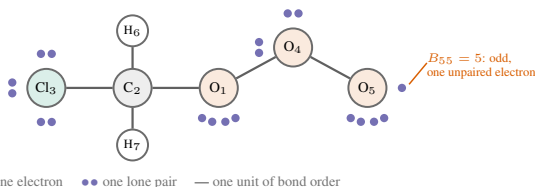
Figure 3 carries one reactant of the dataset through all three descriptions this paper uses at once, the string the file stores, the structure a chemist draws, and the matrix the model is handed. They carry the same information, and the atom map is what makes passing between them a bijection rather than a matching problem. Off-diagonal entries are integers because aromatic rings are resolved into alternating single and double bonds before the matrix is built, the matrix. Measured on the test split, no matrix has a fractional entry and $\max_{i \neq j} B_{ij} = 3$.

(a) text, what the dataset stores

[O:1] ([C:2] ([Cl:3]) ([H:6]) [H:7]) [O:4] [O:5]

Each $:k$ is the atom map. It names the row and column that atom owns in (c), and it is the *same* name on the product side, so no atom is created or destroyed.

(b) graph, the Lewis structure a chemist draws



• one electron •• one lone pair — one unit of bond order

(c) matrix, what the model sees, $B_r \in \text{Sym}_7$

	O ₁	C ₂	Cl ₃	O ₄	O ₅	H ₆	H ₇
O ₁	4	1	0	1	0	0	0
C ₂	1	0	1	0	0	1	1
Cl ₃	0	1	6	0	0	0	0
O ₄	1	0	0	4	1	0	0
O ₅	0	0	0	1	5	0	0
H ₆	0	1	0	0	0	0	0
H ₇	0	1	0	0	0	0	0

■ diagonal: lone-pair electrons, $w_e = 1$

□ off-diagonal: bond order, shared pairs, $w_e = 2$

(d) what (1)–(3) read off that matrix, with no chemistry left in them

$\beta_{O_1} = B_{12} + B_{14} = 1 + 1 = 2$, $c_{O_1} = \frac{1}{2} \cdot 4 + 2 = 4$: a full octet; $\beta_{O_5} = B_{45} = 1$, $c_{O_5} = \frac{1}{2} \cdot 5 + 1 = 3.5$: the half is the radical;

$q_i = V_{a_i} - B_{ii} - \sum_{j \neq i} B_{ij} = 0$ at every atom, so the molecule is neutral; $\mathcal{N}(B_r) = \underbrace{19}_{\text{diagonal}} + 2 \cdot \underbrace{6}_{\text{bond orders}} = 31$.

The valence electrons of the free atoms agree, $3 \cdot 6 + 4 + 7 + 2 \cdot 1 = 31$ ✓.

Figure 3: One molecule of the mechanism dataset, the alkyl trioxide radical $\text{ClCH}_2\text{—O—O—O}^\bullet$, written the three ways this paper uses. Note the deliberate unit mismatch: the shaded diagonal counts *electrons* and everything off it counts *pairs*, which is why $\mathcal{N}$ of (3) weights them 1 and 2. The quantity to keep in view is c_i , the number of pairs touching a nucleus: it is 4 at every closed-shell atom here, and the octet rule is the statement that it cannot exceed that. Every entry is a small non-negative integer, the observation on which §3 turns.

Remark 4 (the dataset makes every hydrogen its own atom). On this dataset every hydrogen is written explicitly and carries an atom map, so it has a row of the matrix like any other atom and $n_i^H = 0$ throughout, and measured over 8000 test reactants, the total number of implicit hydrogens is zero. The term is kept in (1) because the formalism is not specific to this dataset, but no measured quantity depends on it, and in the implementation β_i is a plain row sum.

A.3 ONE STEP, CHECKED ENTRY BY ENTRY

Figure 4 takes the molecule of Figure 3 one step forward and checks $\mathcal{N}(\Delta) = 0$ entry by entry. Exactly three entries change, the two diagonal entries of the atoms involved and the bond entry between them, which is the pattern §5 turns into the model’s move alphabet.

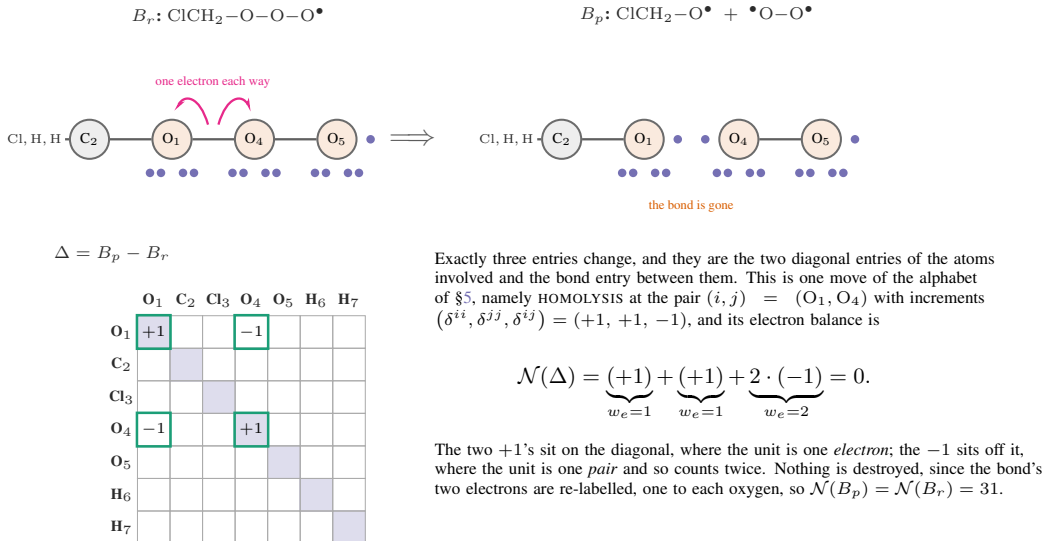


Figure 4: The step whose reactant is Figure 3: the peroxide bond breaks, one electron going to each oxygen. Left, what a chemist draws, two *fish-hooks*, each moving a single electron, because the departure and the two arrivals are one event. Right, the same statement as a matrix: Δ has exactly three non-zero positions and the weighted sum of those three is zero *identically*, not by accident.

B DISCRETE FLOWER, DERIVATION, PARAMETERISATION AND ALGORITHM

This appendix gives the detailed derivation and description of Discrete FlowER, which §3 summarises. It builds the path, derives the generator and the marginal rate, states the loss and relates it to the evidence bound, gives the sampler and contrasts its trajectory with that of Continuous FlowER, and ends with the network parameterisation and the algorithm. Equations (5), (6), (7) and (8) are restated from §3 under their original numbers.

B.1 STATE SPACE, COUPLING AND PATH

State space and coupling. Keep the integrality the data possess and bound each entry by chemistry:

$$\mathcal{V}_{\text{off}} = \{0, 1, 2, 3\}, \quad \mathcal{V}_{\text{dia}} = \{0, \dots, 14\}, \quad \mathcal{V}_e = \begin{cases} \mathcal{V}_{\text{off}}, & e \in \mathcal{O} \\ \mathcal{V}_{\text{dia}}, & e \in \mathcal{D} \end{cases} \quad \mathcal{X} = \prod_{e \in \mathcal{E}} \mathcal{V}_e \quad (5)$$

a finite set of size $4^{N(N-1)/2} \cdot 15^N$. Finiteness is what makes categorical distributions, rate matrices and exact likelihoods available (Austin et al., 2021; Campbell et al., 2022). The coupling is the one place this differs from generative modelling as usually presented, since *the source is the conditioning object*. Generic discrete constructions must choose a prior, either an absorbing MASK state (Austin et al., 2021; Sahoo et al., 2024; Shi et al., 2024), whose reverse process is a masked language model, or a uniform draw over $\mathcal{X}$ (Hoogeboom et al., 2021; Campbell et al., 2022; Lou et al., 2024), because the target is unconditional and there is nothing else to start from. Here the reactant is already a point of $\mathcal{X}$, so $(x_0, x_1) \sim \pi$ with $x_0 = B_r$, $x_1 = B_p$ in $\mathcal{X}$, and the path is a *Markov bridge* pinned at two data points, as in RetroBridge (Igashov et al., 2024) and RetroSynFlow (Yadav et al., 2025) and, in the continuous setting, in data-to-data bridge constructions (Liu et al., 2023b; Shi et al., 2023). No prior appears anywhere below.

Why a process, and why it jumps. Maximum likelihood needs a normalised density, and the decisive observation is about the *data*. By (5) a normalised distribution over a *single entry* of B is a

softmax over at most fifteen classes, which is writable, differentiable, exactly normalised and free. What is unavailable is the *joint* over all $O(N^2)$ entries. A product of per-entry softmaxes normalises trivially but asserts something false, a bond order not being choosable independently of the lone-pair counts on the two atoms it touches, those being tied through (1). Anything richer needs a normalising constant summing over a set exponentially large in N . The escape is the one generative modelling always takes (Ho et al., 2020; Song et al., 2021; Lipman et al., 2023). Do not write the joint, define it as the law of the output of a procedure. If a process $(x_t)_{t \in [0,1]}$ starts at something we already hold and ends distributed as B_p , sampling means simulating it and no normalising constant is formed. What is learned is the process’s *local* behaviour, which is exactly the kind of object a network can fit. Two choices remain. *What it runs on.* We keep the lattice, so the state is an integer matrix at all times, and since between two adjacent lattice points there is nothing at all, such a process must sit at a point, wait, and *jump*. It is therefore a continuous-time Markov chain, and its local behaviour is a matrix of jump *rates*, not a velocity (Campbell et al., 2022; Gat et al., 2024). *What path it takes.* The simplest is to move probability mass rather than a point, so that each entry holds its reactant value for a while and, at a random time, switches once and for all to its product value (Campbell et al., 2024; Gat et al., 2024).

The conditional path. Let $\kappa : [0, 1] \rightarrow [0, 1]$ be a schedule with $\kappa_0 = 0$, $\kappa_1 = 1$, $\dot{\kappa}_t \geq 0$, and we use $\kappa_t = t$. Read κ_t as the probability that a given entry has already switched. The path lets each position independently sit at its reactant value with probability $1 - \kappa_t$ and at its product value otherwise:

$$p_t(x \mid x_0, x_1) = \prod_{e \in \mathcal{E}} \left[(1 - \kappa_t) \delta_{x_e, (x_0)_e} + \kappa_t \delta_{x_e, (x_1)_e} \right] \quad (6)$$

with $\delta_{u,v}$ the Kronecker delta. This is the discrete counterpart of the straight-line interpolation of continuous flow matching (Lipman et al., 2023; Liu et al., 2023a; Albergo & Vanden-Eijnden, 2023). In a vector space one moves a *point* along a segment, here one moves *probability mass* between the two atoms of a two-point measure, independently at each position (Campbell et al., 2024; Gat et al., 2024). To sample it,

$$Z_e \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\kappa_t), \quad (x_t)_e = (1 - Z_e)(x_0)_e + Z_e(x_1)_e, \quad (x_t)_{ji} := (x_t)_{ij}, \quad (18)$$

the draw being made on $\mathcal{E}$ and mirrored, so symmetry is a property of the representation and never has to be projected. Substituting $\kappa_0 = 0$, $\kappa_1 = 1$ gives $p_0 = \delta_{x_0}$, $p_1 = \delta_{x_1}$, so both endpoints are hit *exactly*, with no residual noise scale to trade off. Every entry always holds a value in $\mathcal{V}_e$, so $x_t \in \mathcal{X}$ for every t and every draw, so an intermediate state is a *partially executed mechanism* (Figure 6a).

B.2 THE GENERATOR

On a finite space a continuous-time Markov chain is specified by a rate matrix $u_t(\nu, x)$, the rate of jumping from x to ν , subject to

$$\underbrace{u_t(\nu, x) \geq 0 \ (\nu \neq x)}_{\text{non-negative}}, \quad \underbrace{\sum_{\nu} u_t(\nu, x) = 0}_{\text{columns sum to zero}}, \quad \underbrace{\partial_t p_t(\nu) = \sum_x u_t(\nu, x) p_t(x)}_{\text{Kolmogorov forward equation}}, \quad (19)$$

the standard object of continuous-time discrete generative modelling (Campbell et al., 2022; 2024; Gat et al., 2024; Holderrieth et al., 2025). The second condition forces $u_t(x, x) = -\sum_{\nu \neq x} u_t(\nu, x)$, the total rate of leaving x , and the third is the discrete continuity equation. Because (6) factorises, work at one entry and drop e . Let $\alpha := (x_0)_e$, $b := (x_1)_e$ with $\alpha \neq b$, so $p_t(\nu) = (1 - \kappa_t) \delta_{\nu, \alpha} + \kappa_t \delta_{\nu, b}$. Rather than guess a rate, derive it from the behaviour wanted. From any $\nu \neq b$ jump to b at rate λ_t , and from b never jump. Then the mass at b grows at rate λ_t times the mass not yet there, and requiring that mass to be κ_t gives $\dot{\kappa}_t = \lambda_t(1 - \kappa_t)$. Hence, the jump target being b and the departure from the current value,

$$\lambda_t = \frac{\dot{\kappa}_t}{1 - \kappa_t}, \quad u_t(\nu_e, x_e \mid x_0, x_1) = \lambda_t \left(\delta_{\nu_e, (x_1)_e} - \delta_{\nu_e, x_e} \right), \quad (20)$$

the subtracted term supplying the negative diagonal. Appendix B.3 verifies that this is a legitimate rate matrix and that it generates (6).

B.3 THE CONDITIONAL RATE IS A LEGITIMATE GENERATOR, AND IT GENERATES THE BRIDGE

Fix a position and drop e , then write $\alpha = (x_0)_e$, $b = (x_1)_e$ with $\alpha \neq b$. For $\nu \neq x_e$ the second delta of (20) vanishes, leaving $\lambda_t \delta_{\nu,b} \geq 0$, and $\sum_{\nu} u_t(\nu, x_e) = \lambda_t (\sum_{\nu} \delta_{\nu,b} - \sum_{\nu} \delta_{\nu,x_e}) = \lambda_t (1 - 1) = 0$, so (20) satisfies both conditions of (19). That it generates (6) follows by substitution: using that p_t normalises,

$$\begin{aligned} \sum_{\nu'} u_t(\nu, \nu') p_t(\nu') &= \lambda_t \sum_{\nu'} (\delta_{\nu,b} - \delta_{\nu,\nu'}) p_t(\nu') = \lambda_t [\delta_{\nu,b} - p_t(\nu)] \\ &= \lambda_t [\delta_{\nu,b} - (1 - \kappa_t) \delta_{\nu,\alpha} - \kappa_t \delta_{\nu,b}] \\ &= \lambda_t (1 - \kappa_t) [\delta_{\nu,b} - \delta_{\nu,\alpha}] = \dot{\kappa}_t [\delta_{\nu,b} - \delta_{\nu,\alpha}], \end{aligned}$$

which is exactly $\partial_t p_t(\nu)$ obtained by differentiating (6).

B.4 THE MARGINAL RATE IS A POSTERIOR EXPECTATION

The conditional generator depends on x_1 , which at sampling time is precisely what we do not have. The object a sampler can follow generates the mixture $p_t(x) = \sum_{x_0, x_1} p_t(x | x_0, x_1) \pi(x_0, x_1)$:

$$\begin{aligned} \partial_t p_t(\nu) &= \sum_{x_0, x_1} \pi(x_0, x_1) \sum_x u_t(\nu, x | x_0, x_1) p_t(x | x_0, x_1) && \text{exchange } \partial_t \text{ with a finite sum, then} \\ &&& \text{(19) for the conditional path} \\ &= \sum_x p_t(x) \underbrace{\sum_{x_0, x_1} \pi(x_0, x_1 | x_t = x) u_t(\nu, x | x_0, x_1)}_{=: \bar{u}_t(\nu, x)} && \text{Bayes on } \pi(x_0, x_1) p_t(x | x_0, x_1), \text{ then} \\ &&& \text{swap the sums} \end{aligned}$$

so that, the bracket being by definition a conditional expectation,

$$\bar{u}_t(\nu, x) = \mathbb{E}[u_t(\nu, x | x_0, x_1) | x_t = x], \quad \partial_t p_t(\nu) = \sum_x \bar{u}_t(\nu, x) p_t(x). \quad (21)$$

The rate generating the marginal path is the posterior mean of the rate generating the conditional path. This is the hinge of the method, and it is the discrete instance of the conditional-to-marginal argument of flow matching (Lipman et al., 2023; Campbell et al., 2024; Gat et al., 2024), which generator matching (Holderrieth et al., 2025) shows holds for an arbitrary Markov generator. Inserting (20), and using that only the first delta involves x_1 while the second is a function of x alone,

$$\bar{u}_t(\nu_e, x) = \lambda_t \mathbb{E}[\delta_{\nu_e, (x_1)_e} | x_t = x] - \lambda_t \delta_{\nu_e, x_e} = \lambda_t (p_{1|t}(\nu_e | x) - \delta_{\nu_e, x_e}), \quad (22)$$

an indicator's mean being the probability of its event, where $p_{1|t}(\nu_e | x) := \Pr[(x_1)_e = \nu_e | x_t = x]$ is the *posterior endpoint marginal*. With $\kappa_t = t$, so $\lambda_t = 1/(1-t)$, set this beside the endpoint form of a continuous flow-matching velocity (Lipman et al., 2023; 2024): $u_t(x) = \frac{1}{1-t} (\mathbb{E}[x_1 | x_t = x] - x)$ against $\bar{u}_t(\cdot, x) = \frac{1}{1-t} (p_{1|t}(\cdot | x) - \delta_x)$. The two are identical in shape, a difference of *points* having become a difference of *distributions*, the current position represented by the point mass δ_x . The reformulation is a substitution, not a redesign, which is why a network body designed for one transfers unchanged to the other.

B.5 WHAT TO PARAMETERISE, AND THE LOSS

By (22), knowing $p_{1|t}$ is the same as knowing the generator, so the network predicts the *endpoint*, per position, and is trained by maximum likelihood for that endpoint at states drawn from the bridge, by a plain cross-entropy, as in x_1 -prediction parameterisations of discrete diffusion (Austin et al., 2021; Campbell et al., 2024; Gat et al., 2024):

$$p_\theta(\nu | x, t) = \prod_{e \in \mathcal{E}} p_\theta^e(\nu_e | x, t), \quad p_\theta^e(\nu_e | x, t) = \frac{\exp \ell_\theta(x, t)_{e, \nu_e}}{\sum_{\nu' \in \mathcal{V}_e} \exp \ell_\theta(x, t)_{e, \nu'}}, \quad (7)$$

$$\mathcal{L}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0,1]} \mathbb{E}_{(x_0, x_1) \sim \pi} \mathbb{E}_{x_t \sim p_t(\cdot | x_0, x_1)} \left[- \sum_{e \in \mathcal{E}} \log p_\theta^e((x_1)_e | x_t, t) \right]. \quad (8)$$

In words, corrupt the product back towards the reactant along the bridge, then ask the network to name the product’s value at every position. Nothing is simulated or estimated. Fixing t and x_t and looking at one position, the inner expectation is over x_1 drawn from the posterior, so it equals $H(p_{1|t}) + \text{KL}(p_{1|t} \| p_\theta^e) \geq H(p_{1|t})$ by Gibbs’ inequality, whose first term is free of θ . Hence $p_{\theta^*}^e(\cdot | x, t) = p_{1|t}(\cdot | x)$ and $u_t^{\theta^*} = \bar{u}_t$, the exact marginal generator. Appendices B.6 and B.7 write the identity out and relates (8) to the path-space evidence bound.

B.6 THE MINIMISER OF THE LOSS

Fix t and x_t and look at one position. The inner expectation of (8) is over x_1 drawn from the posterior, so it equals $-\sum_\nu p_{1|t}(\nu | x_t) \log p_\theta^e(\nu | x_t, t)$. Adding and subtracting $\sum_\nu p_{1|t} \log p_{1|t}$ writes this as

$$H(p_{1|t}) + \text{KL}(p_{1|t} \| p_\theta^e) \geq H(p_{1|t})$$

by Gibbs’ inequality, with equality if and only if $p_\theta^e = p_{1|t}$, and the first term is free of θ . Hence $p_{\theta^*}^e(\cdot | x, t) = p_{1|t}(\cdot | x)$ and, by (22), $u_t^{\theta^*} = \bar{u}_t$: the model recovers the exact marginal generator.

B.7 RELATION TO THE EVIDENCE BOUND

Writing the cross-entropy down directly is legitimate but leaves open what it bounds. Let q and p_θ be continuous-time Markov chains on the same finite space with rate matrices u_t^q, u_t^θ and the same initial law. Discretising $[0, 1]$ into steps of length h , expanding the one-step Kullback–Leibler divergence to first order in h and letting $h \rightarrow 0$ gives the path-space divergence

$$\text{KL}(q \| p_\theta) = \mathbb{E}_q \left[\int_0^1 \sum_{\nu \neq x_t} \left(u_t^\theta(\nu, x_t) - u_t^q(\nu, x_t) + u_t^q(\nu, x_t) \log \frac{u_t^q(\nu, x_t)}{u_t^\theta(\nu, x_t)} \right) dt \right]. \quad (23)$$

Taking q to be the bridge conditioned to end at x_1 , the variational argument gives $-\log p_\theta(x_1) \leq \text{KL}(q \| p_\theta) + \text{const}$, and discarding from (23) every term free of θ leaves

$$-\log p_\theta(x_1) \leq \mathbb{E}_q \left[\int_0^1 \sum_{\nu \neq x_t} \left(u_t^\theta(\nu, x_t) - u_t^q(\nu, x_t) \log u_t^\theta(\nu, x_t) \right) dt \right] + \text{const}. \quad (24)$$

The model’s rate is non-zero only for single-entry changes, so with $u_t^\theta(\nu_e, x) = \lambda_t p_\theta^e(\nu_e | x, t)$ for $\nu_e \neq x_e$, and using that each p_θ^e normalises, $\sum_{\nu \neq x_t} u_t^\theta(\nu, x_t) = \lambda_t \sum_e (1 - p_\theta^e((x_t)_e | x_t, t))$, while the bridge moves position e to $(x_1)_e$ at rate λ_t exactly when that position has not yet switched, so that $\sum_{\nu \neq x_t} u_t^q(\nu, x_t) \log u_t^\theta(\nu, x_t) = \lambda_t \sum_e \mathbb{1}[(x_1)_e \neq (x_t)_e] (\log \lambda_t + \log p_\theta^e((x_1)_e))$. Substituting both into (24) and absorbing the θ -free pieces,

$$-\log p_\theta(x_1) \leq \int_0^1 \lambda_t \mathbb{E}_{x_t} \left[\sum_e \mathbb{1}[(x_1)_e \neq (x_t)_e] \left(-\log p_\theta^e((x_1)_e | x_t, t) \right) \right] dt + C(\theta).$$

The bound carries the time weight λ_t and restricts the sum to positions that have not yet switched, whereas the implemented loss (8) weights all times equally and sums over all positions. Both are non-negative combinations of the *same* per-position cross-entropies, each of which is individually minimised at $p_\theta^e = p_{1|t}$, so any strictly positive reweighting leaves the minimiser unchanged. The two objectives differ in the variance and conditioning of the gradient estimator, not in what they fit.

B.8 WHY A PROPER SCORING RULE, DRAWN

Remark 5 (why cross-entropy, and why iterating). Cross-entropy is a *strictly proper scoring rule* (Gneiting & Raftery, 2007), so its optimum is the entire predictive distribution, not a summary. If a reactant admits two mechanisms with $\pi(x_1 | x_0) = \frac{1}{2} \delta_{x_1^{(1)}} + \frac{1}{2} \delta_{x_1^{(2)}}$, the optimum at $t = 0$ is $\frac{1}{2} \delta_{(x_1^{(1)})_e} + \frac{1}{2} \delta_{(x_1^{(2)})_e}$ and sampling selects one of the two, whereas the squared-error objective of Continuous FlowER (Joung et al., 2025) has as its optimum the posterior *mean*, which is neither mechanism and in general not even integral (Figure 5). The head does represent only the

marginals of the posterior, so one step of size $h = 1$ could emit a chimera taking some positions from one mechanism and some from the other. But once a few positions have switched, x_t identifies the branch and $\lim_{t \rightarrow 1} \prod_e p_{1|t}(\nu_e | x_t) = \delta_{\nu, x_1}$. The chain trades a cheap factorised head for many small steps, so the number of integration steps is a modelling parameter, not merely a numerical one.

Let $\pi(x_1 | x_0) = \frac{1}{2}\delta_{x_1^{(1)}} + \frac{1}{2}\delta_{x_1^{(2)}}$ with $x_1^{(1)} \neq x_1^{(2)}$. At $t = 0$ the state carries no information about which branch was taken, so the posterior equals the prior and the minimiser above is $p_{\theta^*}^e(\cdot | x_t, t)|_{t=0} = \frac{1}{2}\delta_{\cdot, (x_1^{(1)})_e} + \frac{1}{2}\delta_{\cdot, (x_1^{(2)})_e}$. Both modes are represented explicitly and sampling selects one rather than averaging them. Under the squared-error objective that Continuous FlowER (Joung et al., 2025) uses, the optimal target is instead $\frac{1}{2}(x_1^{(1)} + x_1^{(2)})$, a matrix that is neither mechanism and in general not even integral, which then has to be rounded. Figure 5 contrasts the two.

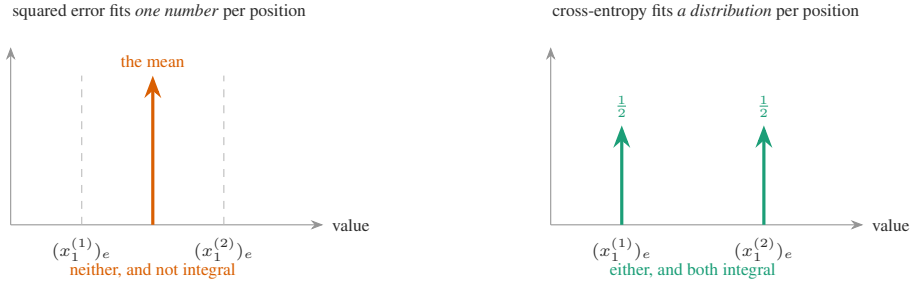


Figure 5: Why the objective matters, at one matrix position of a reactant admitting two mechanisms. A squared-error optimum is the posterior mean and lands between the modes; a cross-entropy optimum is the posterior itself and keeps them apart. Only the right-hand object can be sampled to give either mechanism, and only the right-hand object is supported on the lattice.

B.9 SAMPLING, AND WHAT DECODING COSTS

Write U_t for the marginal rate matrix of (21), the array whose (ν, x) entry is $\bar{u}_t(\nu, x)$. Over an interval h with U_t held fixed the exact transition kernel is the matrix exponential $\exp(hU_t) = I + hU_t + O(h^2)$, and truncating it gives the Euler scheme, the discrete-space analogue of an ODE solver step and a first-order case of the τ -leaping used for continuous-time discrete models (Campbell et al., 2022; Gat et al., 2024). Entrywise, with S steps of size $h = 1/S$ on $t_k = kh$, substituting (22) and collecting the two delta terms,

$$\begin{aligned} \Pr[x_e^{(k+1)} = \nu] &= \delta_{\nu, x_e^{(k)}} + h u_{t_k}^\theta(\nu, x_e^{(k)}) \\ &= (1 - \lambda_k) \delta_{\nu, x_e^{(k)}} + \lambda_k p_\theta^e(\nu | x^{(k)}, t_k), \quad \lambda_k := h \lambda_{t_k} = \frac{1}{S-k}, \end{aligned} \quad (25)$$

which is a distribution, both terms being non-negative and summing to $(1 - \lambda_k) + \lambda_k = 1$. Read as an algorithm, resample position e from the endpoint distribution with probability λ_k , otherwise leave it untouched. Since $\lambda_0 = 1/S$ rises to $\lambda_{S-1} = 1$, early steps are conservative and the last is a forced commitment, the divergence of λ_t as $t \rightarrow 1$ being exactly absorbed by the discretisation. At the last step the mode is taken, so the emitted candidate is $\hat{B}_e = \arg \max_\nu p_\theta^e(\nu | x^{(S-1)}, t_{S-1})$, whose exact log-probability the factorised head supplies at a single sample,

$$s(\hat{B}) = \sum_{e \in \mathcal{E}} \log p_\theta^e(\hat{B}_e | x^{(S-1)}, t_{S-1}) = \log p_\theta(\hat{B} | x^{(S-1)}, t_{S-1}). \quad (26)$$

Decoding is then the identity map, since $\hat{B}$ is already an integer matrix, so no projection, rounding or repair stands between the model and its output. This is the first place where staying on the lattice pays. Continuous FlowER (Joung et al., 2025) ends its trajectory off the lattice and must round. Its field and its noise are both symmetrised, so the trajectory itself never leaves Sym_N , which is why Figure 6(b) draws symmetric matrices throughout, and the symmetry is lost only at the end. The sum-preserving rounding restores $\mathcal{N}$ but acts on the flat array, so it can round an entry and its mirror image apart, and Joung et al. (2025) report exactly that as one of the three modes their invalid outputs fall into. The rounding is not the cause of the other two. Negative entries are already present in the endpoint the field produces, and a matrix that is symmetric, non-negative and correctly

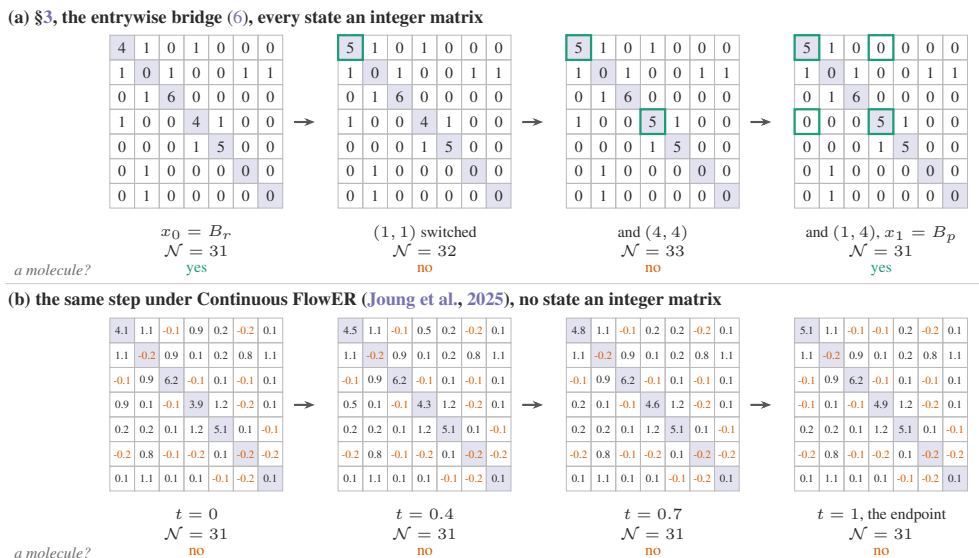


Figure 6: Two ways to move a bond–electron matrix from B_r to B_p , on the homolysis of the peroxide bond of Figure 3. Both rows draw the same seven-by-seven matrix at four points of the same step, in the same columns, so that only the contents differ. Diagonal cells are tinted, off-diagonal cells are plain, and a cell whose value is negative is set in orange. (a) The discrete bridge of §3. The three positions the step changes switch independently, one at a time, the outlined cells are those that have already switched, and the chain never leaves the lattice (5). Independence across positions costs the middle two columns, which carry 32 and 33 electrons, and §4 shows this to be a property of the hypothesis class rather than of the schedule. (b) The continuous path of Joung et al. (2025) on the same step. The drift is centred, so $\mathcal{N}$ is exactly 31 at every t , and the field and the noise are symmetrised, so every matrix drawn here is symmetric. Nothing else survives. Every one of the 49 entries is fractional at every t , entries are negative all over the matrix, and the endpoint is off the lattice and has to be rounded, a rounding that acts on the flat array and so may take the symmetry with it. Only the integrality of (a) makes decoding the identity map, and only the change of elementary event in §5 makes the intermediate columns of (a) molecules as well as matrices.

summed can still seat more pairs on an atom than (2) allows, a condition nothing in that construction represents. Figure 6 contrasts the two paths on one step. Equations (5), (6), (7), (8) and (25) are the whole of **Discrete FlowER**. It is complete, trainable and accurate, and §6 reports it as the direct baseline. Appendix B.10 gives the network parameterisation (the entry codes, the relative-attention body, the two heads and the copy initialisation), and Algorithm 1 in Appendix B.11 collects training and sampling in one place.

B.10 NETWORK PARAMETERISATION

The network must see both where the bridge started and where it currently stands, since the difference is what identifies the positions that have already moved. Both are small integers, so they are packed into a single index per position and embedded with a lookup table:

$$\gamma_{ij} = |\mathcal{V}_{\text{off}}| \cdot (B_r)_{ij} + (x_t)_{ij} \in \{0, \dots, 15\} \quad (i \neq j), \quad \gamma_{ii} := 0, \\ \tilde{\gamma}_i = |\mathcal{V}_{\text{dia}}| \cdot (B_r)_{ii} + (x_t)_{ii} \in \{0, \dots, 224\}.$$

The diagonal is removed from the pairwise code, whose alphabet of $15 \times 15 = 225$ values would overflow the sixteen bond codes, and carried per atom instead, which also reduces its storage from N^2 to N . With $z_i^{(l)} \in \mathbb{R}^d$ the hidden state of atom i at layer l , d the width, $\zeta(t)$ a sinusoidal time embedding and E_a, E_ℓ embedding tables,

$$R_{ij} = \text{GELU}(W_p \mathbf{1}_{16}[\gamma_{ij}] + b_p), \quad z_i^{(0)} = \sqrt{d} \left(E_a[a_i] + E_\ell[\tilde{\gamma}_i] \right),$$

$$\text{bias}_{m,ij}^{(l)} = \langle Q_{m,i}^{(l)} + w_m^{(l)}, [R_{ij}; \zeta(t)]_m \rangle, \quad z^{(l+1)} = \text{Attn}^{(l)}(z^{(l)}; \text{bias}^{(l)}),$$

so that the pairwise code enters as a relative attention bias rather than as an $O(N^2d)$ tensor. The two heads of §3 are

$$\ell_{ij,\nu}^{\text{bond}} = \frac{\langle Q_{i,\nu}, K_{j,\nu} \rangle + \langle Q_{j,\nu}, K_{i,\nu} \rangle}{\sqrt{d_{\text{head}}}} + \beta_{\nu}^{\top} \mathbf{1}_{16}[\gamma_{ij}] + \tau_{\nu}(t), \quad \nu \in \mathcal{V}_{\text{off}},$$

$$\ell_{i,\nu}^{\text{lon}} = (W_o z_i)_{\nu} + \Gamma[\tilde{\gamma}_i]_{\nu} + \rho_{\nu}(t), \quad \nu \in \mathcal{V}_{\text{dia}}.$$

The bond logit is a *symmetrised* bilinear form, so $\ell_{ij,\nu}^{\text{bond}} = \ell_{ji,\nu}^{\text{bond}}$ for all i, j, ν : symmetry is a property of the parameterisation and is never imposed afterwards on a raw asymmetric field. Arrow Flow Matching replaces these two heads by the four low-rank bilinear forms of (15), one per move type, plus a stop logit read from the pooled molecule, and the body above is shared unchanged.

Copy initialisation. Most positions do not change during a step, so the correct output is overwhelmingly “keep the current value”. With $\Lambda > 0$, setting $\beta_{\nu,m}|_{\text{init}} = \Lambda \delta_{\nu, m \bmod |\mathcal{V}_{\text{off}}|}$ and $\Gamma[m]_{\nu}|_{\text{init}} = \Lambda \delta_{\nu, m \bmod |\mathcal{V}_{\text{dia}}|}$ makes the network do exactly that at initialisation, since by construction of the codes $\gamma_{ij \bmod |\mathcal{V}_{\text{off}}|} = (x_t)_{ij}$ and $\tilde{\gamma}_i \bmod |\mathcal{V}_{\text{dia}}| = (x_t)_{ii}$, so the logit at the current value exceeds the others by Λ and $p_{\theta}^e(x_e)|_{\text{init}} = e^{\Lambda}/(e^{\Lambda} + |\mathcal{V}_e| - 1) \approx 0.999$ for $\Lambda = 8$ and $e \in \mathcal{O}$. The rationale is quantitative: the identity map is correct for all but $O(1)$ of the $|\mathcal{E}| = O(N^2)$ positions, while an Adam step moves a logit by at most $O(\eta)$, so discovering it from a random start costs $O(\Lambda/\eta)$ updates spent learning to copy.

B.11 THE ALGORITHM

Algorithm 1 Discrete FlowER, training and sampling.

Training (one step).

- 1: draw a step $(x_0, x_1) = (B_r, B_p) \sim \pi$ and a time $t \sim \mathcal{U}[0, 1]$
- 2: draw $Z_e \sim \text{Bernoulli}(t)$ independently for $e \in \mathcal{E}$ and set $(x_t)_e = (1 - Z_e)(x_0)_e + Z_e(x_1)_e$,
mirrored to the lower triangle ▷ (18)
- 3: form the codes $\gamma, \tilde{\gamma}$ from (B_r, x_t) , run the body, read off $p_{\theta}^e(\cdot | x_t, t)$ ▷ (7)
- 4: take a gradient step on $-\sum_{e \in \mathcal{E}} \log p_{\theta}^e((x_1)_e | x_t, t)$ ▷ (8)

Sampling (one candidate, S steps).

- 5: $x^{(0)} \leftarrow B_r$
 - 6: **for** $k = 0, \dots, S - 1$ **do**
 - 7: $t_k \leftarrow k/S, \quad \lambda_k \leftarrow 1/(S - k)$
 - 8: evaluate $p_{\theta}^e(\cdot | x^{(k)}, t_k)$ for every $e \in \mathcal{E}$
 - 9: **for** $e \in \mathcal{E}$ **do**
 - 10: with probability λ_k resample $x_e^{(k+1)} \sim p_{\theta}^e(\cdot | x^{(k)}, t_k)$, else $x_e^{(k+1)} \leftarrow x_e^{(k)}$ ▷ (25)
 - 11: **end for**
 - 12: mirror the upper triangle onto the lower one
 - 13: **end for**
 - 14: $\hat{B}_e \leftarrow \arg \max_{\nu} p_{\theta}^e(\nu | x^{(S-1)}, t_{S-1})$ and $s(\hat{B}) \leftarrow \sum_e \log p_{\theta}^e(\hat{B}_e | x^{(S-1)}, t_{S-1})$ ▷ (26)
 - 15: **return** $\hat{B}$, which is an integer matrix and needs no decoding
-

C THE OBSTRUCTION IN DETAIL

This appendix expands §4. It says what exactly the entrywise baseline is, why its training path conserves electrons only in mean, what the proof of Theorem 1 looks like as a picture, what the two elementary events do to one and the same step when they are drawn side by side, and why repairing the output at the end is no substitute.

C.1 WHAT EXACTLY THE ENTRYWISE BASELINE IS

Since it is the comparison used throughout, it is worth fixing what *the entrywise baseline* is, and where it factorises. Its state space is the box $\prod_{e \in \mathcal{E}} \mathcal{V}_e$ of (5), so a state is an integer symmetric matrix and nothing more. Its training path (6) is a Markov bridge from B_r to B_p in which each position

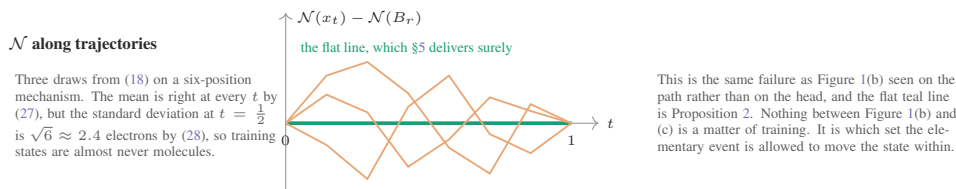


Figure 7: The cost of Figure 1(b) on the training path rather than on the head. The electron count of every draw of the entrywise bridge (18) leaves its reactant value in the interior of the path and returns to it only at $t = 1$, whereas every chain of §5 stays on the flat line.

switches from its reactant value to its product value at its own random time, independently of every other position, having switched by time t with probability κ_t . Its head (7) reads the pair (reactant, current state) and emits at each position *separately* a categorical distribution over $\mathcal{V}_e$ predicting that position’s product value, and the loss (8) is the cross-entropy of those $|\mathcal{E}|$ predictions against B_p . Its sampler (25) resamples each position independently from its own categorical and takes a per-position arg-max at the last step, so decoding is the identity map. The network body is the one Appendix B describes, and the model is accurate. What it never does, at any of those four points, is couple two positions, and that alone settles the question.

C.2 THE PATH IS EXACT IN MEAN AND NOT SURELY

From (18), $(x_t)_e = (x_0)_e + Z_e \Delta_e$ with $Z_e \sim \text{Bernoulli}(\kappa_t)$ independent, so by linearity of $\mathcal{N}$, $\mathcal{N}(x_t) - \mathcal{N}(x_0) = \sum_e w_e Z_e \Delta_e$ and

$$\mathbb{E}[\mathcal{N}(x_t)] - \mathcal{N}(x_0) = \sum_e w_e \mathbb{E}[Z_e] \Delta_e = \kappa_t \mathcal{N}(\Delta) = 0, \quad \text{linearity of expectation, then (3)} \quad (27)$$

$$\text{Var}[\mathcal{N}(x_t)] = \sum_e w_e^2 \Delta_e^2 \text{Var}[Z_e] = \kappa_t(1 - \kappa_t) \sum_{e \in \mathcal{E}} w_e^2 \Delta_e^2 > 0. \quad \begin{array}{l} \text{independence, and} \\ \text{Var}[Z_e] = \kappa_t(1 - \kappa_t) \end{array} \quad (28)$$

Only the *first moment* is correct. A six-position mechanism has $|\Delta_e| = 1$ and $w_e = 2$ at each changed position, so $\sum_e w_e^2 \Delta_e^2 = 24$ and the standard deviation at $\kappa_t = \frac{1}{2}$ is $\sqrt{6} \approx 2.4$ electrons (Figure 7), so the network is trained almost exclusively on states that are not molecules.

C.3 THE PROOF OF THEOREM 1, DRAWN

Figure 8 is the argument of Appendix G as a picture. The two support points the proof constructs differ in one coordinate only, so the linear functional separates them, and the only product support that survives is a single point.

C.4 ONE TRAJECTORY EACH

Figure 9 draws one trajectory of each construction on the same step. The entrywise bridge changes one *entry* at a time, so its intermediate states are matrices carrying the wrong number of electrons, whereas the move bridge of §5 changes one *move* at a time, and every state it passes through carries the right number by Proposition 2. Both are chains on the same lattice, and the only difference is what a single jump is allowed to do.

C.5 REPAIRING AT THE END IS NOT THE SAME AS BEING RIGHT

Imposing the constraint only when decoding, by projecting or filtering the output, delivers a valid candidate at two costs. The constraint plays no part in *producing* the candidate, so it censors a distribution that was never shaped by it, and the intermediate states are not molecules, so training and sampling supports disagree. The alternative, and the subject of the rest of this paper, is to make the constraint a property of the state space and of the moves, so that it holds at every instant by construction.

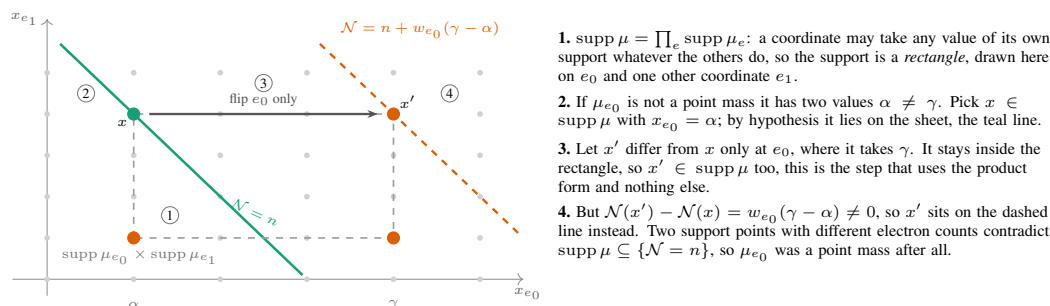
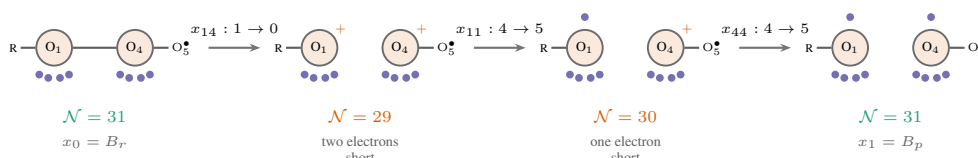
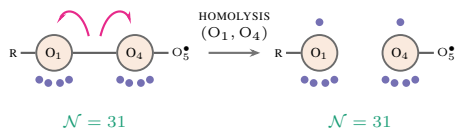


Figure 8: Theorem 1 in four steps, on two coordinates, with $w_{e_0} = 2$ and $w_{e_1} = 1$. Only step 3 uses the hypothesis that μ is a product, and only step 4 uses that $\mathcal{N}$ is linear with non-zero weights; nothing in the argument depends on the number of coordinates, on the alphabets being small, or on the level being n rather than anything else. The picture is the two-coordinate form of Figure 1, whose three coordinates are those of the running example.

(a) the entrywise bridge: the three entries switch *one at a time*, in any of six orders



(b) the move alphabet: the same three increments, as one event



The increments $(\delta^{11}, \delta^{44}, \delta^{14}) = (+1, +1, -1)$ are applied together, so $\mathcal{N}$ never leaves 31: this is the trajectory of Figure 1(a), which stays on the sheet because it moves along it. This particular step is one move and therefore has no interior state at all; a step of several moves does, and those interior states carry the right electron count by Proposition 2, though, as Figure 19 shows, they are not always molecules. The claim here is only about the electron count, and it is the claim the entrywise bridge of §3 cannot make.

Figure 9: One trajectory each, on the step of Figure 4. (a) The entrywise bridge switches positions independently, so it must pass through states in which the bond has gone but the electrons it held have not yet arrived, the reconstruction reads the first of them as a pair of oxygen cations two electrons short of the reactant, and the second as one electron short. The order drawn is one of six and the other five behave the same way, and the eight states reachable this way are the corners of the box of Figure 1(b), and only the two endpoints among them conserve. Averaged over the switch times the count is right, 31 exactly, with standard deviation $\sqrt{1.5} \approx 1.22$ electrons at $\kappa_t = \frac{1}{2}$. (b) The alphabet of §5 performs the same three increments as a single jump, which is what removes the variance rather than centring it.

The two costs are worth separating. Filtering censors a distribution that was never shaped by the constraint, so the model spends capacity on candidates it will never be allowed to emit and the retained mass is not the conditional the training objective fitted. Projection is worse. It maps an inadmissible candidate onto an admissible one by a rule the network never saw, so the emitted law is the pushforward of the model under an untrained map, which is neither the fitted conditional nor a calibrated approximation to it. Neither device touches the intermediate states, so the states the network is trained on and the states a chemist would read remain different sets. The construction of §5 removes both problems at once by moving the constraint into the support of the elementary event.

C.6 THE TWO ENDS OF THE DICHOTOMY

Theorem 1 is a dichotomy and not a bound, so it is worth seeing both of its ends drawn on the same axes. Figure 10 does that on the three positions the step of Figure 6 touches, whose values are $x_{11} = 4$, $x_{44} = 4$ and $x_{14} = 1$ at the reactant and 5, 5 and 0 at the product, so that the sheet within these three coordinates is $x_{11} + x_{44} + 2x_{14} = 10$. Panel (a) is the head the theorem permits, panel (b) is what any spread costs, and panel (c) is the head §5 builds, which is neither of the two because it is not a product measure at all.

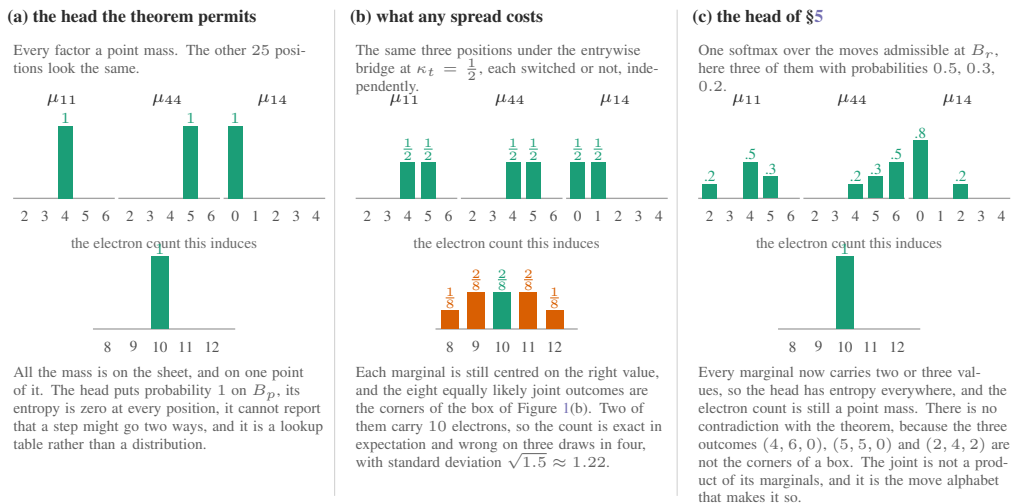


Figure 10: The two ends of the dichotomy Theorem 1 forces, and the third possibility that changing the elementary event opens, on the three positions the step of Figure 6 touches. The diagonal alphabets $\{0, \dots, 14\}$ are drawn truncated and all three electron-count axes are on one scale. (a) The conclusion of the theorem drawn, exact conservation being available to a factorised head at the price of every factor’s spread. (b) What the entrywise construction does at the other place it factorises, its training path. Nothing lies between (a) and (b), since the moment any factor carries two values its support acquires an edge, and the mass on that edge is mass off the sheet. (c) The escape. The head of (15) is a single softmax over moves rather than a product over positions, so its support is a set of matrices that all satisfy $\mathcal{N} = 10$ while its marginals are as spread out as the data require.

D THE MOVE ALPHABET

This appendix expands the first half of §5. It gives the lattice-gas reading in full, writes the arrow identity out on a real hop, shows why the chemist’s third arrow type is not needed, and justifies the two move types that carry single electrons.

D.1 THE LATTICE-GAS DICTIONARY

A bond–electron matrix is a lattice gas. Take the second and third facts of §2 seriously. Electrons are conserved and move in *pairs*, and every nucleus has a finite valence shell, capped by (2). Together they say that a bond–electron matrix is an *occupation-number configuration of a lattice gas*. A position $e \in \mathcal{E}$ is a *site*, with one lone-pair site per atom on the diagonal and one bond site per atom pair off it. A Lewis electron pair is a *particle*, “ $\mathcal{N}$ fixed” is “particle number fixed”, the bound $x_{ij} \leq 3$ is a site capacity, the octet rule (2) is a nuclear capacity, and a curly arrow is a *hop*. A mechanism is then a sequence of hops, a particle leaving one site and arriving at another. That a chemist draws this as *one* arrow with a tail and a head, rather than as two independent annotations, is the content of §5.

Figure 11 draws the correspondence of §5 on the whole reactant of Figure 3. The matrix *is* an occupation-number configuration, the octet rule is a capacity at each nucleus rather than a statement about the matrix as a whole, and a curly arrow is one hop. Three consequences organise everything that follows. A hop is a difference of two unit occupations, so it lies in the kernel of the particle-number functional and a chain built from hops cannot change $\mathcal{N}$ in any moment of its distribution. The constraints are *local* and a hop touches two nuclei, so whether a move keeps the configuration representable is an $O(1)$ test, cheap enough to be a *precondition* of the move rather than a check after the fact. And constrained lattice gases can *jam*, since a hop towards the target may overfill some site even though a simultaneous rearrangement does not, which is what decides how finely a step must be cut.

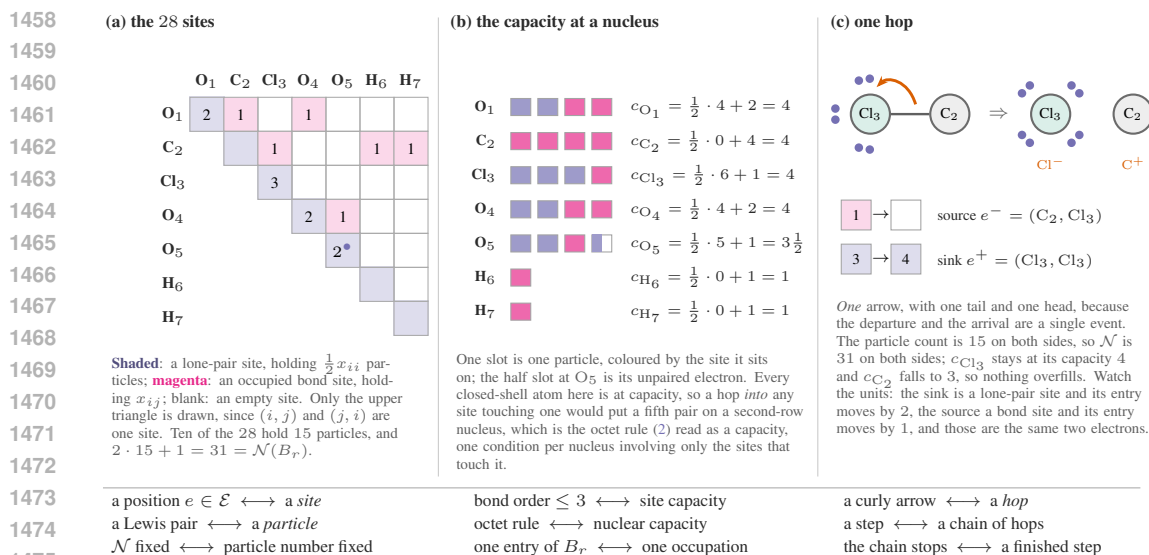


Figure 11: The lattice-gas reading of a bond–electron matrix, on the reactant of Figure 3. (a) The matrix is the occupation-number configuration: one lone-pair site per atom and one bond site per atom pair, 28 in all, and the number in a cell is how many particles sit there. (b) The octet rule is not a statement about the matrix as a whole but a capacity at each nucleus, on the sites that touch it, and its value is exactly the c_i of (1). (c) A mechanism is a sequence of hops, one particle at a time; what makes a hop conserving is not an extra condition but its shape, since by (9) a particle carries two electrons wherever it sits. Everything §5 does is an exercise in the dictionary at the foot.

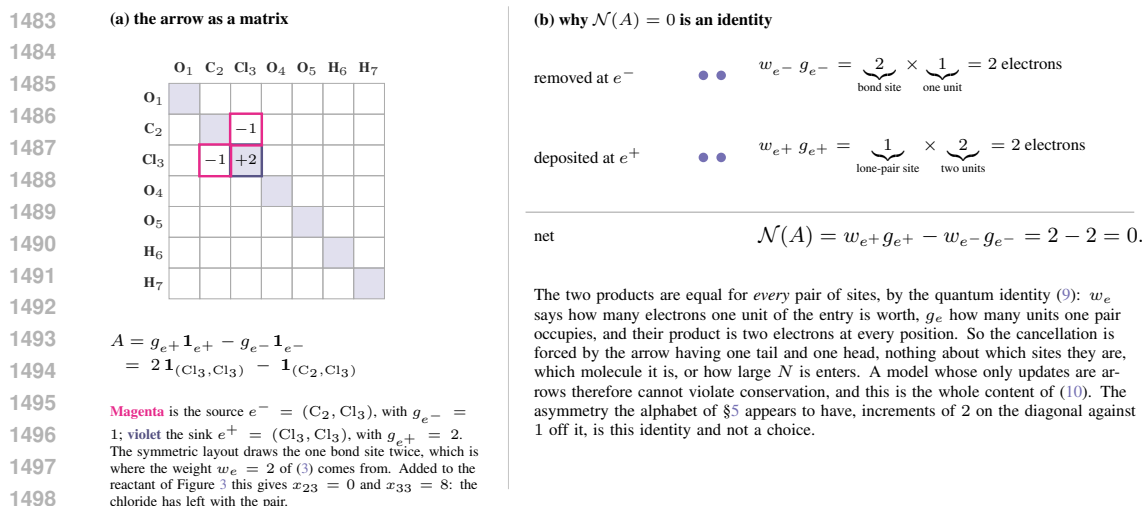


Figure 12: Equation (10) written out, on the hop that Figure 11(c) draws as a curly arrow. (a) An arrow is a matrix: $-g_{e-}$ at the site the pair leaves and $+g_{e+}$ at the site it reaches, and zero at the other 26 positions. (b) Its electron balance is a two-line ledger that closes identically, because the same pair is worth two electrons at both ends. §5 do nothing but repeat this for a chain of arrows.

D.2 THE ARROW IDENTITY, WRITTEN OUT

Equation (10) is short enough to hide what it says, so Figure 12 writes it out on the hop of Figure 11(c). The two sites the arrow touches carry different *units*, the source is a bond site counted in pairs, the sink a lone-pair site counted in electrons, and the increments g_{e-} and g_{e+} differ accordingly. What makes the weighted sum cancel is not that the increments match but that each is the reciprocal of its site's weight, which is the identity $w_e g_e = 2$ of (9). Conservation is therefore a property of the *shape* of an arrow, not a condition imposed on it, and it holds whatever the two sites are.

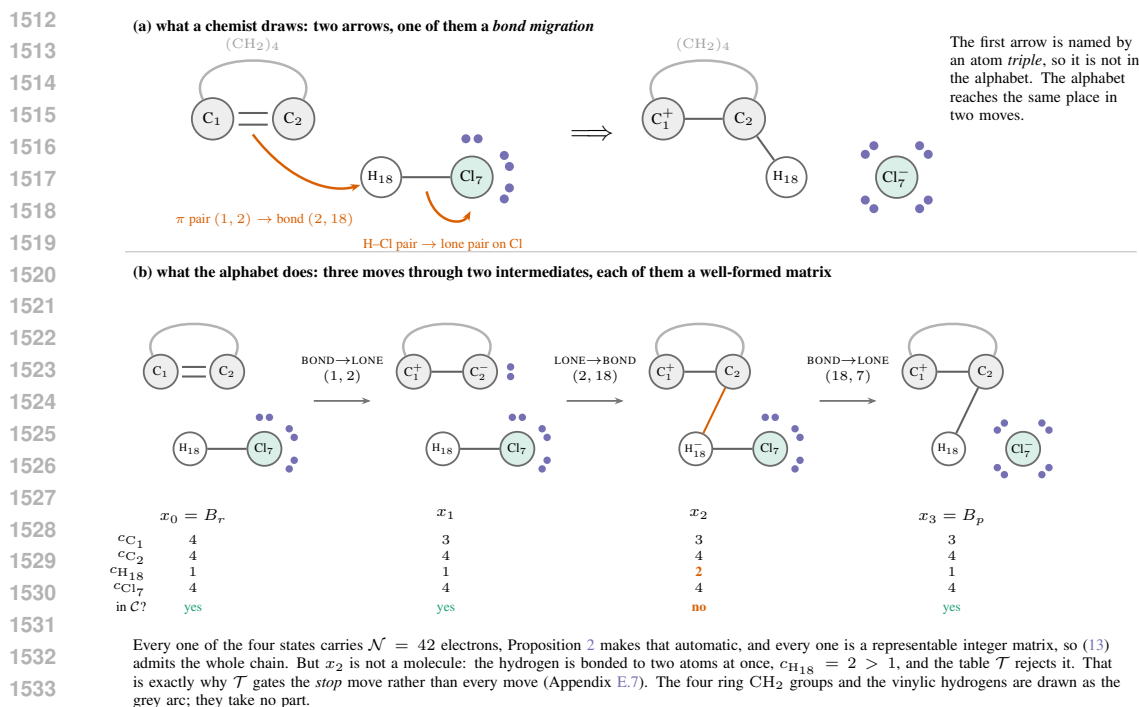


Figure 13: Protonation of cyclohexene by hydrogen chloride, a real test step, showing what the missing bond migration costs. (a) The chemist's first arrow moves a pair from one bond to an adjacent one. (b) The alphabet replaces it by BOND→LONE followed by LONE→BOND at the shared atom; the price is the intermediate x_1 , in which the pair sits on C_2 as a lone pair. The greedy ordering of §5 then produces x_2 , a hydrogen in flight between two partners, a legal bond–electron matrix that is not a legal molecule. Two of the six orders of these three moves keep every intermediate inside $\mathcal{C}$; the ordering is not asked to find one, because nothing in the guarantee depends on it.

D.3 WHAT IS NOT IN THE ALPHABET, AND WHY NOTHING IS LOST

Remark 6 (the bond migration is a composition). A chemist's third arrow type is the *bond migration*, in which a pair moves from the bond (i, j) to an adjacent bond (j, k) , one atom being shared. It is not in the alphabet, because it is named by an atom *triple* and would force the head to score a ragged set of candidates. It is also unnecessary, being the composition of two moves already present,

$$\underbrace{(i, j) \rightarrow (j, j)}_{\text{BOND} \rightarrow \text{LONE}} \text{ then } \underbrace{(j, j) \rightarrow (j, k)}_{\text{LONE} \rightarrow \text{BOND}} \implies \Delta x_{ij} = -1, \Delta x_{jj} = 2 - 2 = 0, \Delta x_{jk} = +1,$$

which is exactly the migration. The price is that the intermediate state, in which the pair sits on j as a lone pair before moving on, has to be a state the chain is allowed to occupy, and the cost in path length is a mean of 2.24 moves per step over the test split. It also buys something the alphabet was not designed for. Appendix E.6 shows that splitting the migration in two is exactly what unlocks the concerted mechanisms, which as single arrows cannot be started at all.

Figure 13 carries a real test step, the protonation of cyclohexene by hydrogen chloride, through both readings, and shows the intermediate the expansion buys.

D.4 SINGLE-ELECTRON MOVES

Pair coordinates would be a tempting simplification, dividing every entry by g_e , and every arrow becomes a unit hop. That is legitimate only when every diagonal entry is even, and on the test split it is not. Measured there, 12.1% of reactant matrices carry an odd diagonal entry, concentrated on P (1379 atoms), Pd (571), Na (34) and Ce (16), which are hypervalent phosphorus and organometallic centres, and 2.17% of steps change some diagonal by an odd amount. Those steps are not miscellaneous. Every one of them, without exception over 4000 test steps, is a single palladium–phosphorus bond breaking with one electron going to each atom, or the same bond forming from two such electrons. That is exactly rows (c) and (d) of the alphabet, so two move types cover the whole radical

sector of this dataset and no general single-electron transfer is needed. Their conservation is the same identity (11): $+1 + 1 + 2(-1) = 0$ and $-1 - 1 + 2(+1) = 0$.

Remark 7 (the alphabet is closed under inversion). Reversing LONE→BOND at (i, j) gives BOND→LONE at (j, i) , and HOMOLYSIS and COLLIGATION are each other’s reverse. So every move can be undone by another move of the alphabet, and that undo is admissible at the state the move produced, since it merely restores a state that was admissible a moment ago. Hence *no reachable state has an empty candidate list*, and the masked softmax (15) always has at least one finite class, which is what makes the mask safe to apply with no fallback branch.

E ADMISSIBILITY, THE TABLE AND THE JAM

This appendix expands the second half of §5. It works the admissibility test (13) through at a real reactant, gives the construction and the result of the capacity sweep behind $\mathcal{T}$, works the Diels–Alder jam out move by move, and shows how the mask is computed on a GPU.

E.1 ADMISSIBLE MOVES, WORKED AT A REAL REACTANT

Everything in (13) follows from one book-keeping fact. A move of type τ at (i, j) changes exactly three entries, hence exactly five of the scalars that appear anywhere in (12):

$$\Delta x_{ii} = \delta_{\tau}^{ii}, \quad \Delta x_{jj} = \delta_{\tau}^{jj}, \quad \Delta x_{ij} = \delta_{\tau}^{ij}, \quad \Delta \beta_i = \Delta \beta_j = \delta_{\tau}^{ij}, \quad \Delta \beta_k = 0 \quad (k \neq i, j). \quad (29)$$

The β increments are the row sums of (1), the only off-diagonal entry the move touches in row i is x_{ij} , and the only one in row j is $x_{ji} = x_{ij}$. Every other atom of the molecule is untouched, whatever N is. This is the locality of the lattice-gas reading, written out, and (13) is one interval test per line of (29).

Worked, at the running reactant. Everything (13) needs is read off two per-atom vectors and one entry of the matrix. For the reactant B_r of Figure 3, the diagonal x_{ii} and the row sums β_i of (1) are

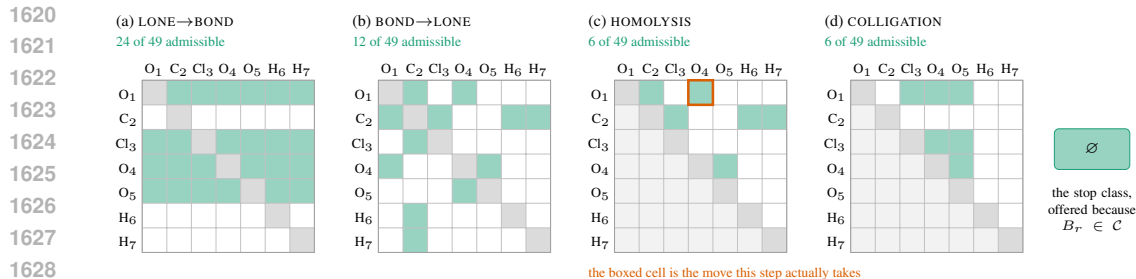
	O ₁	C ₂	Cl ₃	O ₄	O ₅	H ₆	H ₇
x_{ii} (lone-pair electrons)	4	0	6	4	5	0	0
β_i (bonding pairs)	2	4	1	2	1	1	1

Take LONE→BOND at the ordered pair $(i, j) = (O_1, C_2)$, whose increments are $(\delta^{ii}, \delta^{jj}, \delta^{ij}) = (-2, 0, +1)$. The five tests, in the order they are written in (13), are

$$\begin{array}{ccc} 0 \leq \underbrace{x_{ii} - 2 = 2}_{4} \leq 14, & 0 \leq \underbrace{x_{jj} + 0 = 0}_0 \leq 14, & 0 \leq \underbrace{x_{ij} + 1 = 2}_1 \leq 3, \\ \text{O}_1 \text{ has a lone pair to give} & \text{C}_2 \text{'s diagonal is untouched} & \text{the bond may grow to a double} \\ \\ 0 \leq \underbrace{\beta_i + 1 = 3}_2 \leq 12, & 0 \leq \underbrace{\beta_j + 1 = 5}_4 \leq 12; & \\ \text{O}_1 \text{ gains a bonding pair} & \text{C}_2 \text{ gains one too} & \end{array}$$

the value each test starts from is written under its symbol. All five pass, so the cell (O_1, C_2) of Figure 14(a) is admissible. Now the *same* move at the reversed pair (C_2, O_1) , where the first test reads $0 \leq 0 - 2 = -2$, which fails, because C_2 has no lone pair to give away. One failure is enough, and the whole C_2 row of that panel is empty for the same reason. This is why the panels are not symmetric, since a move is indexed by an *ordered* pair, and (i, j) asks something of i that (j, i) asks of j .

The four counts printed in Figure 14 are this arithmetic done 49 times per panel, and each has a one-line reading. LONE→BOND needs $x_{ii} \geq 2$, which holds at the four atoms O_1, Cl_3, O_4, O_5 ; each may point at any of the other six atoms, giving $4 \times 6 = 24$. BOND→LONE needs $x_{ij} \geq 1$, so it is offered exactly on the six bonds of the molecule, in both directions, giving $6 \times 2 = 12$. HOMOLYSIS needs the same $x_{ij} \geq 1$ but is symmetric in its two atoms, so it is offered on the upper triangle only, giving 6. COLLIGATION needs a spare electron at *both* ends, $x_{ii}, x_{jj} \geq 1$, which the same four atoms have, so it is offered on the $\binom{4}{2} = 6$ pairs among them. In every case the mask has reproduced a chemical precondition without being told one, having only ever compared integers.



How to read a panel. One panel per move type; the row is the move’s first atom i and the column its second atom j , so a cell ij is a candidate move (τ, i, j) . **Teal:** all five tests of (13) pass, so the move is offered. **White:** at least one fails, so the class is set to $-\infty$ before the softmax. **Grey:** not a move at all, the diagonal $i = j$, which no move indexes, and the lower triangle of HOMOLYSIS and COLLIGATION, which are symmetric in their two atoms and so are offered once per unordered pair. Across the four panels 48 of the $4 \times 49 = 196$ cells are teal; with the stop class that is 49 live classes against 148 dead ones.

Figure 14: The admissible set $\mathcal{A}(x)$ at the reactant of Figure 3, drawn in full, every move the model may consider at that one state. The counts are worked out in Appendix E.1; what the picture shows is that five integer comparisons reproduce the chemistry unaided. BOND→LONE and HOMOLYSIS need a bond to break, so their panels *are* the molecule’s adjacency matrix. LONE→BOND needs a lone pair to donate, so the rows of C2, H6 and H7, which have none, are empty, while their *columns* are not, those atoms being able to receive. COLLIGATION needs a spare electron at both ends, so it appears only among the four atoms whose diagonal is non-zero. Nothing here is a filter applied after the fact: these are the only classes the softmax of (15) ever sees.

E.2 THE TABLE $\mathcal{T}$, ONE INEQUALITY AND 65 NUMBERS

$\mathcal{T}$ is a set of triples (a, d, b) : element, lone-pair electrons, bonding pairs. Its content is a single inequality, already written in (2), with the constant 4 replaced by a per-element capacity C_a :

$$(a, d, b) \in \mathcal{T} \iff \frac{1}{2}d + b \leq C_a \quad \text{that is} \quad (a_i, x_{ii}, \beta_i(x)) \in \mathcal{T} \iff c_i(x) \leq C_{a_i}. \quad (30)$$

So the N local clauses of (12) are N integer comparisons against a lookup of one number, and $\mathcal{C}$ is a set of lattice points cut out by one linear equation and $O(N^2)$ linear inequalities. Nothing is built or written to decide membership. That leaves exactly one quantity the algebra does not supply, the 65 numbers C_a , one per element. Chemistry answers $C_C = 4$ and $C_H = 1$, and if (4) were a statement about chemistry that would end the matter. It is not. Its right-hand side is $\Pr[\hat{B} \text{ is chemically valid}]$, a property decided by the valency checker the experiments grade against, and a capacity asserted on paper would guarantee validity only against itself.

E.3 HOW THE CAPACITIES ARE MEASURED

Fix one element a and ask which pairs (d, b) it may wear. The question is finite, since d runs over $0, \dots, D = 14$ and b over $0, \dots, \bar{\beta} = 12$. So sweep the whole grid, all $65 \times 15 \times 13 = 12675$ cells, and decide each one by building the smallest molecule that wears it and seeing whether the evaluation gives it back unchanged. Figure 15(c) draws the five steps on one cell.

Step 1, list the neighbourhoods. The number b says how many bonding pairs the atom holds in total but *not* how they are shared out among its neighbours. Three pairs may be three single bonds, or one double and one single, or one triple bond, the same $b = 3$ but three different molecules. Write such a sharing-out as

$$b = o_1 + o_2 + \dots + o_L, \quad 1 \leq o_\ell \leq 3,$$

where L is how many neighbours the atom has and o_ℓ is the bond order to neighbour ℓ , capped at three because that is all the representation can express. They are not interchangeable to the evaluation, and the model may produce any of them, so all of them must be tested.

Step 2, build one probe per splitting. A splitting is turned into a molecule by giving the atom exactly L neighbours, one per o_ℓ , and nothing else. This is a bond–electron matrix like any other, of size $1 + L$:

$$\underbrace{P_{00} = d}_{\text{the lone pairs being tested}}, \quad \underbrace{P_{0\ell} = P_{\ell 0} = o_\ell}_{\text{the bond to neighbour } \ell}, \quad \underbrace{P_{\ell\ell} = 8 - 2o_\ell}_{\text{the neighbour's own lone pairs}}. \quad (31)$$

Every neighbour is a fluorine, chosen because it raises the fewest objections of its own. It is univalent, so it can never quietly attach itself to a second atom and change the neighbourhood being tested,

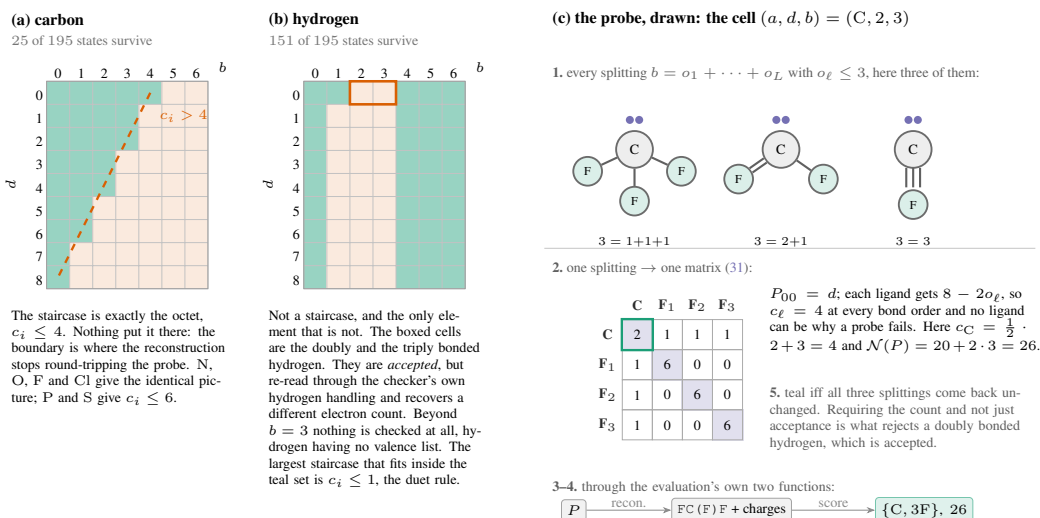


Figure 15: How the capacities C_a of (30) are measured. A cell is one local state, $d = x_{ii}$ lone-pair electrons down the side, $b = \beta_i$ bonding pairs across the top, and teal means the reconstruction and the valency check both returned the probe unchanged. (a) For carbon the sweep returns exactly the staircase $\frac{1}{2}d + b \leq 4$, so $C_C = 4$ is read off rather than assumed; N, O, F and Cl give the same picture, and P and S give $C_a = 6$. (b) Hydrogen is the single element of the 65 whose teal set is not a staircase, and its shape is an artefact of the checker rather than a fact about chemistry (Appendix E.4). (c) The procedure of Appendix E.3 on one cell. Both of its quantifiers were found by testing. With a single-splitting probe and no electron count, an untrained roll-out emitted 54 non-conserving candidates in 3200; with this one it emits none.

its capacity is a plain octet, and its own lone-pair count can be tuned to whatever the bond order demands. That last point is what $P_{\ell\ell} = 8 - 2o_\ell$ does. Reading c_ℓ off (1), $c_\ell = \frac{1}{2}(8 - 2o_\ell) + o_\ell = 4$, so a neighbour sits at a full octet whether the bond to it is single, double or triple. *No neighbour can ever be the reason a probe is rejected*, and the cell is therefore decided by the probed atom alone.

Steps 3 and 4, run the evaluation and demand the probe back. P is handed to the two checks the evaluation itself performs and to no others, the reconstruction, which turns a matrix into a molecule, and the valency check, which reads that molecule back and reports an atom multiset and a valence-electron count. Both must return what went in. Demanding the count and not merely a successful reconstruction is what makes the measurement mean something. The valency rules are stated only for the main-group elements that have them and wave through everything else, metals and hydrogen included, so a doubly bonded hydrogen is accepted while the electron count recovered for it differs from the one P specified, its formal charge having been read differently. Such a state would pass a naive stop mask and fail the metric, which is exactly the gap (4) forbids.

Step 5, accept the cell only if every splitting survived. A cell records a property of the pair (d, b) , and the model may realise that pair by any of the splittings of step 1, so the cell is accepted only when all of them come back intact. One failure among them condemns the whole cell. Both quantifiers earn their place. With the naive probe, one splitting and no electron count, an untrained roll-out emitted 54 non-conserving candidates in 3200, and with the procedure above it emits none.

E.4 WHAT THE SWEEP RETURNS

The sweep accepts 9580 of the 12675 cells, and its result is almost entirely summarised by (30): for 64 of the 65 elements the accepted set is *exactly* the staircase $\frac{1}{2}d + b \leq C_a$, so the capacity is well defined, the closed form is exact, and C_a takes only four values.

C_a	elements	why
4	C, N, O, F, Cl, Ge, Br, Sn, Pb	the octet (2), recovered and not assumed
5	Si	
6	P, S, As, Se, Sb, Te, I, Bi	the expanded shell of a third-row atom
∞	the remaining 45: every metal, and He, Li, Be, B, ...	no valence list is checked, so the only cap is the extent of the grid

The two finite groups are exactly the elements the checker states a valency rule for, and $C_a = \infty$ means the whole grid is accepted, so for those elements the table imposes nothing beyond the representable ranges (13) already enforces.

The exception, and what it costs. Hydrogen alone returns something that is not a staircase (Figure 15(b)): $b = 0$ accepted at every d , $b = 1$ accepted only at $d = 0$, $b \in \{2, 3\}$ rejected everywhere, and $b \geq 4$ accepted everywhere again. The rejections are real, being the doubly and triply bonded hydrogens of step 4. The acceptances beyond them are not a chemical statement at all. The checker states no valency rule for hydrogen, so past the point where the checker’s own hydrogen handling stops disagreeing, it agrees with anything. The largest staircase that fits inside hydrogen’s accepted set is $c_i \leq 1$, the duet rule, and no hydrogen of the test split leaves it.

Remark 8 (the enumeration audits the algebra rather than replacing it). What the construction needs is (30) plus 65 capacities, and the sweep is how those capacities are *audited* against the checker that will grade the output, and the audit passes wherever the checker has an opinion. Adopting $C_H = 1$ for the one element where it has none would make $\mathcal{T}$ the closed form at every element and shrink it from 9580 rows to 9433, forbidding 147 states that are inside what the checker accepts and outside anything the test split contains. The enumerated table is shipped because Proposition 3 is stated against (4) and the enumerated table is the largest set for which it is true; the algebraic table is a subset, and is the choice if the guarantee is to be stated against chemistry instead.

The capacities the data uses. $\mathcal{T}$ is deliberately looser than chemistry needs, because it must admit every state the checker accepts and not merely every state the training set contains. For comparison, the largest c_i any atom of each element attains over the test split is 0 for Ca and Cs; $\frac{1}{2}$ for Ce and Na; 1 for H, Li and K; 3 for Fe; 4 for B, C, N, O, F, Al, Cl, Br and I; 5 for Si and Ag; 6 for Cu, P and S; and 7 for Pd and Ru, the half-integers being radicals. Every entry is at or below its C_a , as it must be, since every reactant and product is a molecule.

E.5 A THIRD CORRECTION, IN THE RECONSTRUCTION ITSELF

Steps 1 and 4 above tighten the table. A third defect could not be fixed by any table, because it is not local at all. The reconstruction rebuilds a molecule by copying the reactant as a template, deleting its bonds, and installing the bonds the predicted matrix specifies. The template was produced by resolving aromatic rings into alternating single and double bonds, but the reconstruction leaves its *atoms* still flagged as aromatic. A bond–electron matrix knows nothing of aromaticity, and it is a Kekulé structure, and those flags belong to the template’s bonds, not to the ones now being installed. Left in place, the writer emits an aromatic-looking string joined by explicit single and double bonds, and re-reading that string fails whenever the step changed a ring. The consequence is that a state can satisfy every local condition of (12) and still fail the read-back half of (4), so *the local characterisation of the sector is false against the reconstruction as it stood*. Measured on unconstrained roll-outs, 7 of 128 candidates failed this way, and clearing the flags removes all of them. The fix is one line and changes no molecule that already reconstructed, so it can only raise a validity number. On the entrywise baseline it raises validity from 0.9754 to 0.9761 and moves top-1 accuracy by 10^{-4} , which is why it went unnoticed. For a model whose claim is exactly one, it is the difference between a theorem and a near miss.

E.6 WHAT ONE MOVE DOES TO A CAPACITY, AND THE JAM

Suppose the table were imposed at *every* move rather than only at termination. By (30) that means keeping $c_i \leq C_{a_i}$ at every atom throughout, so the question is what a move does to c . Substituting (29) into $c_i = \frac{1}{2}x_{ii} + \beta_i$ answers it completely:

$$\Delta c_i = \frac{1}{2}\delta_\tau^{ii} + \delta_\tau^{ij}, \quad \Delta c_j = \frac{1}{2}\delta_\tau^{jj} + \delta_\tau^{ij}, \quad \Delta c_k = 0 \quad (k \neq i, j), \quad (32)$$

and reading the four rows of the alphabet off it gives its whole effect: LONE→BOND has $(\Delta c_i, \Delta c_j) = (0, +1)$, since the pair already touched i and now also touches j ; BOND→LONE has $(-1, 0)$; and HOMOLYSIS and COLLIGATION have $(-\frac{1}{2}, -\frac{1}{2})$ and $(+\frac{1}{2}, +\frac{1}{2})$, the shared pair becoming half a pair each and back again. Summing and using $\delta_\tau^{ii} + \delta_\tau^{jj} = -2\delta_\tau^{ij}$ from (11) gives $\Delta c_i + \Delta c_j = \delta_\tau^{ij}$, so a move that forms bond order adds one unit of coordination in total and a move that breaks it removes one. And, which is the point, *a move that raises c at an atom does nothing to*

1782
1783
1784
1785
1786
1787
1788
1789
1790
1791
1792
1793
1794
1795
1796
1797
1798
1799
1800
1801
1802
1803
1804
1805
1806
1807
1808
1809
1810
1811
1812
1813
1814
1815
1816
1817
1818
1819
1820
1821
1822
1823
1824
1825
1826
1827
1828
1829
1830
1831
1832
1833
1834
1835

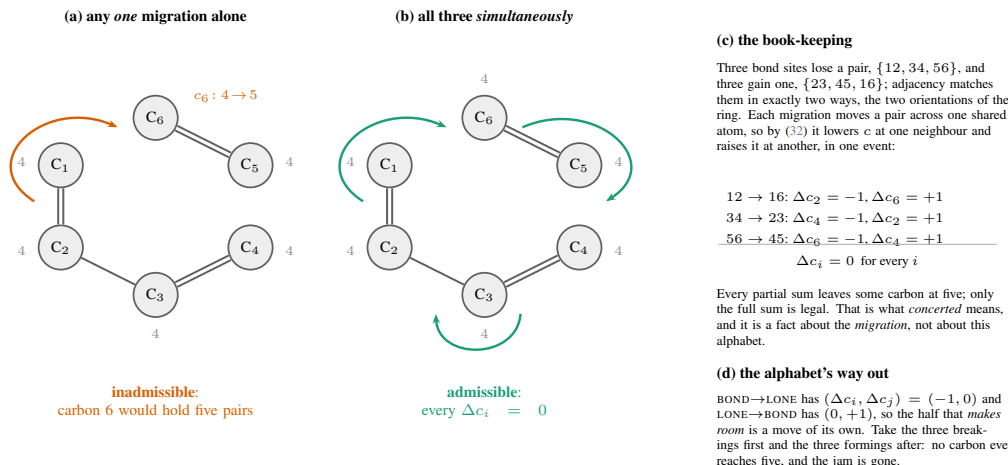


Figure 16: The Diels–Alder is jammed *as arrows*, and not as moves. Butadiene and ethylene close a six-membered ring: three bonds lose a pair and three gain one, and every carbon starts and ends with a full octet, $c_i = 4$. (a) Any single migration towards the product raises some carbon to five pairs before it has lost one; the same happens for all six migrations of both orientations, so there is no admissible *first* arrow and hence no admissible ordering of arrows at all. (b) The three applied together change no coordination number anywhere, which is what *concerted* means. (c)–(d) But the migration is not in the alphabet (Remark 6): expanded into its two moves, its lowering half becomes a move that can be taken on its own, and the jam disappears. Measured over 4000 test steps, 5.06% of non-identity steps admit no ordering of arrows through $\mathcal{T}$, and 0% admit no ordering of moves: every one of the 169 arrow-level jams is dissolved by the expansion on the test split.

make room there first: the two atoms move independently, and a pair move raises c at one of them while leaving the other alone. So an atom already at its capacity blocks every move that would raise it, whatever the rest of the chain is trying to do. This is the octet rule acting as a *dynamical* selection rule, and Figure 16 shows it locking the Diels–Alder reaction completely.

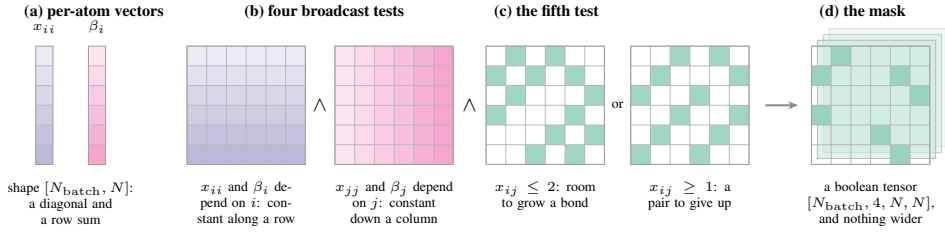
The canonical instance is the Diels–Alder reaction, in which all six carbons sit at $c_i = 4$ on both sides while three bond sites lose a pair and three gain one, so every single first hop raises some c_i to 5, and there is no admissible ordering at all, though the three applied *together* give $\Delta c_i = 0$ everywhere. The configuration is *jammed*, and chemistry’s name for this is that the Diels–Alder is a *concerted* reaction. Jams are not rare, since every pericyclic step is one, and the blocks they need contain an odd number of moves, the jams being the odd alternating cycles of pericyclic chemistry (a twentieth of non-identity steps, needing three or five moves and never two or four).

E.7 WHERE THE TABLE IS CHECKED

The construction imposes $\mathcal{T}$ on the *stop* move alone, as (13) and Proposition 3 do, and this is a choice rather than a necessity. Imposed at every move, an admissible ordering still exists for every step of the test split and the greedy procedure still finds it. The reason to impose it only at the end is that (4) asks for nothing more. It constrains the emitted candidate, and a condition on the states passed through on the way is structure the theorem does not use. The cost is that the ordering of §5 is greedy under (13) only, so its paths are not the ones that would stay inside $\mathcal{T}$. An intermediate is a well-formed matrix carrying the right electrons but need not be a molecule. Figure 13 passes through a hydrogen bonded to two atoms and Figure 19 through a carbon holding four and a half pairs. These are *book-keeping* intermediates, not claims about a reaction coordinate.

E.8 REALISATION ON A GPU

The state-dependent alphabet is the one thing that could have destroyed throughput, and it does not, because every move is named by a type and an ordered atom pair. The candidate set is a fixed $(4, N, N)$ grid plus one class, the same shape as the entrywise model’s own head, and the mask is a boolean tensor of that shape. Four of the five tests in (13) read a *per-atom* quantity, so they are $[N]$ vectors broadcast along the axis they do not depend on, and the fifth collapses to a choice between two boolean grids, its inequality’s direction being fixed by sign δ_{τ}^{ij} . Building $\mathfrak{A}(x)$ costs



The state-dependent alphabet is the one thing that could have destroyed throughput, and it does not. Because every move is named by a type and an ordered atom pair, the candidate set is a fixed grid; because four of the five range tests of (13) read a per-atom quantity, they are $[N_{\text{batch}}, N]$ vectors broadcast along the axis they do not depend on; and because the fifth reads x_{ij} only through an inequality whose direction is fixed by sign δ_r^{ij} , it collapses to a choice between two boolean grids. No integer tensor of size $[N_{\text{batch}}, 4, N, N]$ is ever materialised, there are no dynamic shapes, and there is no host synchronisation anywhere in the roll-out.

Figure 17: How the mask of Figure 14 is actually computed. The grids are schematic and $N = 6$; what matters is which axis each factor depends on, shown by the shading. Building $\mathfrak{A}(x)$ costs one diagonal, one row sum, four broadcast comparisons and one select, against a network body that is quadratic in N and unchanged from the entrywise model. *The constraint that makes the guarantees possible is also the cheapest part of the model.*

Table 2: The two constructions. Only the elementary event changes, and the network body, the bridge idea, the proper scoring rule and the iterate-and-condition sampler are shared.

	Discrete FlowER (§3)	Arrow Flow Matching (§5)
state space	the box $\prod_e \mathcal{V}_e$ (5)	$\mathcal{C}$, the physical sector (12)
elementary event	a position flip	a move in $\ker \mathcal{N}$
head	$ \mathcal{E} $ softmaxes over ≤ 15 classes	one softmax over $4N^2 + 1$ masked classes
loss	cross-entropy on the endpoint (8)	cross-entropy on the next move (16)
conservation	mean exact, variance > 0 (28)	surely, every moment (Prop. 2)
validity	not guaranteed (Thm. 1)	surely (Prop. 3)
score; beam	factorised (26); no beam	exact trajectory (17); beam

one diagonal, one row sum, four broadcast comparisons and one select, against a body quadratic in N , with no dynamic shapes and no host synchronisation anywhere in the roll-out. *The constraint that makes the guarantees possible is also the cheapest part of the model.* The stages are the following.

stage	shape	cost
network body (unchanged)	$[N_{\text{batch}}, N, d_{\text{model}}]$, N^2 atten- tion	dominant, unchanged
move logits, bilinear per type	$[N_{\text{batch}}, 4, N, N]$	one matmul
stop logit, from pooled atoms	$[N_{\text{batch}}]$	negligible
per-atom x_{ii} and β_i	$[N_{\text{batch}}, N]$	a diagonal and a row sum
move mask , five range tests	$[N_{\text{batch}}, 4, N, N]$	four of the five broadcast over one axis
stop mask , one lookup per atom	$[N_{\text{batch}}, N]$	an <code>index_select</code> , then <code>all</code>
state update	$[N_{\text{batch}}, N, N]$	four scatter-adds

The table $\mathcal{T}$ is a flat boolean tensor indexed by (a_i, x_{ii}, β_i) , resident on device, a few thousand entries, and the padding element’s only admissible state is the all-zero one that padded atoms hold, so padding never blocks the stop class. The fifth range test, the only one involving a quantity varying over both atom axes, reduces to a choice between two boolean grids, room to grow a bond or a pair to give up, so no integer tensor of size $[N_{\text{batch}}, 4, N, N]$ is ever materialised. Figure 17 draws the broadcast structure that makes this true.

F THE BRIDGE, THE HEAD AND DECODING

This appendix expands the training path and the decoder of §5. It covers how a step is cut into arrows and the arrows into moves, how the resulting multiset is ordered, what the head computes, the two ways a decoder rather than a parameterisation can go wrong, and why the whole construction is kept in discrete time.

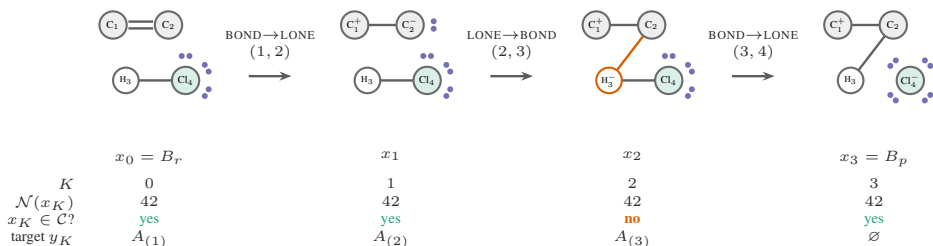


Figure 18: The bridge (14) on one real step, the protonation of an alkene by hydrogen chloride, $m = 3$ moves, four states, four targets. The chemist’s first arrow is a bond migration, expanded by Remark 6 into two moves, so H_3 crosses from Cl_4 to C_2 by way of a hydride intermediate. Every state carries $\mathcal{N} = 42$ by Proposition 2 rather than by luck, but x_2 is not a molecule ($c_{H_3} = 2 > 1$), so $\mathcal{T}$ refuses it the $\emptyset$ class, which is all the construction needs. Training draws one column uniformly and scores its target under (15).

F.1 DECOMPOSITION, CUTTING A STEP INTO ARROWS

Read as a picture, $\Delta = B_p - B_r$ marks every position it touches with a sign. An off-diagonal $\Delta_{ij} < 0$ is a bond *broken*, $\Delta_{ij} > 0$ a bond *formed*, a diagonal $\Delta_{ii} < 0$ an atom *donating* a lone pair and $\Delta_{ii} > 0$ an atom *accepting* one. Cutting Δ into arrows means threading those marks together, and the rule is *alternation*: an arrow takes a pair out of one place and puts it into the next, so along a chain of atoms the bonds must read break, form, break, form, and so on. The marks are grouped into walks

$$i_0 \xrightarrow{\pm} i_1 \xrightarrow{\mp} i_2 \xrightarrow{\pm} \dots$$

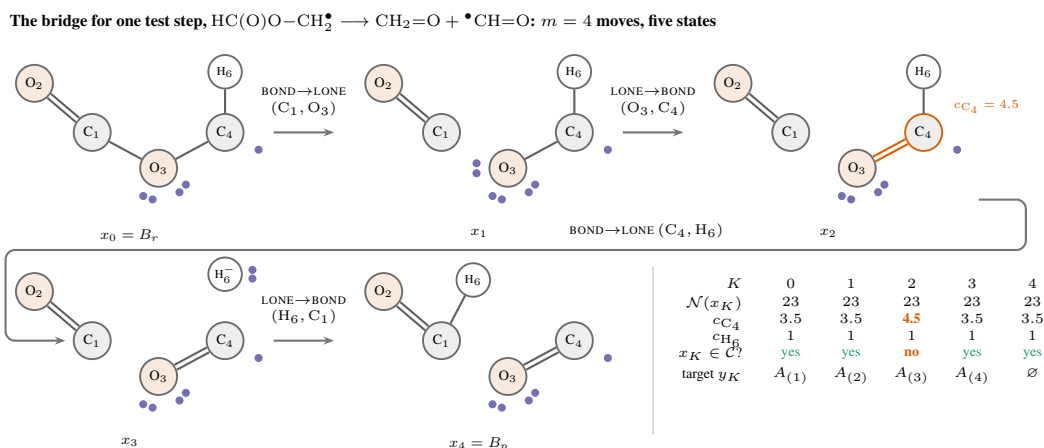
whose consecutive bond changes alternate in sign and whose two ends are settled by the diagonal, with a walk that *begins* with a formation has to start at an atom donating a lone pair, since something must supply the new bond, and a walk that *ends* with a break has to end at an atom accepting one, since the broken pair must land somewhere. A walk that closes into a cycle has no ends, so it neither donates nor accepts, and that is the pericyclic case, the Diels–Alder of Appendix E.6 is exactly one such six-atom cycle. Every consecutive break-and-form pair along a walk is one curly arrow, and leftover donations matched to leftover acceptances with no bond change between them are lone-pair transfers and become single arrows of their own. Each arrow is then rewritten in the alphabet, an arrow into or out of a diagonal is already one of the two pair moves, an arrow from bond to bond is a migration and expands into the two moves of Remark 6, and a step of odd parity is one of the two single-electron moves of Appendix D.4. That such a decomposition always exists is a theorem about degree-preserving transformations (Flamm et al., 2023).

F.2 ORDERING

The m moves of a step are a *multiset*, not a sequence, and their sum is Δ whatever order they are applied in. Not every order is usable, though, because every intermediate matrix must satisfy (13), and a step that both forms and breaks the same bond can drive its order to four en route if the two are taken the wrong way round. Order them greedily, at each point taking any remaining move that is admissible at the running state. Greedy is a heuristic here, not a theorem, because a step may contain two moves acting on the same entry in opposite directions and the counts therefore need not evolve monotonically. It is however a heuristic with no observed failures: over 2000 test steps it finds an admissible order for every step, including the 0.05% whose unordered factorisation would have produced a bond of order four. A step for which it failed would carry no usable training target and would simply be dropped from training, and none is. The ordering is deterministic per step, so it is computed once, offline, and cached with the rest of the featurisation. Figure 19 shows the whole of this and the preceding subsection on one step, the decomposition, the order, the states of the path, and the target at each K , and Figure 18 shows the same on a shorter step.

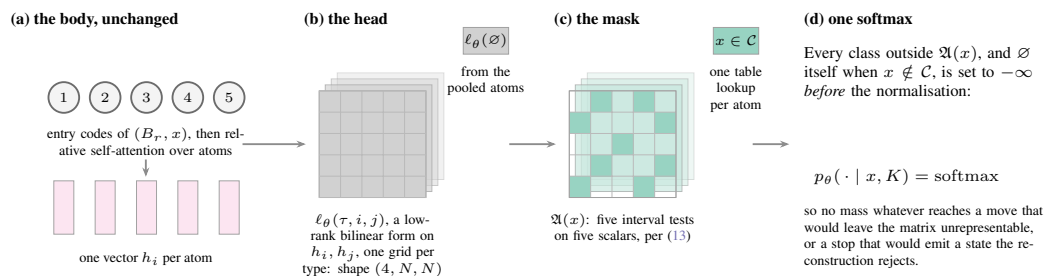
F.3 THE HEAD, STAGE BY STAGE

Figure 20 draws the four stages of (15), the body producing one vector per atom, the four bilinear grids of move logits and the pooled stop logit, the mask (13) together with the stop gate $x \in \mathcal{C}$, and the single softmax over what survives. No probability mass whatever is assigned to a move that would leave the matrix unrepresentable, or to a stop that would emit a state the reconstruction rejects,



Training draws $K \sim \text{Uniform}\{0, \dots, 4\}$, forms x_K , and scores the class in the bottom row of the table under the masked softmax (15). Four of the five targets are moves and the fifth is $\emptyset$, so the model learns *when the mechanism is over* by the same mechanism it learns what to do next; every target is admissible at its own state, so the loss is finite at every draw. Every state on the path carries $\mathcal{N} = 23$ electrons, by Proposition 2 rather than by luck. But x_2 is not a molecule, since C_4 would hold four and a half pairs, and neither is x_3 , which puts a lone pair on a hydrogen. Both are legal bond-electron matrices, and neither is a state the chain may stop in, which is all the construction needs. The spectator hydrogens H_5 on C_1 and H_7 on C_4 , and O_2 's two lone pairs, are not drawn.

Figure 19: The bridge, on the smallest multi-move step of the test split. The chemist’s factorisation of this step is three arrows, one of them a bond migration; expanded by Remark 6 and ordered greedily by §5 it becomes the four moves shown, and the whole H_6 atom crosses from C_4 to C_1 by way of a hydride intermediate. The path is computed once, offline, and cached, so training sees exactly the states the sampler will visit, which is the point of building it: the network is never asked at sampling time about a state it was never trained on. On the test split a step needs 2.24 moves on average and never more than sixteen, and 18.5% of steps have $m = 0$ and contribute only the $\emptyset$ target.



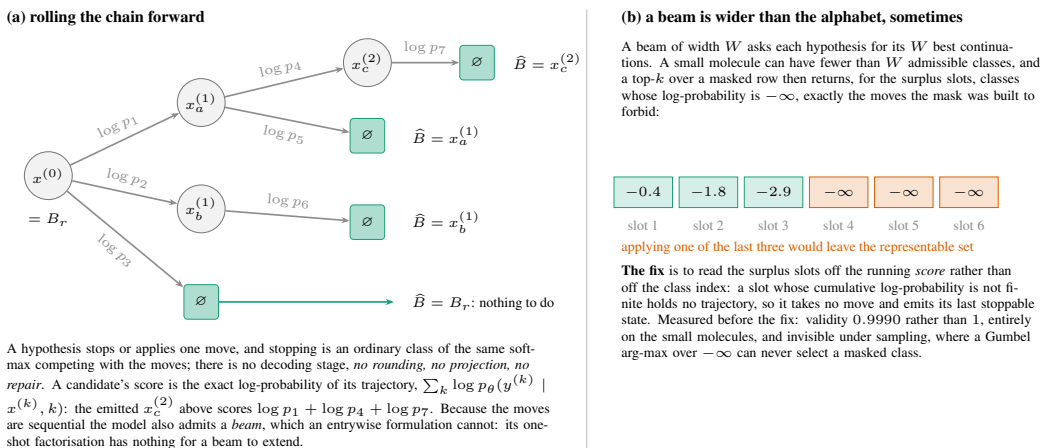
Nothing in (a) changes from the entrywise formulation; only (b)–(d) do. The candidate set has no ragged shape and no dynamic size, it is the fixed grid of (b) with the boolean tensor of (c) laid over it, so the constraint that makes Propositions 2 and 3 true is also the cheapest part of the model. By Remark 7 the admissible set is never empty, so (15) is always a well-defined distribution and the mask needs no fallback branch.

Figure 20: The head. Four grids of move logits, one per type and each indexed by an ordered atom pair, plus one stop logit from the pooled molecule; a boolean tensor of the same shape, plus one boolean for stopping; and a single softmax over what survives. The two guarantees hold at the level of the *parameterisation*, not of the sampler: an inadmissible move is not a move given low probability, it is a class the distribution does not contain.

and by Remark 7 the admissible set is never empty, so (15) is always a well-defined distribution. Figure 14 is what the third stage contains at a real state.

F.4 TWO WAYS A DECODER CAN GO WRONG

Why a beam, and why only here. The model of §5 admits a *beam* decoder, and the entrywise model of §3 does not, for the reason a beam always needs. A beam extends partial hypotheses and compares their partial scores, which requires the output to be factorised as an ordered chain of conditionals, each conditioned on the prefix already committed to. Here it is, because (17) is that chain rule, $x^{(k)}$ is the prefix, and truncating the sum at k gives a partial score that every hypothesis of the same length is comparable on. In (25) nothing is committed. The head is re-formed from scratch at each of the S Euler steps, the whole matrix is resampled, and the emitted candidate is read off the last step alone, so the intermediate distributions are not the conditionals of any factorisation



The mask makes the guarantee true of the distribution; the decoder has to be careful not to sample outside its support. The same principle answers the roll-out budget: a hypothesis that has not stopped within S positions emits the last state at which it was allowed to stop, which always exists because the chain starts at $B_r \in \mathcal{C}$. The guarantee therefore holds for every budget, $S = 0$ included; $S = 12$ is used, and at a mean of 2.24 moves per step the roll-out usually terminates after three or four.

Figure 21: Sampling. The log-probabilities are symbols, not measurements: what matters is that every edge leaving a state is a class of one masked softmax, that $\emptyset$ competes with the moves on equal terms, and that a candidate's score is a plain sum along its trajectory, so this model can be ranked by score, by frequency or by summed probability, where the two other flow models fill only one or two of the three protocols. Panel (b) records the one place where a correct parameterisation was not enough, and it cost a day to find.

of $p_\theta(\hat{B} | B_r)$ and there is no prefix, no partial score and nothing for a beam to extend. Figure 21 draws both decoders, and the rest of this subsection records the two ways a decoder can go wrong.

A beam is wider than the alphabet, sometimes. There is one way a decoder can break a guarantee that the parameterisation holds. A beam of width W asks each hypothesis for its W best continuations. A small molecule can have fewer than W admissible classes, and a top- k over a masked row then returns, for the surplus slots, the indices of classes whose log-probability is $-\infty$, which are exactly the moves the mask was built to forbid. Applying one takes the state out of the representable set and the candidate does not reconstruct. Measured before the fix: validity 0.9990 rather than 1, entirely on the small molecules, and invisible under sampling, where a Gumbel arg-max over $-\infty$ can never select a masked class. The fix is to read the surplus slots off the running *score* rather than off the class index, a slot whose cumulative log-probability is not finite holds no trajectory, so it takes no move and emits its last stoppable state. *The mask makes the guarantee true of the distribution, and the decoder has to be careful not to sample outside its support.*

When a hypothesis has nothing it may emit. Two things can leave a hypothesis with no state to hand over, and both are answered the same way. It falls back to a state it is *known* to be able to leave from, which always exists because the chain starts at $B_r \in \mathcal{C}$. The first is the roll-out budget. The chain is capped at $S = 12$ positions, a budget the mean step length of a couple of moves leaves ample room under, and a hypothesis that has not stopped by then emits the last state at which it was allowed to stop, so the guarantee holds for every budget, including $S = 0$, and the cap is a compute knob and not a correctness assumption. The second is a limitation of the checker rather than of the construction, and is the residue Appendix J.3 reports. Of 400,000 beam candidates from a trained model, 32 were rejected by the evaluation and all 32 lay inside $\mathcal{C}$. What happens to them is that the checker perceives one of their rings as aromatic and then cannot assign that ring a Kekulé structure, so the molecule is inconsistent in the checker's own model and cannot be written around either. The test that decides the fallback therefore has to be the one the metric applies, and that is not the reconstruction. On these states the reconstruction *succeeds* and returns a complete aromatic string, and only the canonicalisation that follows fails, so a fallback keyed on the reconstruction is a no-op on the entire failure mode. Figure 21 draws a roll-out and the place where this can happen.

F.5 WHY NOT A CONTINUOUS-TIME FORMULATION

A continuous-time version of §5 is available. Give each move a non-negative rate and integrate the path-space divergence (23) against a bridge whose not-yet-applied moves fire at rate λ_t . It is strictly more machinery for the same content. The importance weight $\lambda_t(m - K_t)$ has unbounded variance as $t \rightarrow 1$; "stop" becomes a limit rather than a class, so the model no longer learns termination by the same mechanism it learns what to do next, and there is no per-candidate likelihood to rank by without integrating. The discrete counting process $K \sim \text{Uniform}\{0, \dots, m\}$ of (14) avoids all three.

G PROOFS

Theorem 1 (a product measure cannot live on the conservation sheet). *Let $\mu = \bigotimes_{e \in \mathcal{E}} \mu_e$ be a product measure on $\prod_e \mathcal{V}_e$ with $\text{supp } \mu \subseteq \mathcal{H}_n$ for some fixed n . Then every μ_e is a point mass.*

Proof. The support of a product measure is the product of the supports, $\text{supp } \mu = \prod_e \text{supp } \mu_e$, since a configuration lies in it exactly when each coordinate lies in its own factor’s support, independently of the others. Suppose some coordinate e_0 has two distinct values $\alpha \neq \gamma$ in $\text{supp } \mu_{e_0}$. Choose $x \in \text{supp } \mu$ with $x_{e_0} = \alpha$, which is possible because each remaining coordinate is free, and let x' agree with x except $x'_{e_0} = \gamma$, so that $x' \in \text{supp } \mu$ too. But $\mathcal{N}$ is linear and x, x' agree off e_0 , so $\mathcal{N}(x') - \mathcal{N}(x) = w_{e_0}(\gamma - \alpha) \neq 0$, since $w_{e_0} \in \{1, 2\}$. Two support points then carry different electron counts, contradicting $\text{supp } \mu \subseteq \{\mathcal{N} = n\}$. Figure 8 in Appendix C.3 draws the four steps of this argument on two coordinates, and Figure 1(a,b) draws the two shapes they compare. $\square$

Proposition 2 (conservation). *Let $(x^{(k)})_{k \geq 0}$ be any chain whose every jump is a move of the alphabet. Then $\mathcal{N}(x^{(k)}) = \mathcal{N}(x^{(0)})$ surely.*

Proof. Induction on k , with a trivial base case. For the step, $x^{(k+1)} = x^{(k)} + A_\tau$ for some type τ and some ordered pair, and $\mathcal{N}$ is linear, so $\mathcal{N}(x^{(k+1)}) = \mathcal{N}(x^{(k)}) + \mathcal{N}(A_\tau) = \mathcal{N}(x^{(k)})$, since $\mathcal{N}(A_\tau) = 0$ by (11) for each of the four types. No property of how the jump was chosen is used, so the identity holds pathwise and therefore for every moment of the law of $\mathcal{N}(x^{(k)})$. $\square$

Proposition 3 (validity). *Let a chain start at $x^{(0)} = B_r$, take jumps in $\mathfrak{A}(x^{(k)})$, and be permitted to terminate only at a state x with $(a_i, x_{ii}, \beta_i(x)) \in \mathcal{T}$ for every i . Then the state at which it terminates lies in $\mathcal{C}$.*

Proof. Non-negativity and $x_{ij} \leq 3$ hold at every k by induction, holding at B_r and being exactly what (13) tests at each jump, on exactly the three entries the jump changes. $\mathcal{N}(x^{(k)}) = \mathcal{N}(B_r)$ by Proposition 2. The N local conditions hold at the terminal state by hypothesis. Those are the three clauses of (12). Figure 14 in Appendix E.1 shows the set the first clause of the hypothesis ranges over at a real reactant, and Figure 18 a chain that stops where the third permits it to. $\square$

H RELATED WORK

Two literatures meet here, generative modelling on discrete spaces and data-driven reaction prediction. The first two paragraphs below cover the second and the narrow intersection with the first, the literature this paper competes with. The last two cover the literature it is built out of, the general methodology of flow matching on continuous and discrete spaces, and the standard devices for making a generative model respect a constraint.

Mechanistic reaction prediction. Data-driven reaction prediction divides by what the model emits. Sequence models emit a product string, and the analogy with machine translation was proposed by Nam & Kim (2016) and Schwaller et al. (2018) and refined into the Molecular Transformer (Schwaller et al., 2019), with graph encoders replacing the reactant string later (Tu & Coley, 2022). Graph-edit models emit a change to the molecular graph, either by ranking candidate

products generated at a predicted reaction centre (Jin et al., 2017; Coley et al., 2017; 2019a) or by generating the edits themselves as a sequence (Sacha et al., 2021). The same decomposition drives retrosynthesis models that predict a graph edit and then complete the fragments (Somnath et al., 2021; Chen & Jung, 2021), which together with search (Segler et al., 2018) is what a synthesis planner consumes. None of these represents electrons, and Sabanza Gil et al. (2023) show how easily aggregate accuracy hides the resulting failures. Mechanistic models instead emit the elementary steps. Kayala & Baldi (2012) rank polar, pericyclic and radical elementary reactions between approximate molecular orbitals. ELECTRO (Bradshaw et al., 2019) generates a reaction as a sequence of electron-path steps and is the closest antecedent of our move alphabet, though its paths are linear and alternate removal with addition. Reactron (Chen et al., 2025) predicts a product by generating the arrow-pushing diagram of each mechanistic step, and Neukomm et al. (2025) teach a language model an annotated SMILES encoding of electron flow. That arrow-pushing is expressive enough to be a complete formalism, rather than a convenient notation, is a theorem. Every atom-to-atom map decomposes into cyclic electron-pushing diagrams (Flamm et al., 2023). Joung et al. (2025) assembled the mechanistic dataset all of these can now be measured on.

Flow matching in mechanistic reaction prediction. FlowER (Joung et al., 2025) is the direct ancestor of this paper, transporting the reactant bond–electron matrix to the product matrix by continuous flow matching, with a zero-sum training target and zero-sum noise, so that the drift conserves electrons exactly. Two gaps remain there, and closing them is what the present paper does. The guarantee covers the drift and not the rounding that returns the off-lattice endpoint to the integers, which restores the total but not the symmetry, the non-negativity or the per-atom capacities, and it covers the single global functional $\mathcal{N}$ and not the N local per-atom conditions, which are nowhere represented. Elsewhere in chemistry the same machinery appears with a different state space. Tuo et al. (2026) use conditional flow matching with deterministic optimal transport for transition states and reaction products in three dimensions. Duan et al. (2025) pose transition-state generation as an optimal-transport problem between reactant and product geometries, having previously solved it by an object-aware diffusion model (Duan et al., 2023), and Kim et al. (2024) generate transition-state geometries from 2D graphs by diffusion. On the retrosynthesis side, RetroBridge (Igashov et al., 2024) and RetroSynFlow (Yadav et al., 2025) make the same choice of coupling we do, pinning the path at two data points rather than at a prior, but over molecular graphs and synthons rather than over an electron budget, and neither carries a conservation law. What distinguishes the present work from all of these is that the constraint is enforced by the *support of the elementary event* rather than by the geometry of the state space or the target of a regression.

Flow matching. Continuous flow matching regresses a velocity field onto the time derivative of a conditional interpolation between a source and a target sample, so that a single simulation-free regression yields a probability path between two distributions (Lipman et al., 2023; Liu et al., 2023a; Albergo & Vanden-Eijnden, 2023). The three constructions were introduced independently and were shown to be instances of one stochastic-interpolant framework (Albergo et al., 2023) which also contains score-based diffusion (Ho et al., 2020; Song et al., 2021) as a special case. The practical variants differ mainly in how the source and target are coupled, minibatch optimal transport straightening the paths and reducing the number of solver steps (Tong et al., 2024), and the recipe carries to Riemannian manifolds (Chen & Lipman, 2024) and to large-scale image synthesis (Esser et al., 2024). A recent tutorial (Lipman et al., 2024) collects the design choices. On a discrete space there is no velocity, and the field’s role is taken by a rate matrix. Categorical diffusion supplied the first such models, multinomial diffusion (Hoogeboom et al., 2021) and the discrete-state denoising processes of Austin et al. (2021), whose absorbing-MASK and uniform corruptions are still the two default priors, and Campbell et al. (2022) put them in continuous time as Markov chains. Discrete flow matching proper (Campbell et al., 2024; Gat et al., 2024) then carries the conditional-to-marginal argument over: a bridge between two points induces a conditional rate, and the rate generating the marginal path is its posterior mean, exactly (21). Subsequent work widens the design space with arbitrary discrete probability paths under a kinetic-optimality criterion (Shaul et al., 2025), score-entropy parameterisations (Lou et al., 2024), the simplified masked objectives of Sahoo et al. (2024) and Shi et al. (2024), and continuous relaxations onto the simplex (Stark et al., 2024), while generator matching (Holderrieth et al., 2025) shows all of them, jump processes included, to be instances of one construction over an arbitrary Markov generator. Graph-structured discrete diffusion (Vignac et al., 2023) is the closest of these to our state space. Our Discrete FlowER (§3 and Appendix B) is this recipe instantiated on bond–electron matrices, written out in full precisely

because §5 is the same recipe with one substitution. What is new there is not a path or a loss but the *elementary event*, and Theorem 1 is the reason to change it.

Constraints in generative models. The usual devices are a penalty, a rejection filter, or a projection applied to the output. Diffusion and flow models have been adapted to constrained domains by changing the noising process, using reflected stochastic differential equations (Lou & Ermon, 2023), barrier and reflected metrics on manifolds (Fishman et al., 2023) or mirror maps into a dual space (Liu et al., 2023c), or by making sampling itself a constrained optimisation (Christopher et al., 2024). These handle convex sets and smooth manifolds well, but a level set of a linear functional intersected with a lattice is neither, and none of them makes the *intermediate* states of the process legal. Neuro-symbolic layers instead compile a logical constraint into the output distribution, either as a differentiable penalty (Xu et al., 2018) or as a tractable circuit that renormalises over the satisfying assignments (Ahmed et al., 2022), which is exact but costs that grows with the constraint’s circuit size. The device used here is the cheapest member of that family, a *masked support*, in which violating classes are removed before the softmax is normalised, exactly as in lexically constrained and grammar-constrained decoding (Hokamp & Liu, 2017; Post & Vilar, 2018; Willard & Louf, 2023). What makes it applicable is not the mask but the choice of elementary event, which puts the global constraint in the kernel of $\mathcal{N}$ and leaves a mask that is $O(1)$ to evaluate and provably non-empty (Remark 7).

I EXPERIMENT SETTINGS

I.1 DATA

The mechanistic dataset of Joung et al. (2025) is used exactly as released, with no filtering, deduplication or resplitting of our own. Its splits hold 1,933,750 training, 21,723 validation and 218,997 test elementary steps, and 31,189 test pathways of which 3000 seeded ones are rolled out for the recursive pathway metric. A pathway is terminated by an identity step, and 18.5% of steps are identity steps. Every hydrogen is mapped explicitly (Remark 4). Off-diagonal entries are integers because aromatic rings are Kekulé-resolved before the matrix is built, and $\max_{i \neq j} B_{ij} = 3$ over the whole test split. Preprocessing for Arrow Flow Matching, the decomposition of every training step into moves and the greedy admissible ordering of §5, is deterministic, run once offline and cached. The chemical validity of a candidate, in the sense of (4), is decided throughout by one fixed implementation of the valency rules, RDKit (Landrum et al., 2024) writing and rereading a SMILES string (Weininger, 1988), and the same implementation is what the capacities of $\mathcal{T}$ are enumerated against in Appendix E.3. Which implementation it is matters nowhere in the construction. It matters only that the model is graded and the table is built against the same one.

I.2 TRAINING PROCEDURE

Every model is trained for at most 100 epochs with AdamW, gradient clipping at 1.0, and early stopping on validation loss with patience 25. The learning-rate schedule is a linear warmup followed by cosine decay to a floor of 0.1 of the peak rate (a tenth of that for the models noted in Table 3). The inverse-square-root Noam decay used by the originating papers is replaced because its rate collapses to a small fraction of the peak within a few warmup lengths and keeps falling, which under-trained the transformer baselines. Optimiser defaults follow the AdamW convention $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, and the transformer-family models (Continuous and Discrete FlowER, Arrow Flow Matching, the Molecular Transformer, Graph2SMILES) override $\beta_2 = 0.998$ and $\varepsilon = 10^{-9}$, per their published settings. The checkpoint with the best validation loss is the one evaluated. Each model is trained at five widths, and every reported parameter count is measured on the instantiated model rather than nominal.

I.3 DECODING BUDGETS

Every model emits at most ten candidates per step. Arrow Flow Matching rolls out at most $S = 12$ positions per hypothesis and Continuous FlowER integrates its ODE for 10 Euler steps, as does the entrywise chain of §3 ($S = 10$). The “marginal” ranking rule divides each trajectory’s log-probability by the GNMT length penalty $((5 + L)/6)^{0.6}$ before summing over trajectories that

reached the same product, and ELECTRO normalises its own scores and therefore uses exponent 0.

Table 3: Architecture and optimiser hyper-parameters at the largest width. The width knob names the quantity varied across the five sizes, and the ladder is given below the table. The column S is the number of positions a candidate is generated in, the Euler steps of (25) for the two entrywise flow models and the roll-out length of (17) for the arrow model, with a dash where a model has no such parameter.

model	architecture at the base width	S	width knob	peak lr	warmup	batch
Molecular Transformer	6+6 layers, $d=256$, 8 heads, $d_{\text{ff}}=2304$	—	d, d_{ff}	1×10^{-3}	1000	256
Graph2SMILES	6+6 layers, $d=384$, 8 heads, $d_{\text{ff}}=2304$	—	d, d_{ff}	4×10^{-4}	2000	256
MechT5	2 enc. + 8 dec. layers, $d=512$, 6 heads, $d_{\text{ff}}=768$	—	d, d_{ff}	3×10^{-4}	1000	512
MechLLaMa	6 layers, $d=512$, 4 heads, $d_{\text{ff}}=1088$, RoPE	—	d, d_{ff}	3×10^{-4}	1000	512
ELECTRO	4 GGNN layers, $d=1024$, path length ≤ 10	—	d	1×10^{-4}	2000	128
Reactron	3 message-passing + 3 attention layers, $d=608$	—	d	2×10^{-4}	2000	128
Continuous FlowER	6 layers, $d=512$, 16 heads, $d_{\text{ff}}=2048$, $\sigma=0.15$	10	d, d_{ff}	3×10^{-4}	4000	96
Discrete FlowER	the same body, entrywise head	10	d, d_{ff}	3×10^{-4}	4000	96
Arrow Flow Matching	the same body, masked move head	12	d, d_{ff}	3×10^{-4}	4000	96

The width ladder. The five sizes nano < micro < mini < small < base are set by the knobs above so that every model spans roughly 1.2M to 21M parameters. The three flow models share one ladder, $(d, d_{\text{ff}}) \in \{(128, 512), (128, 1536), (256, 1024), (384, 1536), (512, 2048)\}$, so their curves may be read against each other directly, and the sequence models use $(64, 512) \dots (256, 2304)$ for the Molecular Transformer and $(128, 768) \dots (384, 2304)$ for Graph2SMILES, the two mechanistic language models $(128, 256) \dots (512, 768)$ and $(128, 320) \dots (512, 1088)$, and the graph models $d \in \{256, 384, 512, 768, 1024\}$ for ELECTRO, and $d \in \{160, 256, 384, 512, 608\}$ for Reactron. Two adjustments were forced by numerical stability rather than chosen. Reactron diverges at its default peak rate once its width exceeds about four times the published one, so its two largest sizes use 2×10^{-4} with a 2000-step warmup, and Continuous FlowER’s ODE produces non-finite states for embedding widths whose per-head width is not a multiple of eight, so its micro point keeps $d = 128$ and grows d_{ff} instead of taking an intermediate width.

J FURTHER EXPERIMENTS

J.1 WHAT MAKES THE COMPARISON FAIR, AND WHAT THE FIVE REGIMES MEASURE

A candidate list is produced by a *decoder*, beam search or sampling, and ordered by a *ranking rule*, either the model score, the marginal probability over trajectories, or how often a candidate was drawn, and not every model can fill every cell. Continuous FlowER (Joung et al., 2025) has neither. Its ODE is a one-shot map from noise, with nothing for a beam to extend, and every draw leaves with weight one, so there is no score to order by. Discrete FlowER (§3) has a score (26) but still no beam, its endpoint head being one-shot as well. Arrow Flow Matching (§5) has both, its moves being sequential, as do the sequence and graph baselines. The single cell that *every* model can fill is therefore sampling with frequency ranking, and that is the protocol of §6. Any other choice would either grant Arrow Flow Matching a decoder its two closest competitors do not possess, or drop a model from the comparison. The five regimes reported below score the same trained models on the same test split, and what changes between them is how candidates are generated, how they are ordered, and what counts as the right answer, never the model or its weights. What Arrow Flow Matching scores under each of the other four regimes is reported in Appendices J.4 and J.5, which also record that its two guarantees and its accuracy are insensitive to the choice.

Beam. Beam search over the model’s own decoder, each distinct product ranked by the single highest-scoring trajectory that reached it. This is the protocol the other beam regimes are read against.

Sampling, by score. Candidates drawn independently from the model rather than searched for, then ranked exactly as above. Against the first regime this isolates what the search itself is worth.

Sampling, by frequency. The same draws as the previous regime, ranked instead by how many of them landed on each product. The model’s own scores are discarded, which is the only rule

available to a decoder that computes none, and therefore the only regime in which all ten models are comparable. This is the regime of §6.

Beam, by marginal probability. The same candidates as the first regime, reordered. A product is reachable by many trajectories, most visibly by the orderings of one and the same multiset of moves, so its probability is the sum over them, not the largest one, and taking the largest discards that mass, and discards more of it from the products with more routes in. Each trajectory is first divided by a length penalty, since a score that sums one negative term per decision otherwise prefers short trajectories for no reason beyond their being short. That the sum is the right quantity does not make it the better ranking, and measured here it is not uniformly one. It pays where a decoder really does reach one product by many distinct routes, and costs where a beam instead holds near-duplicates of one route, whose mass it then double-counts.

Beam, branch-aware ground truth. The predictions and the ordering of the first regime, scored differently. The dataset’s ground truth is set-valued, since 20.2% of test steps sit on a reactant that some other step carries to a different product, and the default metric reads it as if it were single-valued, so a deterministic decoder is marked wrong for answering a branch point with the wrong one of the data’s own answers. Here a step counts as correct when the model returns a product the split records for that reactant. Validity and conservation are properties of a molecule, not of the ground truth, so those panels repeat the first regime exactly, and only the accuracy panels and the overall score move.

The ceiling drawn at $k = 1$. The dashed segment at $k = 1$ in every accuracy panel is the highest top-1 a deterministic decoder can reach in that panel. A decoder is a function of its input, so where the data record several products for one reactant it can satisfy only one of them. On this split, of the 31,189 test pathways, 12,367 record two or more different products for one and the same mapped reactant, and 20.2% of test steps sit on such a reactant, which caps top-1 step accuracy at 0.8840 and the teacher-forced conjunction at 0.6035. The roll-out metric is capped at 1.0000, needing only its endpoint to match. The bound constrains $k = 1$ only. A top-2 number may exceed it, and a *sampling* decoder may exceed it even at $k = 1$, by answering two occurrences of one reactant differently, which is inconsistency rather than accuracy.

The budget. Every regime is given the same budget of ten candidates per test step, a beam of ten in the beam regimes and ten independent draws in the sampled ones, and duplicates are merged afterwards, so no model is charged for candidates its competitors were not allowed either. Appendix I.3 gives the per-model roll-out lengths.

J.2 THE MAIN COMPARISON AND THE SIZE SWEEP, PLOTTED

Figure 22 plots the numbers of Table 1, and Figure 23 repeats the comparison at all five widths, both under the sampling decoder with frequency ranking of §6.

J.3 TWO QUALIFICATIONS

What the guarantee is a guarantee about. It is a guarantee against a specific validity checker, which is why $\mathcal{T}$ is enumerated against exactly that one. A residue of 8×10^{-5} is bought by declining to answer. Of 400,000 beam candidates from a trained model, 32 were rejected and all 32 lay inside $\mathcal{C}$, the checker perceiving one of their rings as aromatic and then failing to assign it an alternating-bond structure. The guarantee is untouched and $\mathcal{C}$ is not leaky. What fails is the identification of “lies in $\mathcal{C}$ ” with “is chemically valid”, which no per-atom table can close, because no per-atom table can see a ring. Such a hypothesis falls back to the last state at which it was allowed to stop. That is not repair, since nothing is rounded or projected and the substitute is a state the chain actually occupied, but it is a substitution and should be read as one.

A defect in the reconstruction. The enumeration of $\mathcal{T}$ exposed a defect in the reconstruction itself, which rebuilt a molecule by copying the reactant as a template and installing new bonds while leaving the template’s atoms flagged as aromatic, so a step that changed a ring produced a string that could not be read back. Clearing those flags changes no molecule that already reconstructed, and it raised the entrywise baseline’s validity only from 0.9754 to 0.9761, which is why it had gone unnoticed, but for a model whose claim is exactly one it is the difference between a theorem and a near miss. All numbers reported here are measured after that correction, for every model.

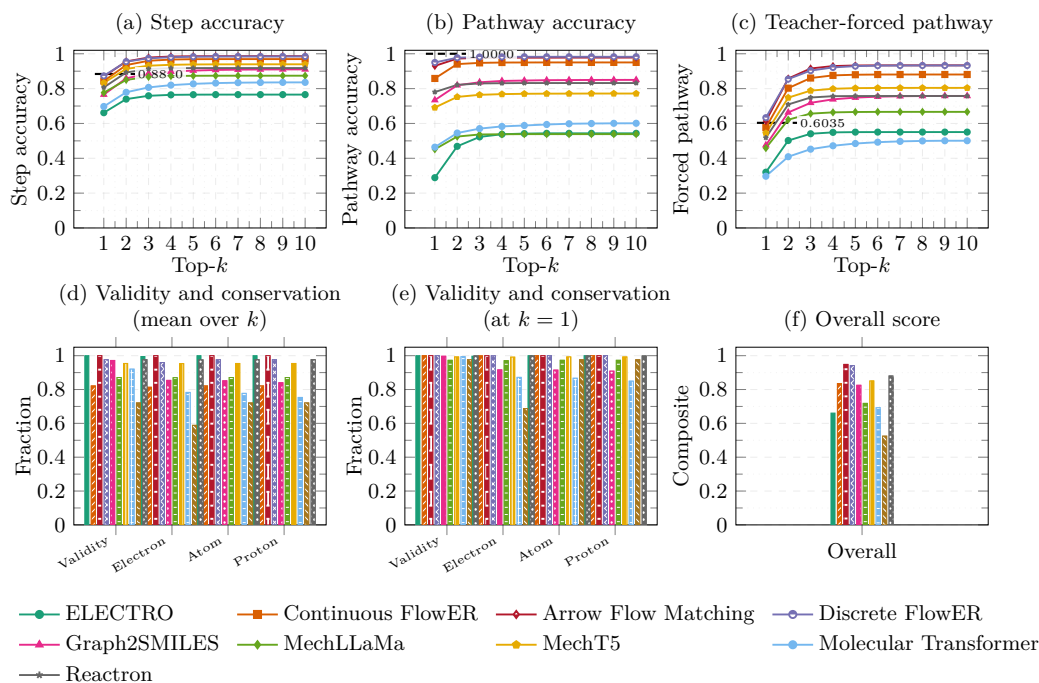


Figure 22: All nine models at the smallest width, the nano rung of the ladder at roughly 1.3M parameters, under the one protocol every model can fill, with candidates drawn by sampling and ranked by frequency and scored against the recorded product. The key gives each model its short name. Continuous FlowER carries no trajectory score and Discrete FlowER no beam, so wherever they appear they are decoded by sampling with frequency ranking, as Appendix I.3 records. (a)–(c) accuracy against the candidate budget k , with the dashed segment at $k = 1$ marking the ceiling that branching in the data imposes on a deterministic decoder. (d)–(e) validity and the three conservation checks, averaged over k and at $k = 1$. (f) the overall score. Arrow Flow Matching is the only model at 1.0000 on all four checks of (d) while matching the best accuracy curves, and its overall score in (f) is the highest in the study, at this width and at every other. ELECTRO reaches validity 1 by construction but falls well below on accuracy, and the continuous flow model is the weakest on validity of the nine. The same numbers are tabulated in Table 1.

J.4 THE OTHER FOUR REGIMES AT THE SMALLEST WIDTH

Figure 24 is beam search ranked by trajectory score, Figure 25 sampling ranked by trajectory score, Figure 26 beam search ranked by marginal probability over trajectories, and Figure 27 beam search scored against the branch-aware ground truth. Each repeats Figure 22 panel for panel under one of the four regimes the main text does not report. Arrow Flow Matching’s four guarantee panels read exactly 1 in every one of them, as they must, validity and conservation being properties of the emitted molecule that no decoder or ranking rule can change. Its accuracy is insensitive to the choice as well, the beam decoder, which its two closest competitors do not possess, buying about half a point at $k = 10$ and nothing at $k = 1$. The two entrywise flow models are absent from the beam regimes for the reason given above.

J.5 THE SIZE SWEEP UNDER THE OTHER FOUR REGIMES

Figure 28 is beam search ranked by trajectory score, Figure 29 sampling ranked by trajectory score, Figure 30 beam search ranked by marginal probability over trajectories, and Figure 31 beam search scored against the branch-aware ground truth, each repeating Figure 23 under one of the four regimes. The picture does not move. Arrow Flow Matching’s validity and its electron, atom and proton conservation are flat at exactly 1 across the whole $15\times$ range of capacity under every regime, its accuracy is flat too, and its overall score at the smallest width stays above every point any other model reaches at any width.

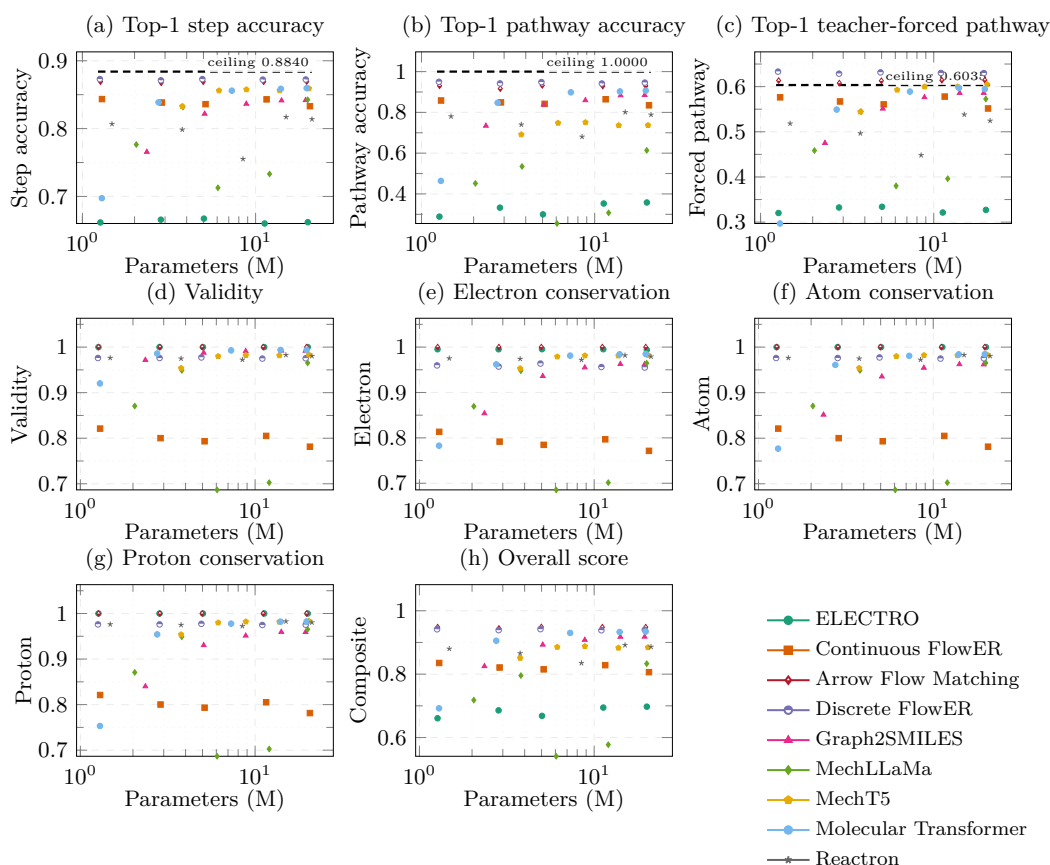


Figure 23: Every model at five widths spanning 1.2–21M parameters, under sampling with frequency ranking and scored against the recorded product. As in Figure 22, Continuous FlowER carries no trajectory score and Discrete FlowER no beam, so both are always decoded this way. Arrow Flow Matching’s four guarantee panels (d)–(g) are flat at exactly 1 across the whole range, and its accuracy panels are flat too, so the guarantee is not bought with capacity. Panel (h) is where the inductive bias shows in the budget. Its overall score at the smallest width, 0.948 at 1.3M parameters, is above every point every other model reaches anywhere on the ladder, the best of them being 0.942 for Discrete FlowER at 1.3M and 0.934 for the Molecular Transformer at 19.7M. The continuous flow model’s validity and electron conservation *fall* as it grows. The remaining models improve with width in the way sequence and graph models normally do, and none of them reaches 1 on any conservation panel except ELECTRO, whose construction fixes the molecule rather than predicting it.

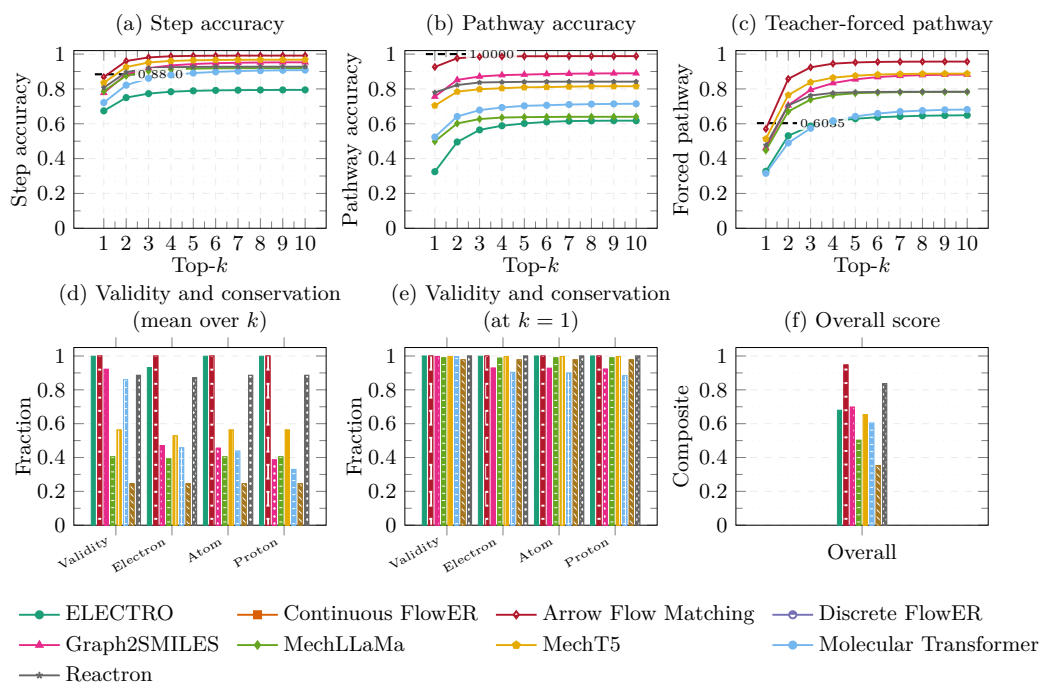


Figure 24: The comparison at the smallest width under beam search ranked by trajectory score, panel by panel as in Figure 22. Continuous and Discrete FlowER are absent, neither having a sequential factorisation for a beam to extend.

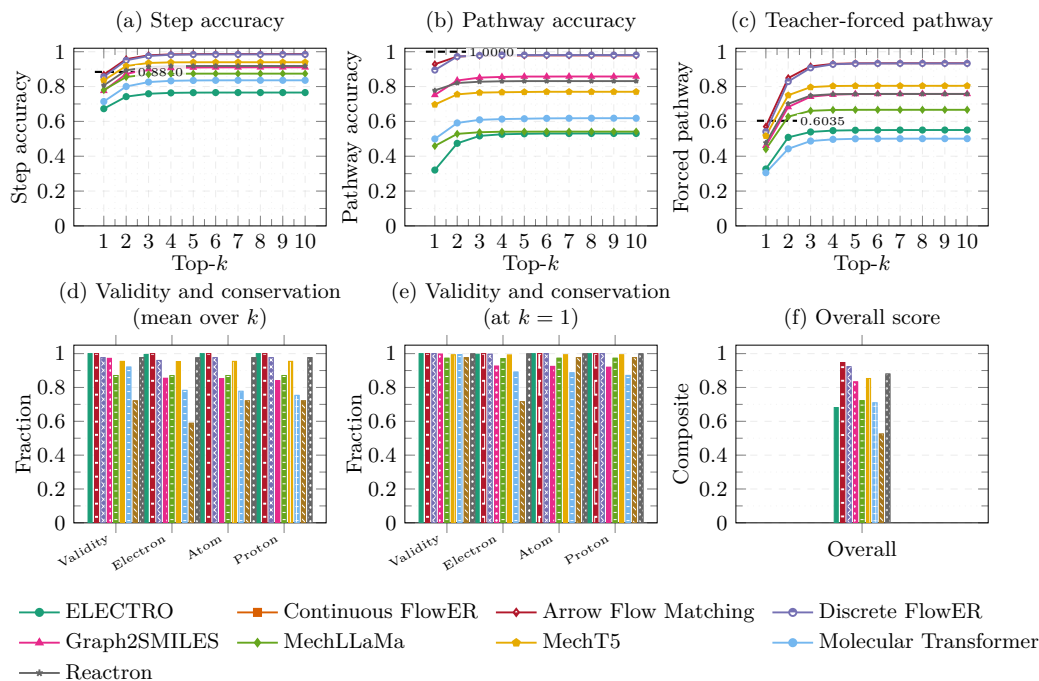


Figure 25: The comparison at the smallest width under sampling ranked by trajectory score. Continuous FlowER is absent, all its ODE draws leaving with the same weight, so it computes no score to rank by. Read against Figure 22, this isolates what the ranking rule is worth at a fixed set of draws.

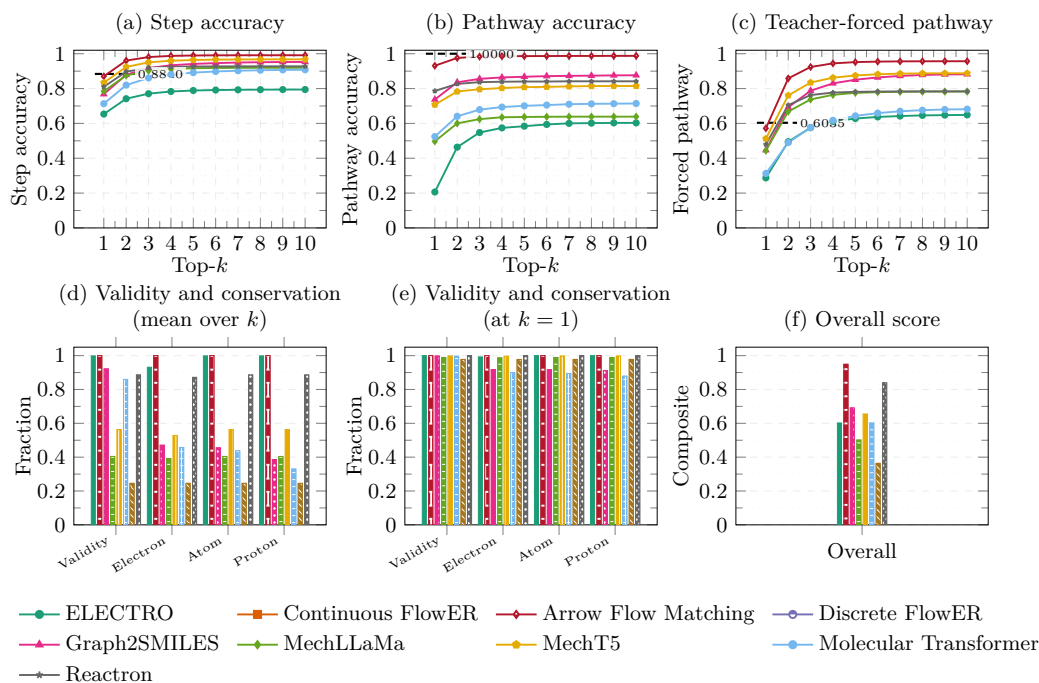


Figure 26: The comparison at the smallest width under beam search ranked by marginal probability over trajectories. Read against Figure 24, this isolates the ranking rule at a fixed set of candidates.

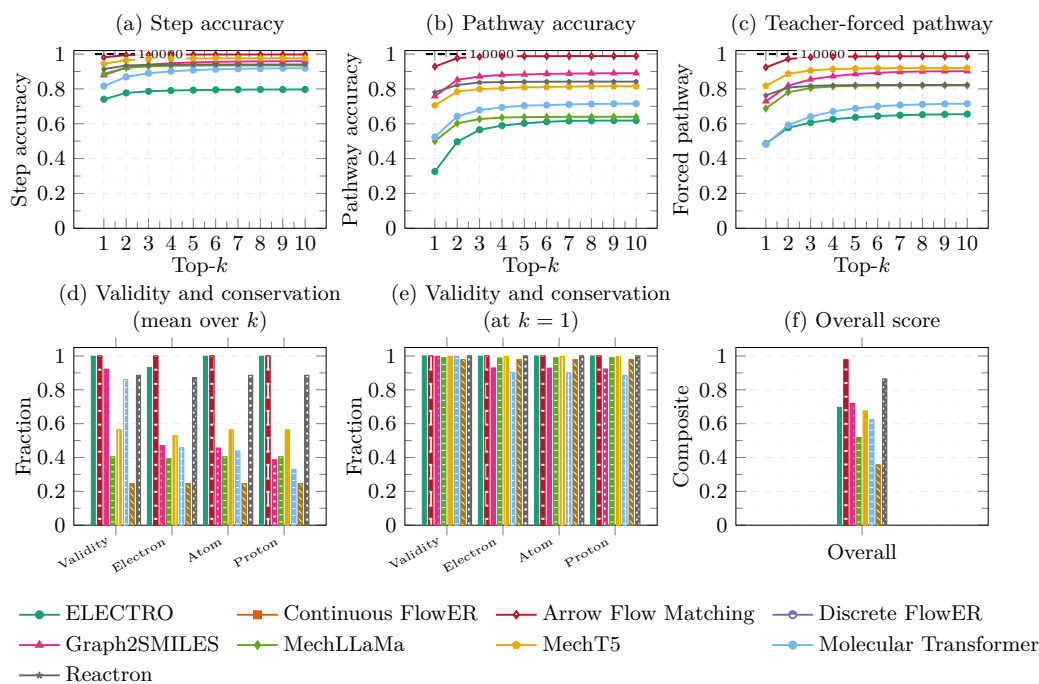


Figure 27: The comparison at the smallest width under beam search scored against the branch-aware ground truth. The predictions and the ordering are those of Figure 24, and only what counts as correct changes, so the validity and conservation panels repeat that figure exactly.

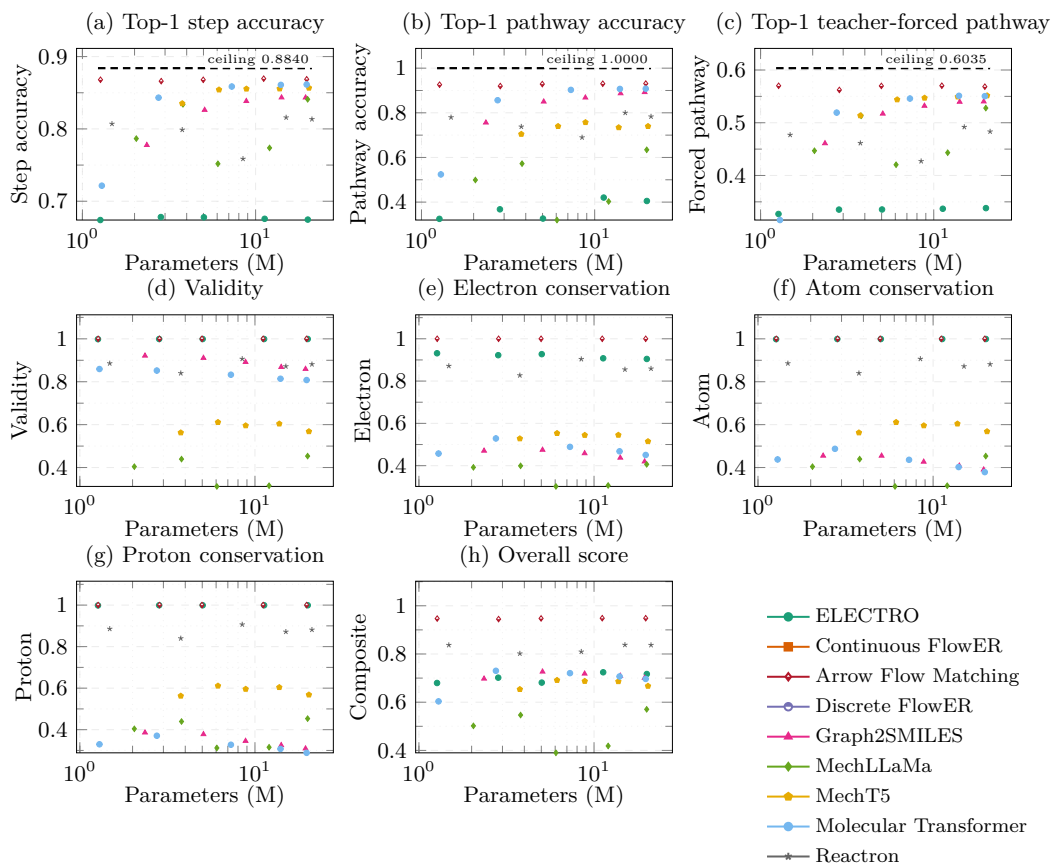


Figure 28: The size sweep under beam search ranked by trajectory score.

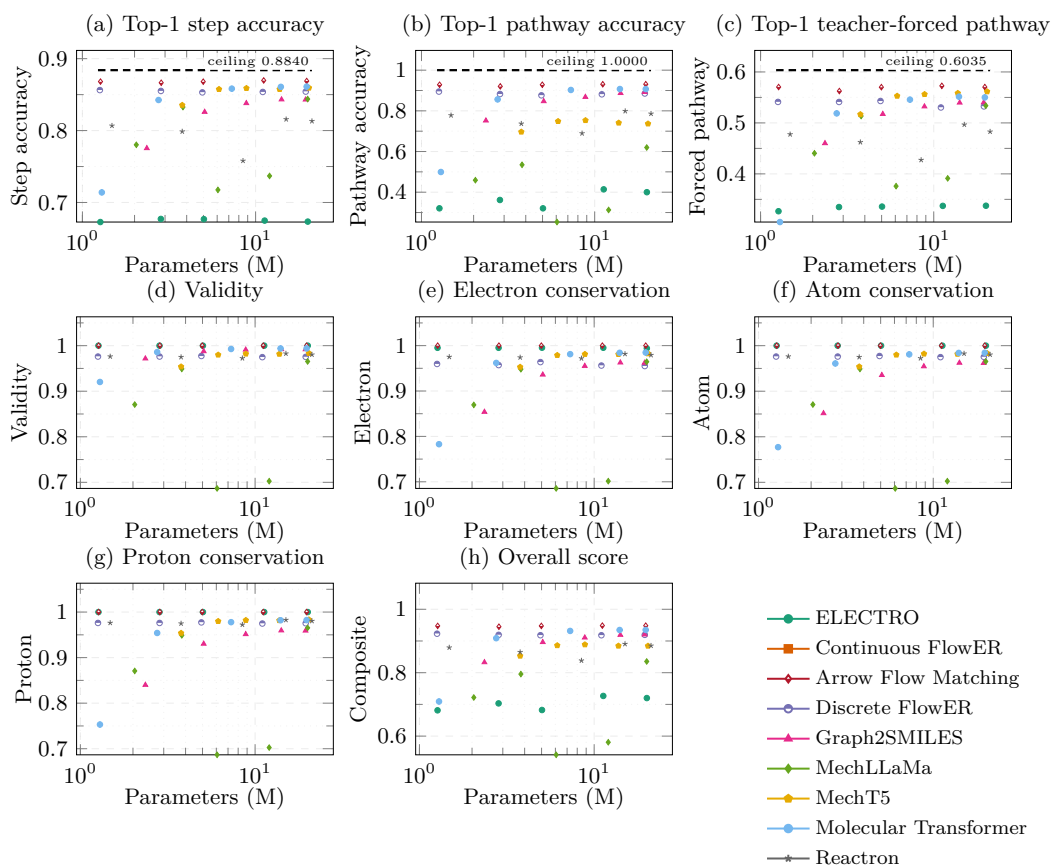


Figure 29: The size sweep under sampling ranked by trajectory score.

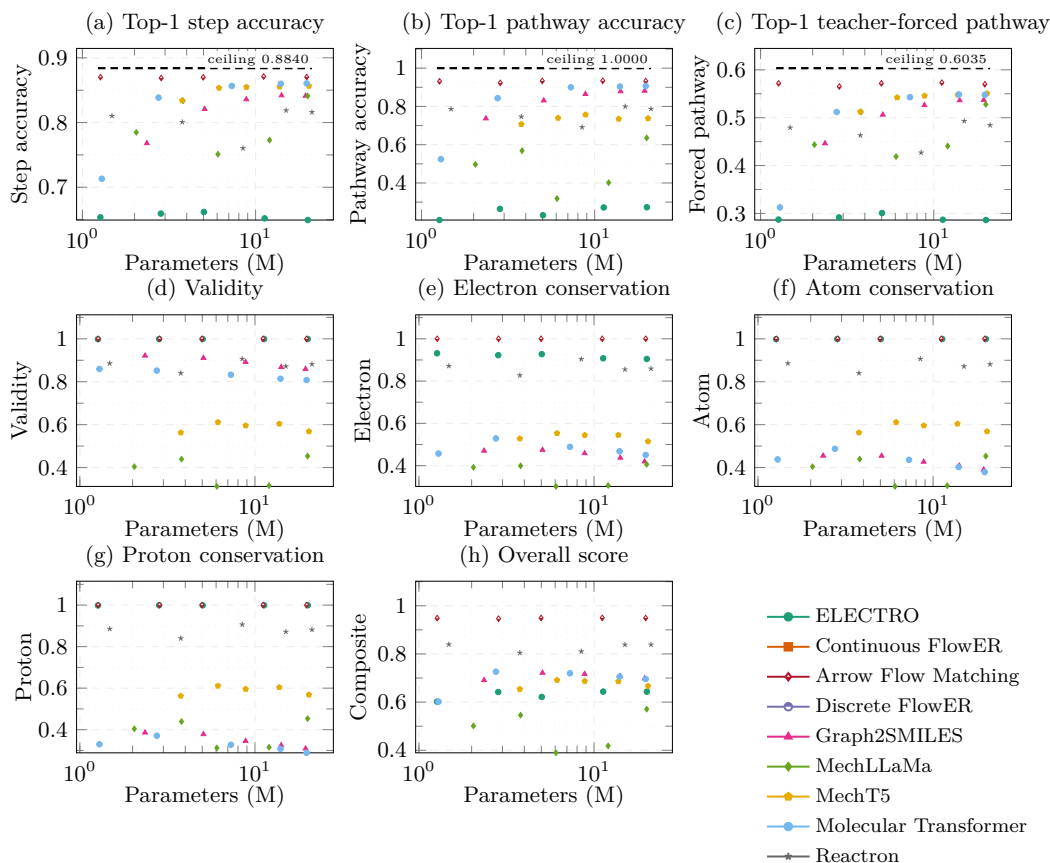


Figure 30: The size sweep under beam search ranked by marginal probability over trajectories.

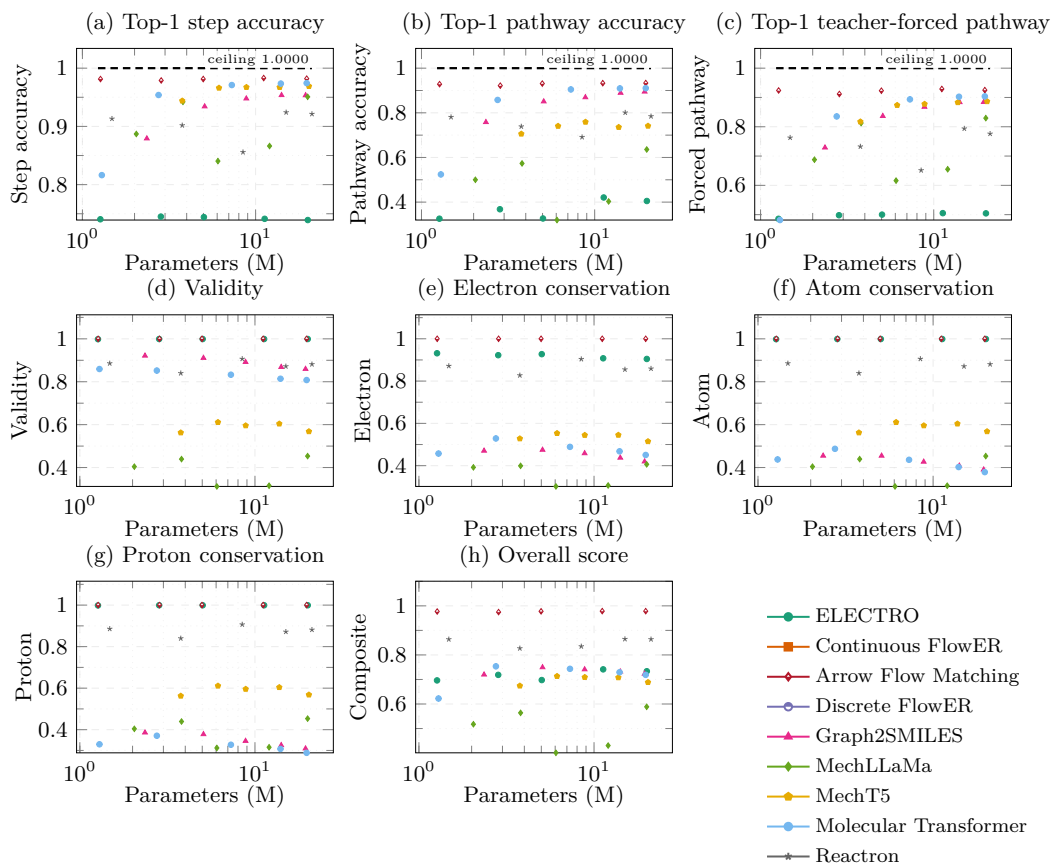


Figure 31: The size sweep under beam search scored against the branch-aware ground truth.